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# 飞行时间二次离子质谱在冬虫夏草识别上的应用


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# 为什么要用TOF-SIMS分析鉴定冬虫夏草

冬虫夏草是一种名贵药材。在青藏高原高寒地带，由蝙蝠蛾幼虫遇上了冬虫夏草菌，生长而成。有显著的治病、保健和抗衰老作用，与人参、灵芝等齐名,统被誉为“仙草”。

吴仪洛（1757年）《本草从新》，记有：“冬虫夏草四川嘉定府所产最佳，云南、贵州所产者次之。冬在土中，身活如老蚕，有毛能动，至夏则毛出之，连身俱化为草”。

清代中叶（1723年），法国人巴拉南来华采购冬虫夏草带往巴黎，经过真菌学家分析后，首次提出冬虫夏草是由“冬虫”和“夏草”两部分组成的复合体，其中“草”也不是草而是某一种真菌。






- 冬虫夏草很长时间被用作滋补、保健药品。
- 冬虫夏草具有医治疾病的功能：如甘平保肺，益肾止血，化痰止咳的功效。同时人们还发现冬虫夏草还具有抗癌、免疫调节、抗菌、镇静、催眠等功效。



- 无论是从鉴别伪劣的冬虫夏草，维护消费者利益，还是应当积极地开展基础生物学研究,科学地培育出人工冬草夏草,以作为天然资源的补充,都是是从事冬草夏草研发的科技工作者所面临的一项紧迫任务。



冬虫夏草
Cordyceps sinensis
(～30,000 USD/kg)

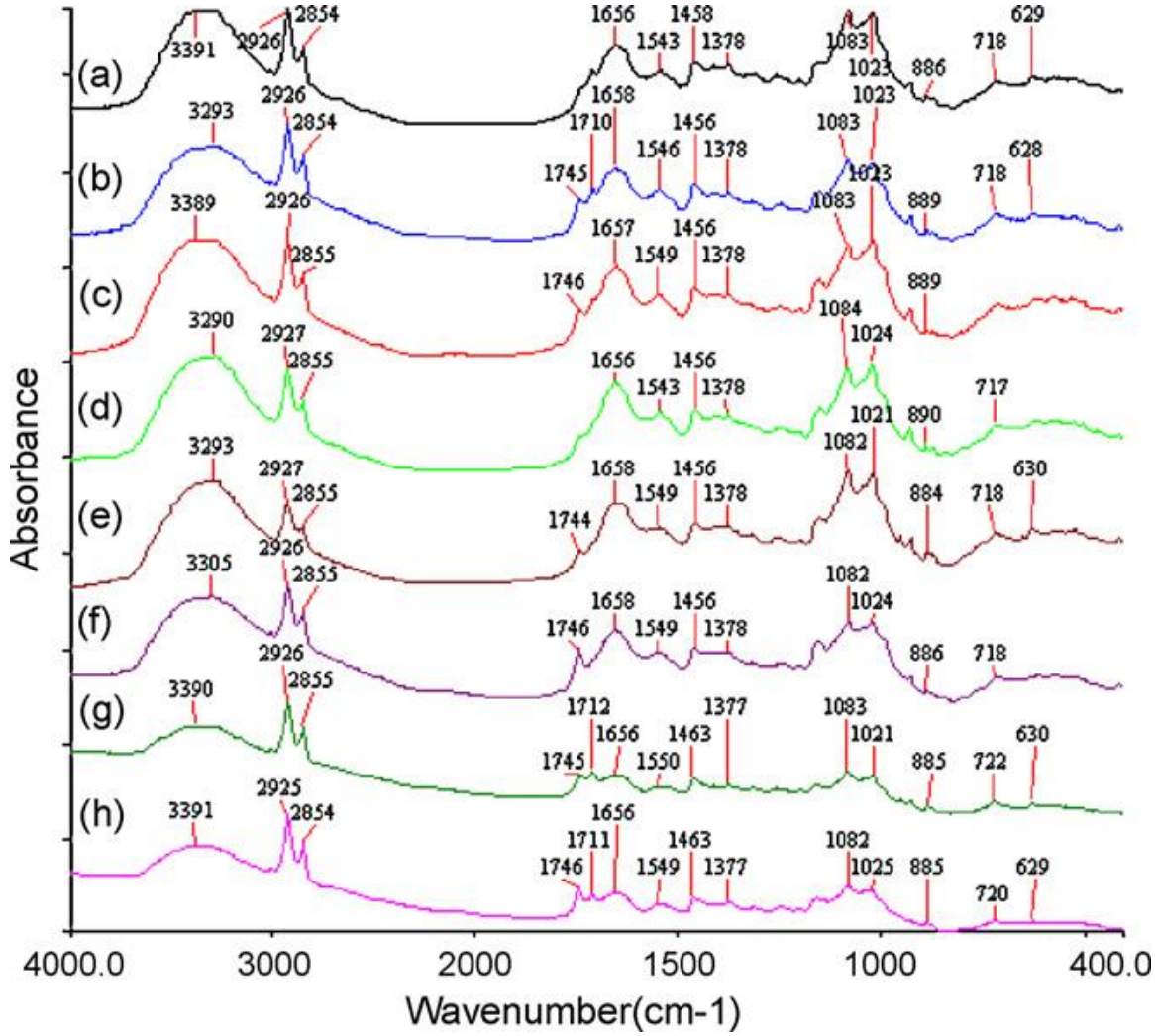


菌丝体
Cordyceps mycelium
(～100 USD/kg)



西藏白草
Pennisetum
(～1 USD/kg)

# 冬虫夏草和伪劣品的红外光谱

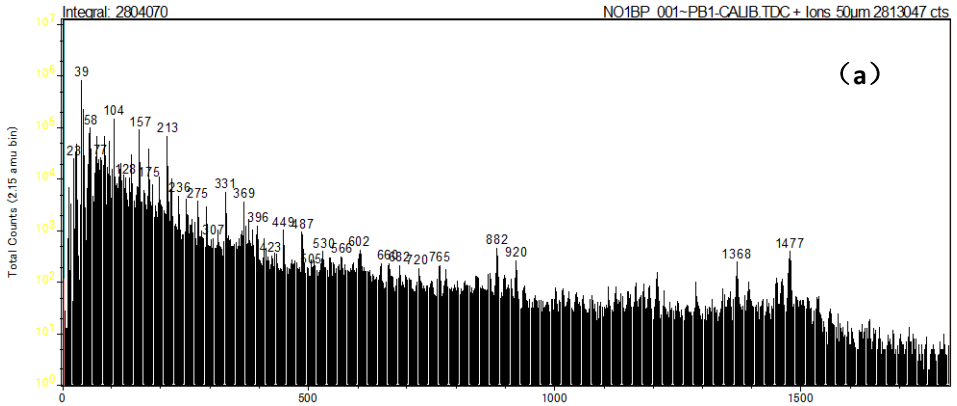


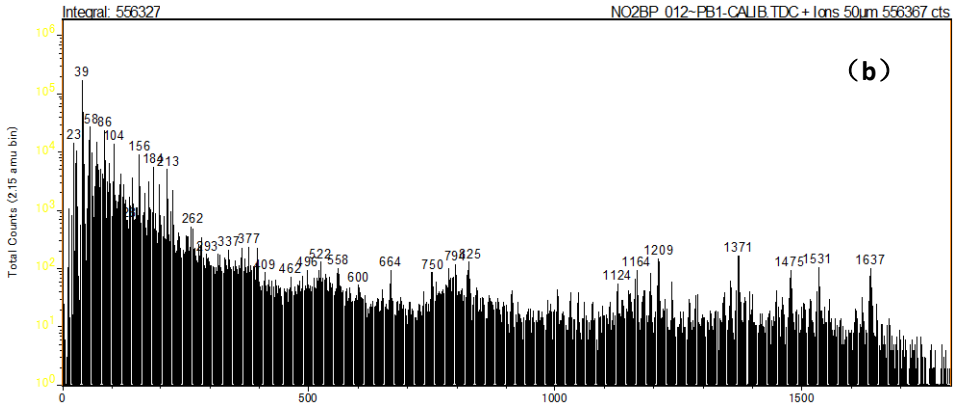

冬虫夏草不同产地红外光谱

(a) 标准样品(中国药品生物制品检定所);
(b) 青海兴海地区;
(c) 西藏那曲地区;
(d-f) 青海玉树、黄南、祁连地区;
(g-h) 伪劣、仿制品.

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# 冬虫夏草、仿制品、伪劣品的TOF-SIMS质谱图



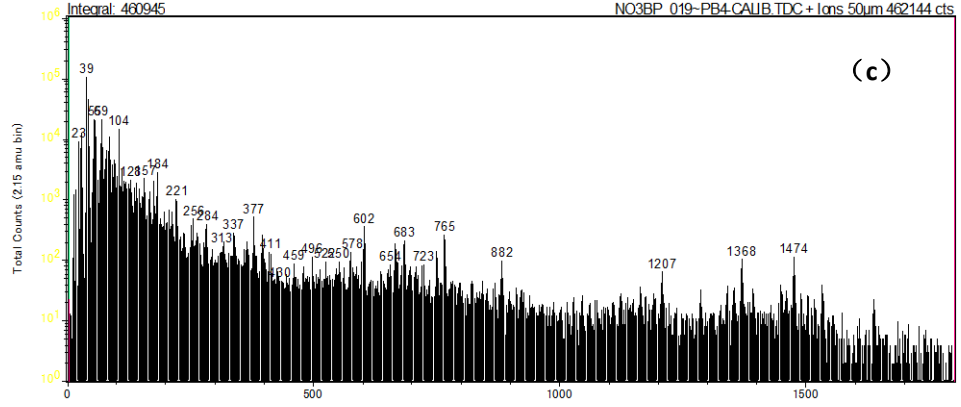
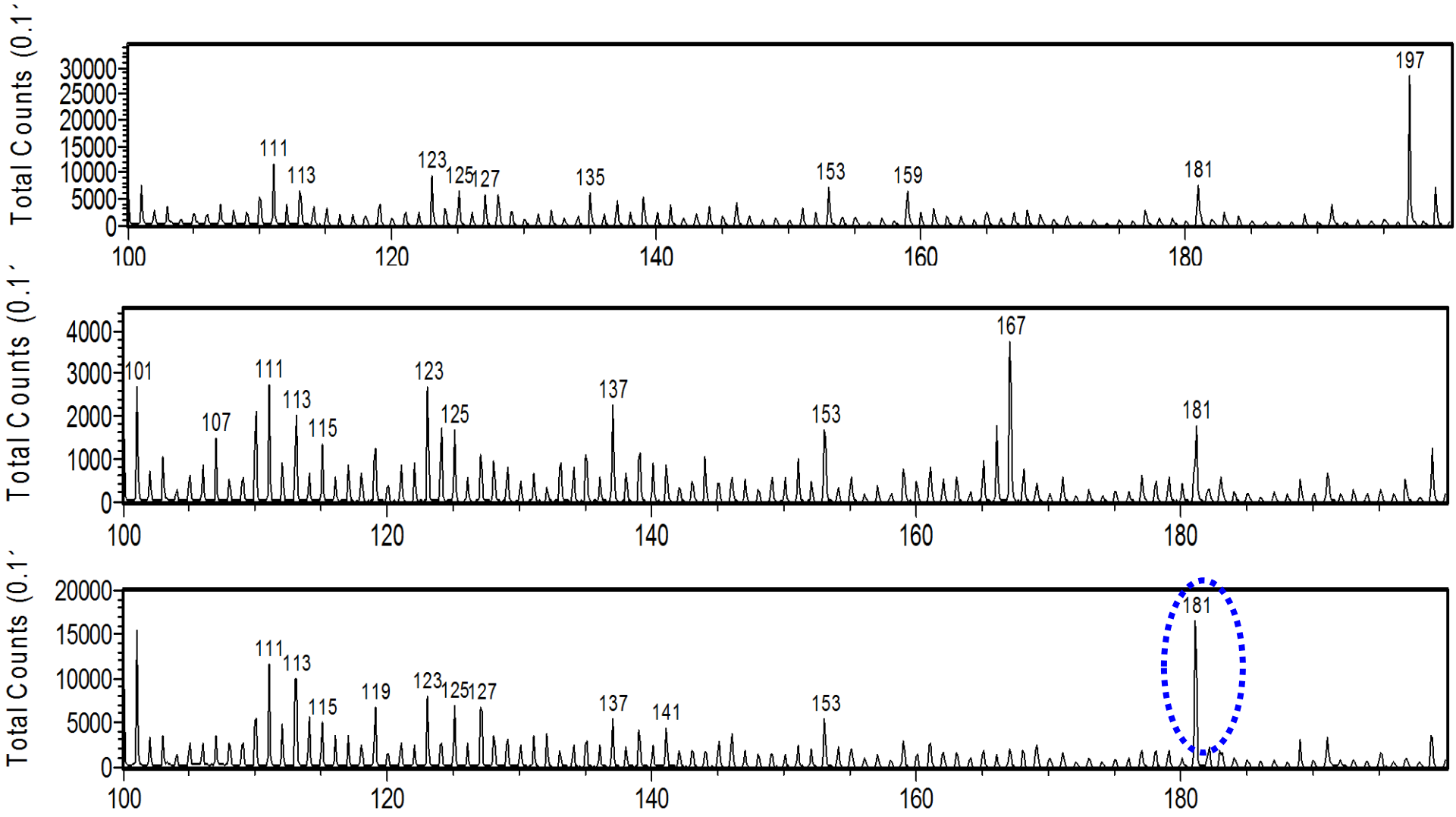




图1 TOF-SIMS 0～1800amu 正离子质谱图：(a)白草 (b)菌丝体 (c)冬虫夏草

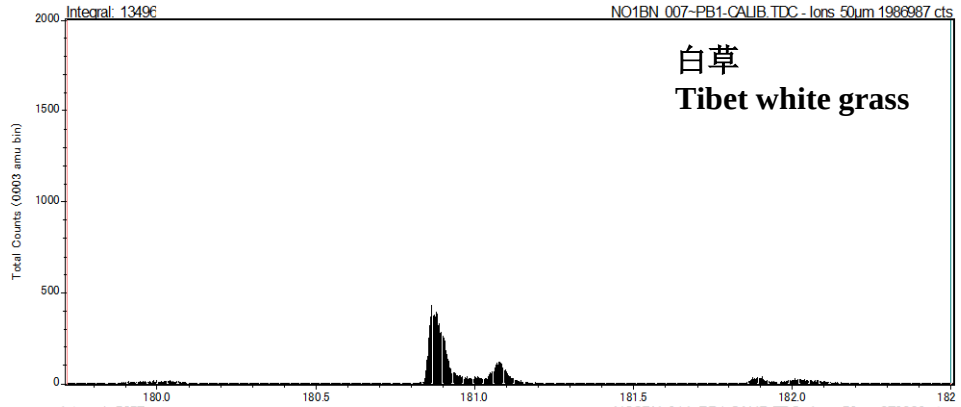
Fig.1 The positive TOF-SIMS mass spectra (0-1800amu) of Tibet white grass (a), Cordyceps sinensis mycelium (b) and Cordyceps sinensis (c)

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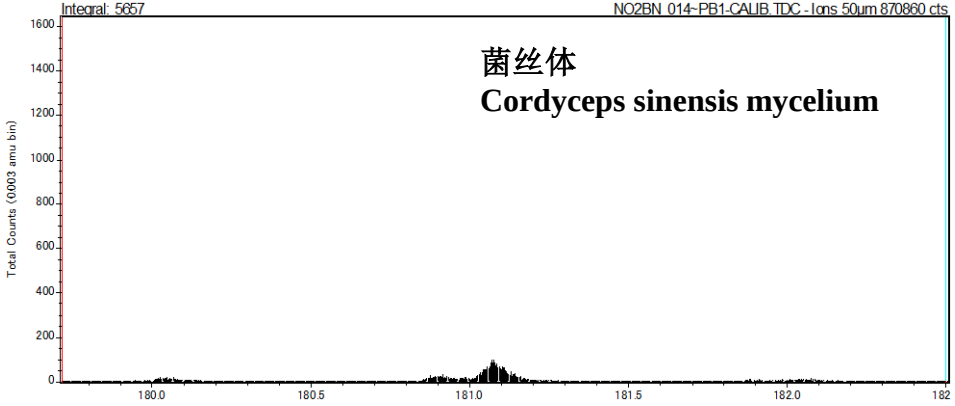
# TOF-SIMS质谱图(-Ion)：100-200amu



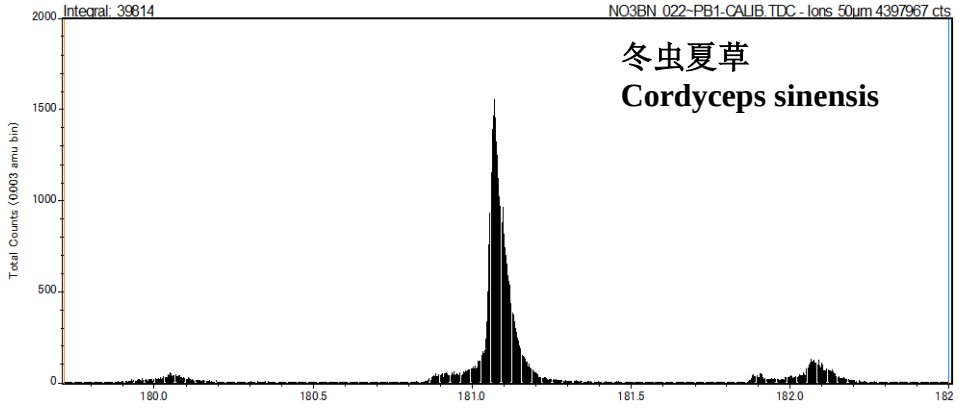
# 181amu 碎片离子(-Ion)




西藏白草
Pennisetum
(Tibet white grass)



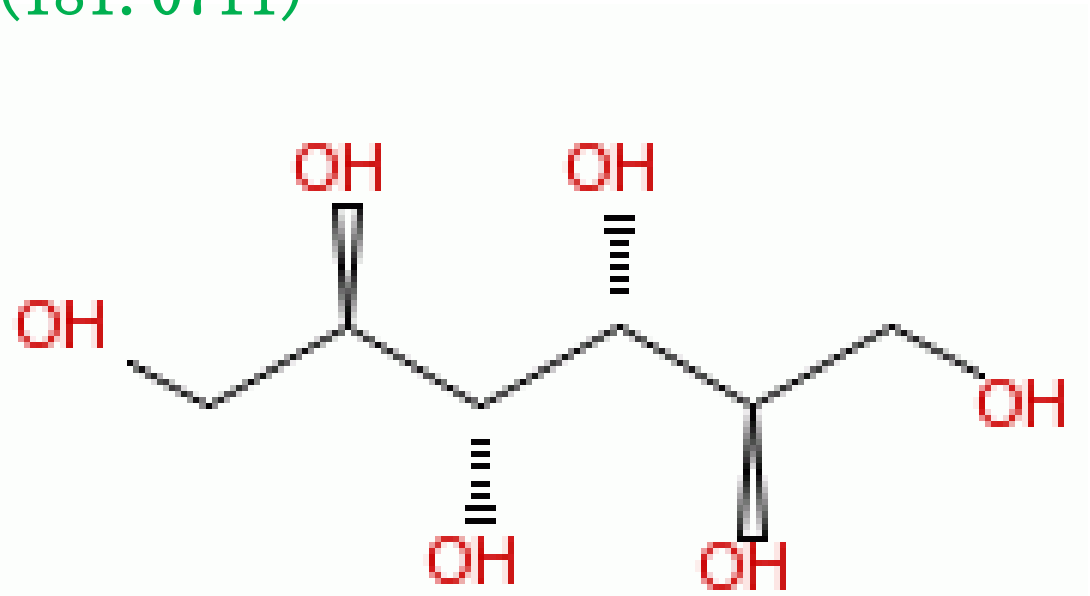

菌丝体
Cordyceps mycelium




冬虫夏草
Cordyceps sinensis

# 181amu(181.0759) 碎片离子的推定的分子式

$C_6H_{13}O_6$(181.0711)




甘露醇

Mannitol



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# TOF-SIMS分析技术

## Time-of-Flight Secondary Ion Mass Spectrometry

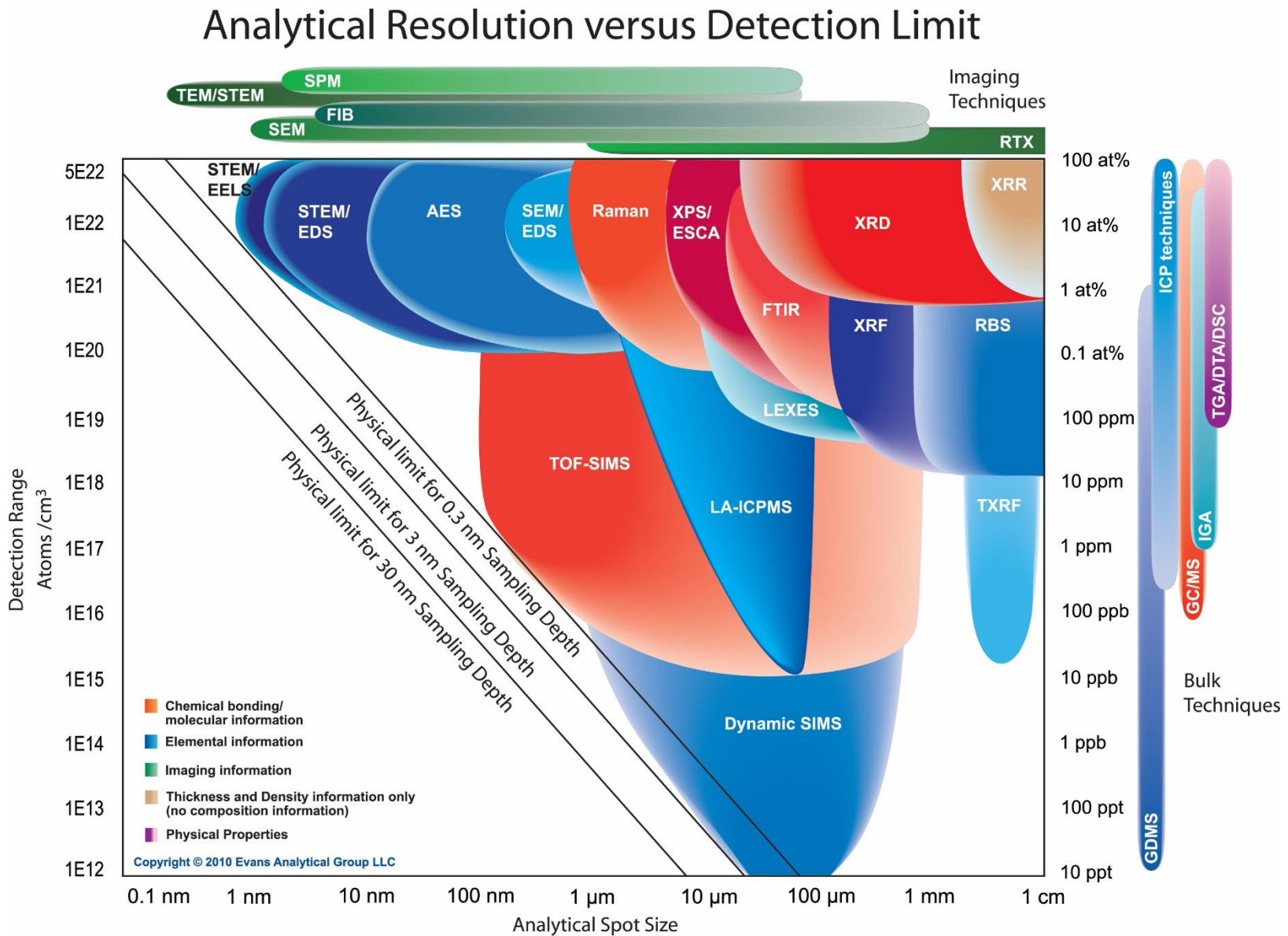
**ToF-SIMS or TOF-SIMS**

# 表面分析手段/表征表面的手段已达100多种

表面分析方法及其所获取的信息

| 分类 | 要获取的信息 | 适用的表面分析方法 |
|---|---|---|
| 构造 | 表面原子·分子排列 | STM, AFM, LEED, RHEED, ICISS, APFIM, SXRD, XSWM |
| | 微区构造(Micro-Domain) | REM, TEM, u-RHEED |
| | 表面点缺陷 | STM, ICISS |
| | 局域构造 | EXAFS, SXAFS, EELFS, NMR |
| | 结晶·超晶格构造 | TEM, REM, TED, STEM, RHEED |
| | 表面振动构造 | HREELS, RRS |
| | 吸附原子·分子的振动状态 | HREELS, IR, RRS, SERS, PAS, SFG |
| 形态 | 表面形状 | SEM, OM, XCRT, STM, AFM |
| | 宏观区域构造(Macro-Domain) | ECP, PEEM, REM, u-RHEED, CL, PL |
| | 超薄膜的膜厚 | ELL, SEM, TEM |
| | 位错，晶格缺陷，损伤 | TEM, SRT, AM, EE |
| | 界面特性 | EBIC, Strobo-SEM |
| 组成 | 元素的定性、定量 | AEG, EPMA, XPS, SIMS, ISS, PIXE, TRXRF, NRA, AA |
| | 二维元素分布 | AES, EPMA |
| | 深度剖面分析 | AES, SIMS, RBS, GDMS, APFIM |
| | 杂质分布 | SIMS, PL, CL |
| | 表面分子种类 | LIMS, FABMS, TD, EDS, PSD, IR, SPA, RDS |
| 电子状态 | 化学结合状态 | XPS, XANES |
| | 价带构造 | UPS, PIES, PYS, IPES, EELS, SEES, XPS |
| | 禁带中的局域构造 | CL, PL, DLTS |

# 分析技术 泡泡图


Analytical Resolution versus Detection Limit
Imaging Techniques
SPM
TEM/STEM
FIB
SEM
RTX
STEM/ EELS
STEM/ EDS
AES
SEM/ EDS
Raman
XPS/ ESCA
XRD
XRR
FTIR
XRF
RBS
LEXES
TOF-SIMS
LA-ICPMS
TXRF
Dynamic SIMS
Physical limit for 0.3 nm Sampling Depth
Physical limit for 3 nm Sampling Depth
Physical limit for 30 nm Sampling Depth
Chemical bonding/ molecular information
Elemental information
Imaging information
Thickness and Density information only (no composition information)
Physical Properties
Copyright © 2010 Evans Analytical Group LLC
Detection Range Atoms /cm³
5E22
1E22
1E21
1E20
1E19
1E18
1E17
1E16
1E15
1E14
1E13
1E12
100 at%
10 at%
1 at%
0.1 at%
100 ppm
10 ppm
1 ppm
100 ppb
10 ppb
1 ppb
100 ppt
10 ppt
ICP techniques
TGA/DTA/DSC
IGA
GC/MS
GDMS
Bulk Techniques
0.1 nm
1 nm
10 nm
100 nm
1 µm
10 µm
100 µm
1 mm
1 cm
Analytical Spot Size

# TOF-SIMS 特点与XPS、AES比较

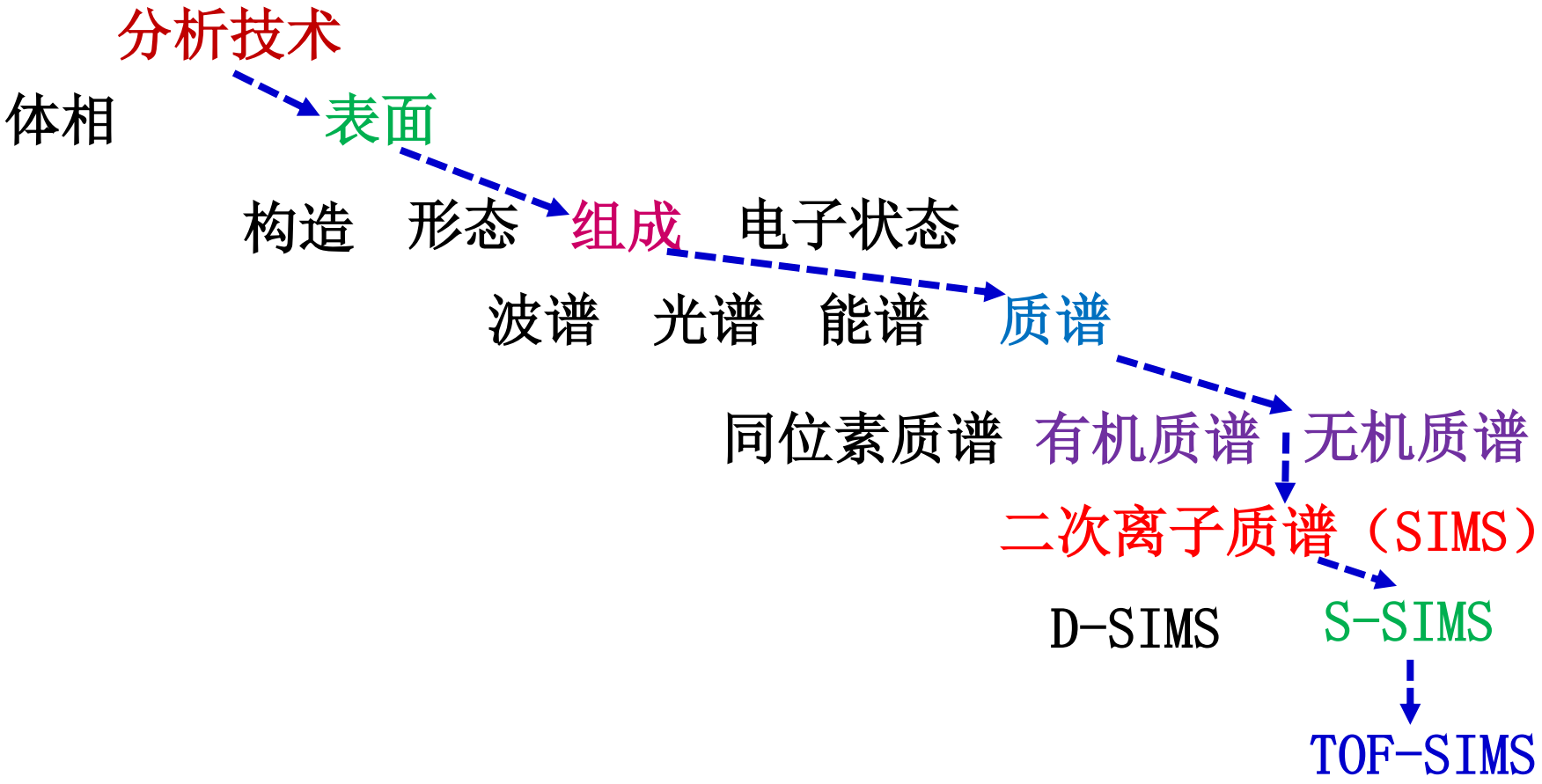
TOF-SIMS具有以下几个特点：

1）能检测出包括氢在内的所有元素及其同位素，
XPS、AES不能检测H，且不能分辨出元素的同位素。
2）对所有元素具有ppm～ppb(因元素而异)的检测灵敏度，
XPS、AES的检测极限为0.1%～0.01%。
3）能测定从表面到深至数十μm的微量元素及化合物浓度布，
AES只能分析元素，XPS最多只能分析元素的化学状态的深度分布。并且检测极限在0.1%。
4）能观测各种元素及化合物的二维以及三维的分布，
AES、XPS不能观察三维分布。
5）能鉴定有机化合物分子，
AES不能鉴定、XPS只能鉴定到元素的化学状态（价态）。

# 表面分析技术比较

| | AES | ESCA/XPS | TOF-SIMS | D-SIMS |
|---|---|---|---|---|
| 有机分析 | * | **** | **** | * |
| 灵明度– 检测极限 | ** | ** | ***** | **** |
| 化学键 | *** | **** | ** | * |
| 分子重量 | | | **** | |
| 空间分辨率 | ***** | *** | **** | ** |
| 绝缘物的分析 | ** | ***** | ***** | *** |
| 深度剖析灵明度 | ** | ** | **** | ***** |
| 深度剖析速度 | * | * | *** | ***** |
| 定量 | **** | ***** | * | * |

# TOF-SIMS 在分析技术中的节点

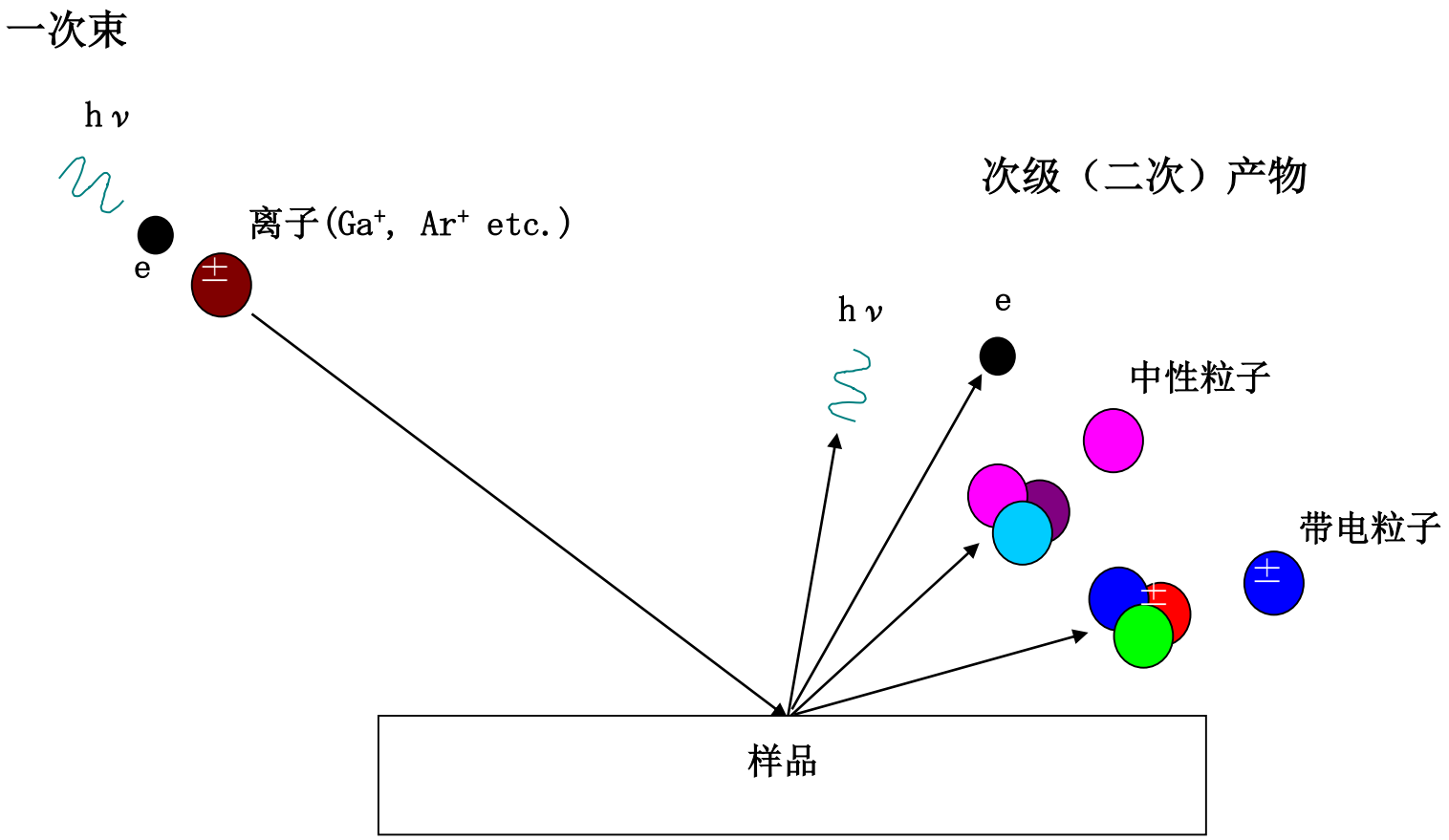


# 为什么能实现表面分析？

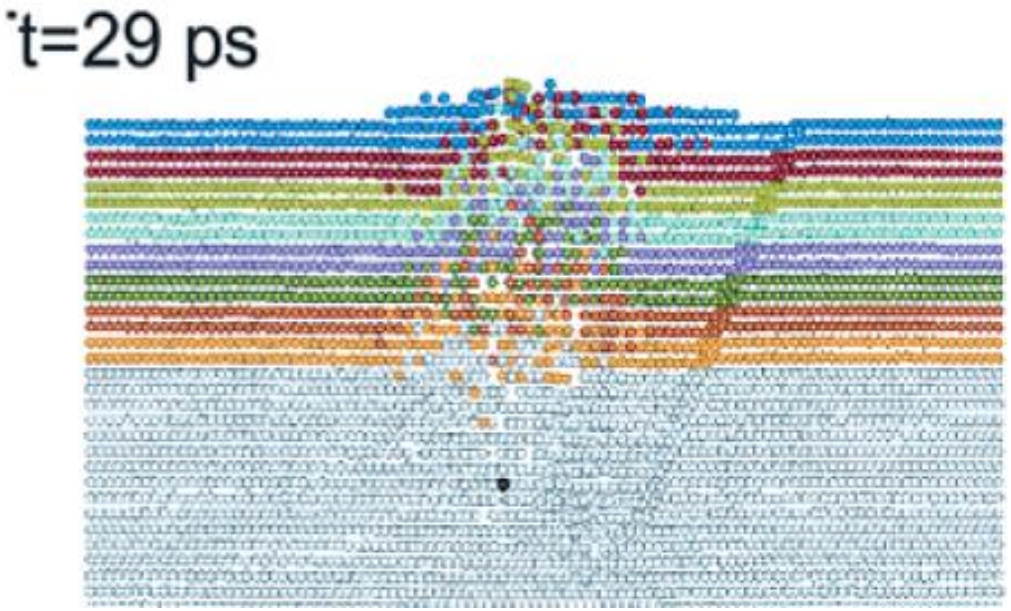
…

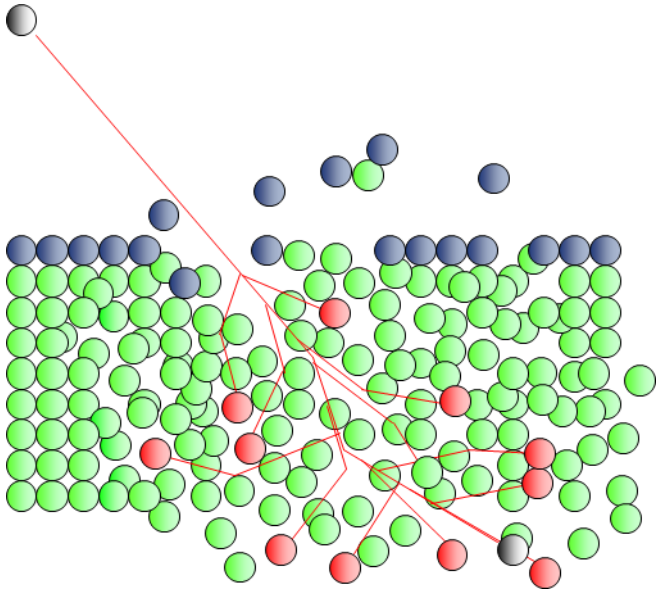
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# 一次束与二次（次级）产物



# 实现表面分析的依据



Z. Postawa, B. Czerwinski, M. Szewczyk, E. J. Smiley, N. Winograd and B. J. Garrison., J. Phys. Chem. B108, 7831 (2004).

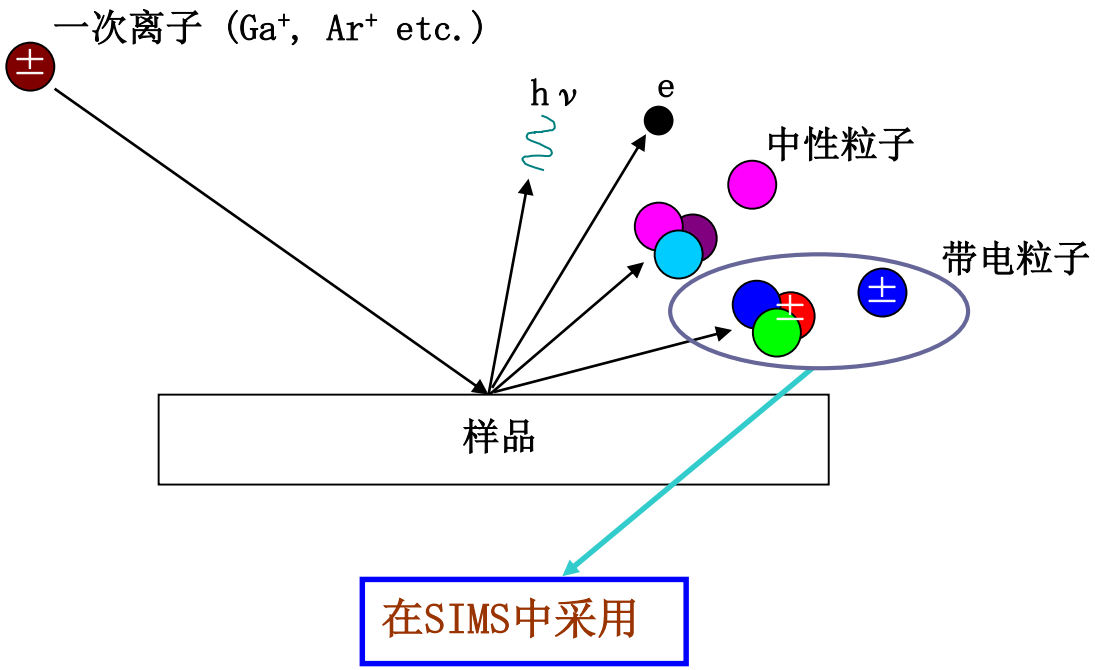
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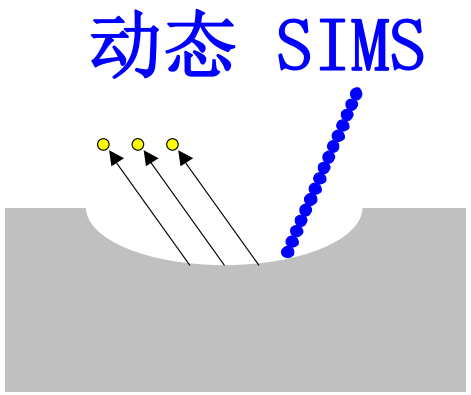
# 二次离子质谱（次级离子质谱）

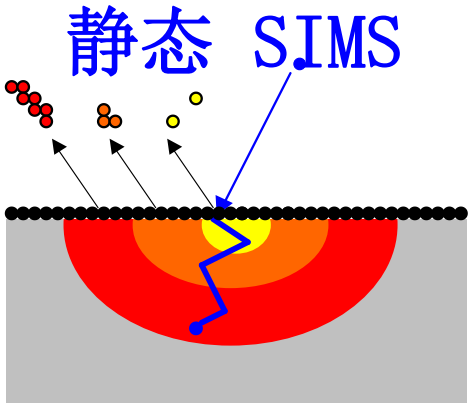
# Secondary Ion Mass Spectroscopy（SIMS）



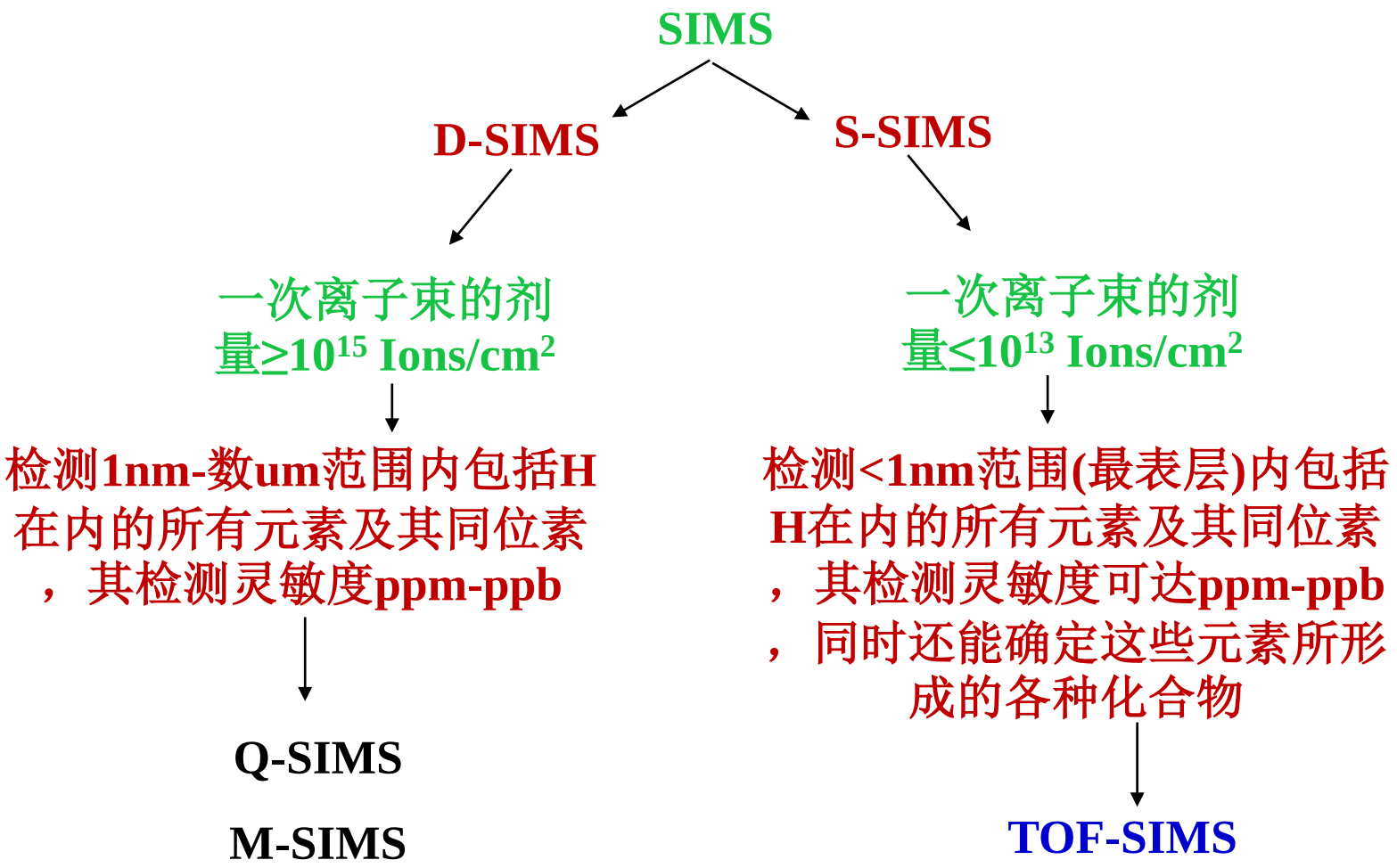
# 二次离子质谱的原理



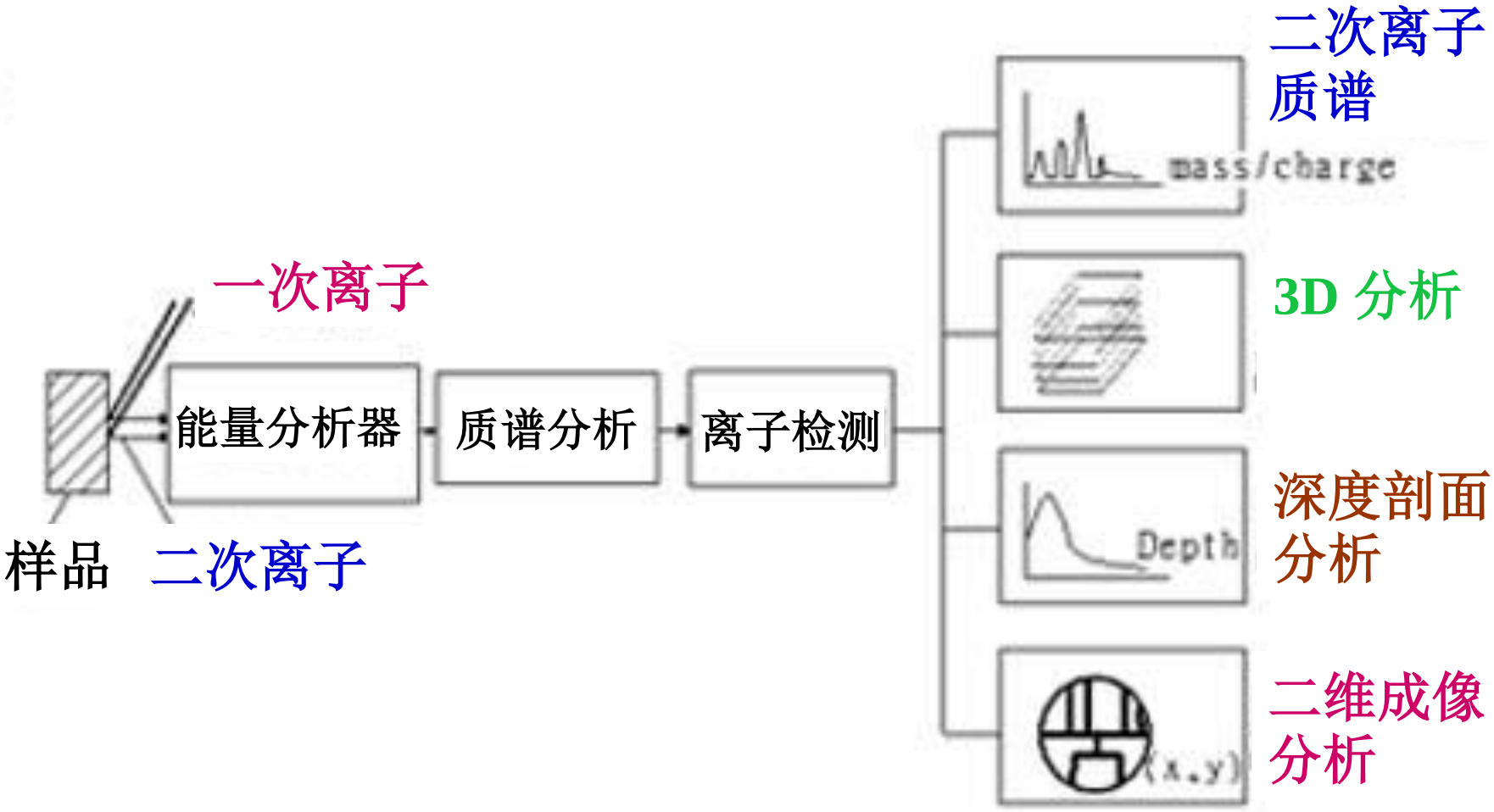



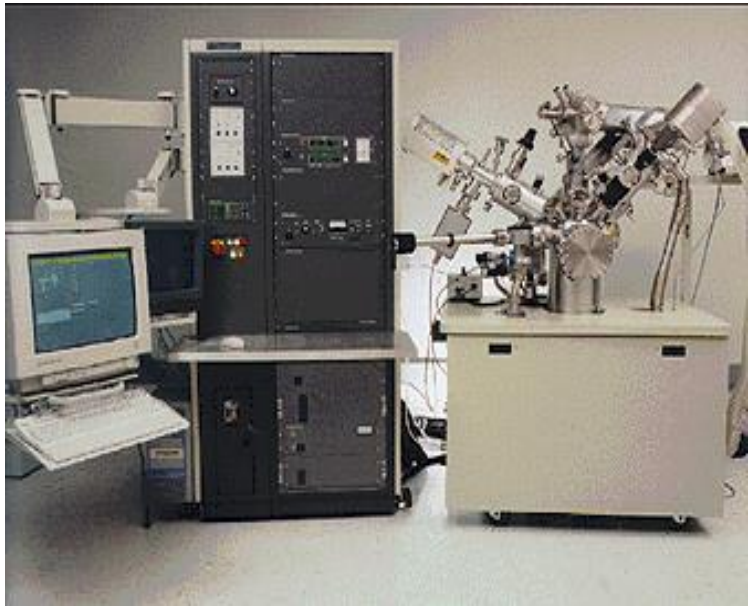
# 二次离子质谱的分类


SIMS
D-SIMS
S-SIMS
一次离子束的剂量≥10^15 Ions/cm^2
一次离子束的剂量≤10^13 Ions/cm^2
检测1nm-数um范围内包括H在内的所有元素及其同位素，其检测灵敏度ppm-ppb
检测<1nm范围(最表层)内包括H在内的所有元素及其同位素，其检测灵敏度可达ppm-ppb，同时还能确定这些元素所形成的各种化合物
Q-SIMS
M-SIMS
TOF-SIMS

# 二次离子质谱概略图


一次离子
能量分析器
质谱分析
离子检测
样品
二次离子
mass/charge
Depth
(x,y)
二次离子质谱
3D 分析
深度剖面分析
二维成像分析

# 二次离子质谱装置



PHI ADEPT-1010



Ion-TOF TOF.SIMS-V



CAMECA IMS 7f



PHI TRIFT-V

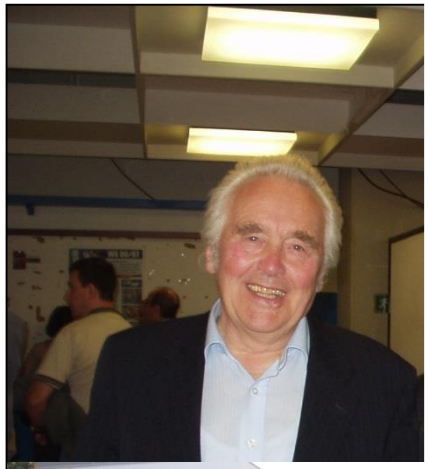
# 二次离子质谱的起源

1）起源可以追溯到上世纪初的1910年英国物理学家汤姆逊（电子的发现者，J.J. Thomson)。汤姆逊用Kanalstrahlen阴极射线打到金属板上时发现有“次级射线”向各个方向出射，它们其中大部分是中性粒子，其中的一小部分是带正电的。

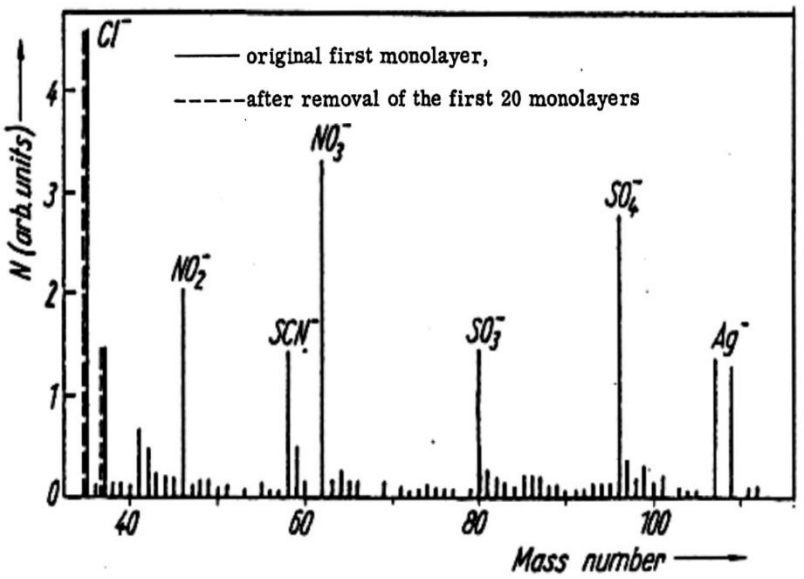
2）1931年Woodcock观察了第一幅的质谱图（ 在样品NaF、$CaF_2$ 上得到的负离子质谱)

3）1969年德国的Benninghoven教授开发了一种全新的分析手段，Benninghoven教授把这种分析手段命名为静态次级离子（二次离子）质谱（Static-SIMS），而把通常进行深度剖面分析的SIMS称作动态次级离子（二次离子）质谱（Dynamic-SIMS），

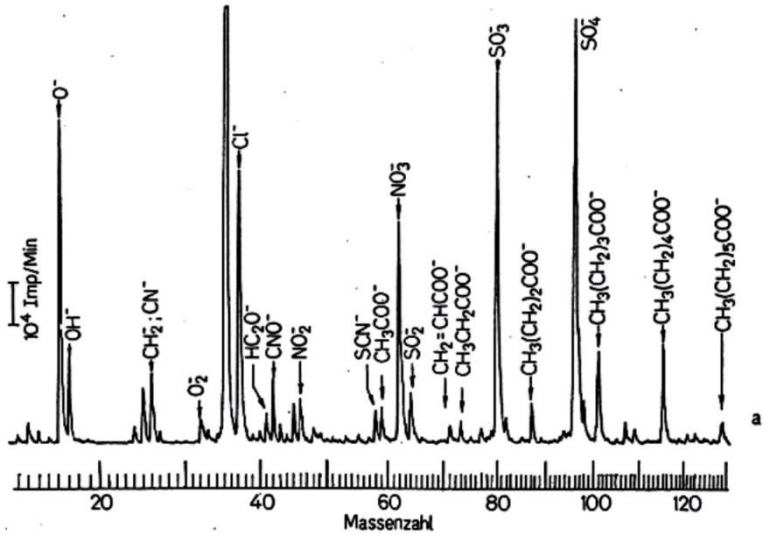
# 1969年德国的Benninghoven教授得到的最初的静态SIMS质谱图








On Ag catalytic agent



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# TOF-SIMS I

**1982, 1 mm, 10 ns, Static SIMS only**



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# TOF-SIMS装置



Ion-TOF TOF.SIMS 5

PHI TRIFT-V



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# 清华大学于2013年8月引入一台目前最新型号的TOF-SIMS

型号：　　TOF.SIMS 5-100

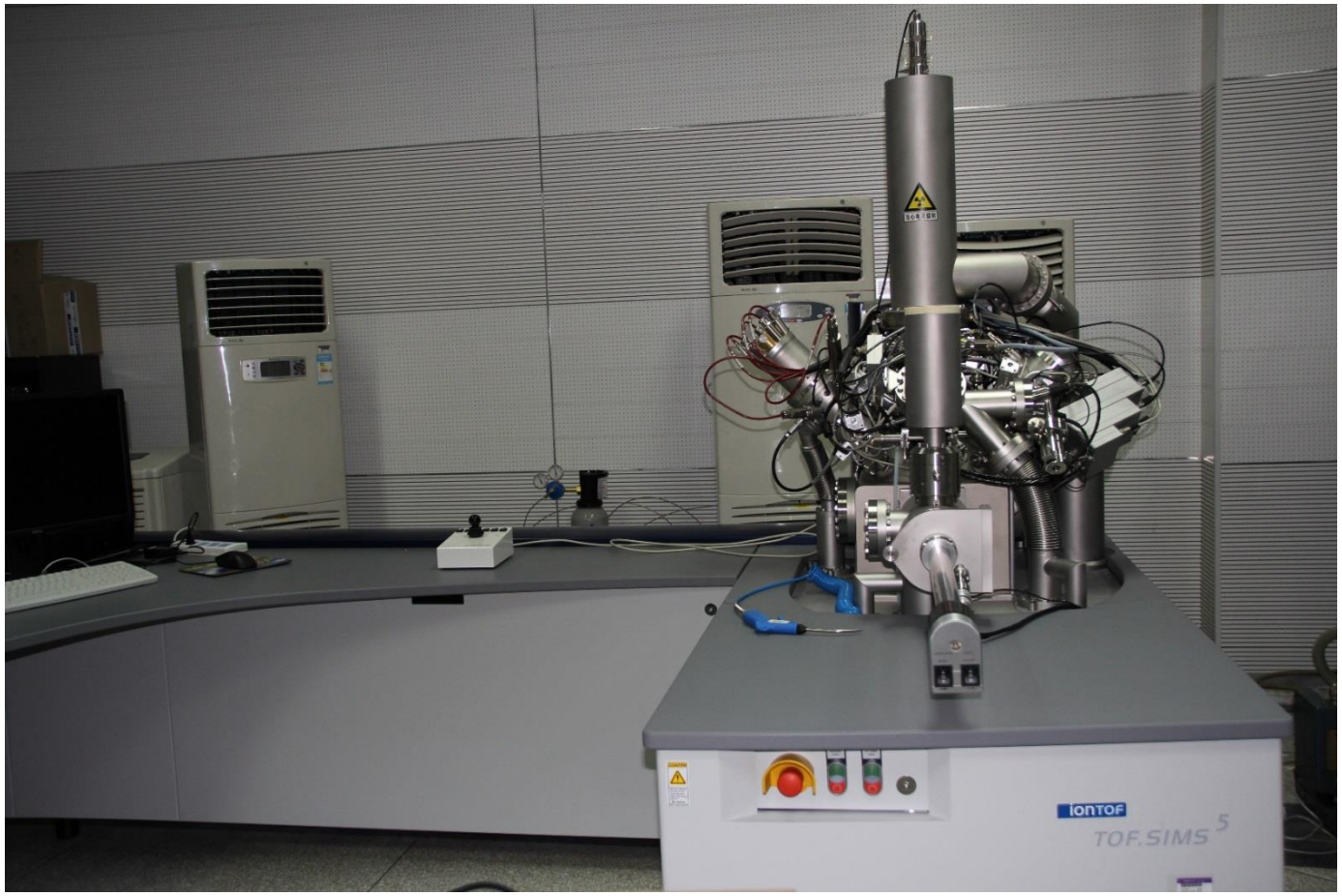
厂家：　　ION-TOF GmbH（德国）



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# TOF.SIMS 5-100

## 放置地点：理科楼 D-104



# TOF-SIMS 原理

不仅要分析

1）H, C, O, N, CN, Si, $C_3H_5$, Fe, ……

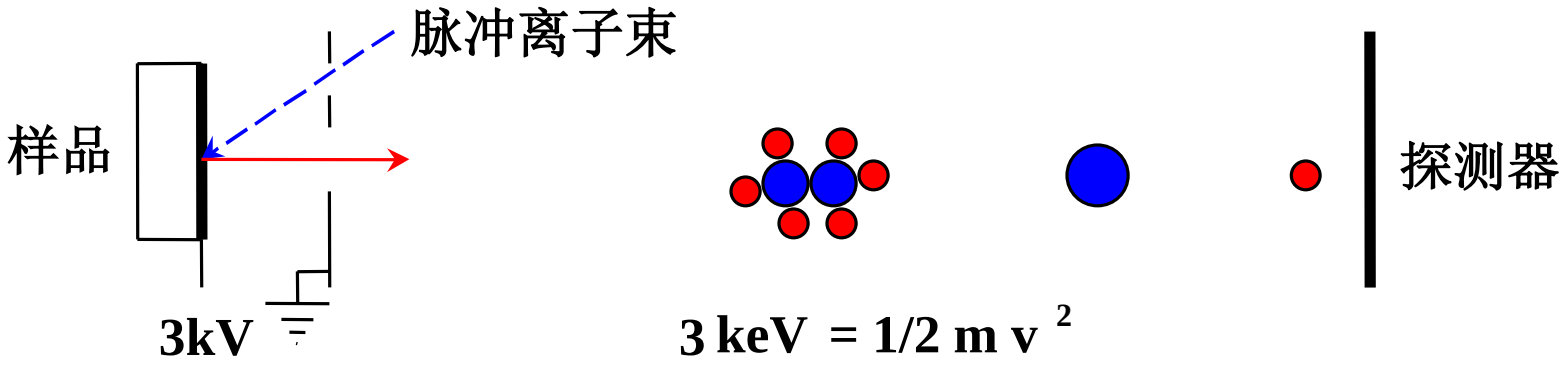
还要分析

2）Fe, COSi, $SiCH_2N$, $C_3H_4O$, $C_3H_6N$, $C_4H_8$, ……

……



# TOF-SIMS 原理



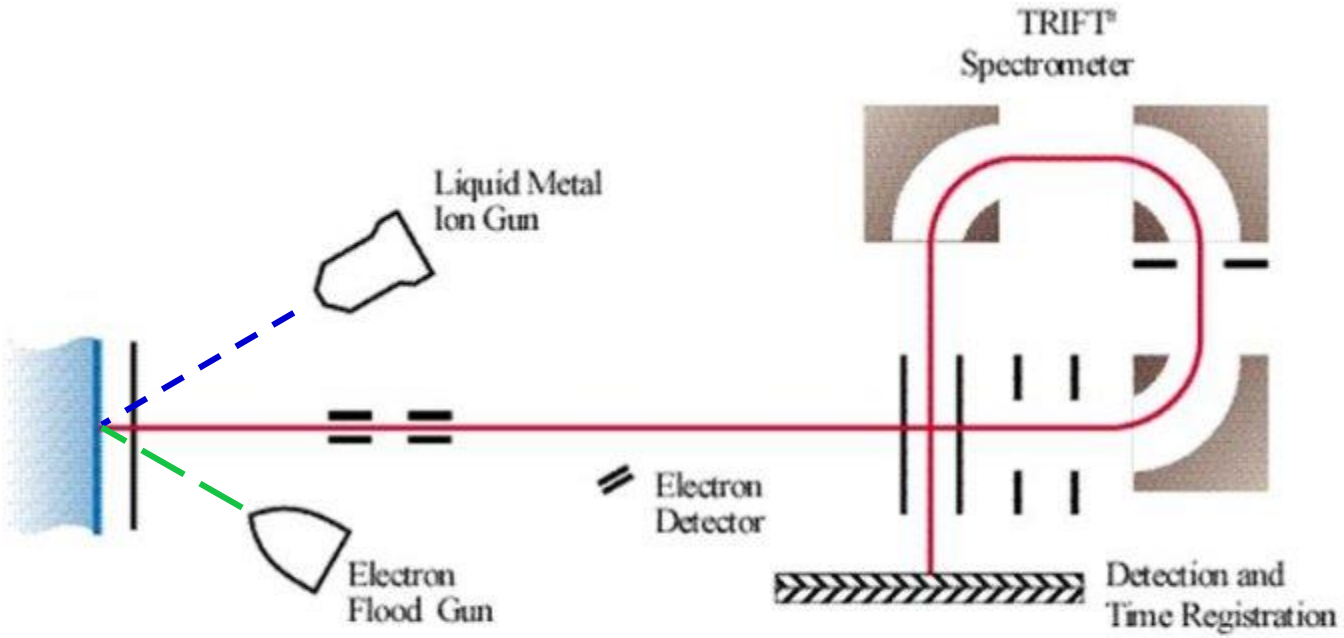

质量为m的离子的飞行时间： $t = k(m)^{1/2}$

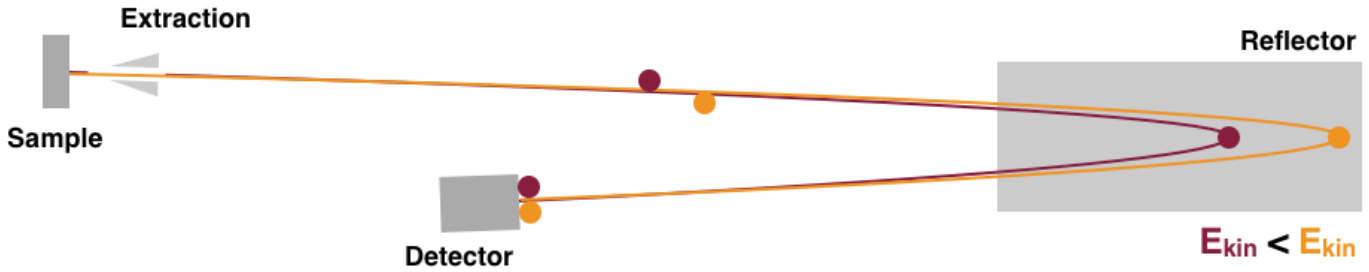
飞行时间转换成质量： $m = a\,t^{2} + b$

选取2个以上已知质量的谱峰用于校准

通过最小2乘法精确算出a、b

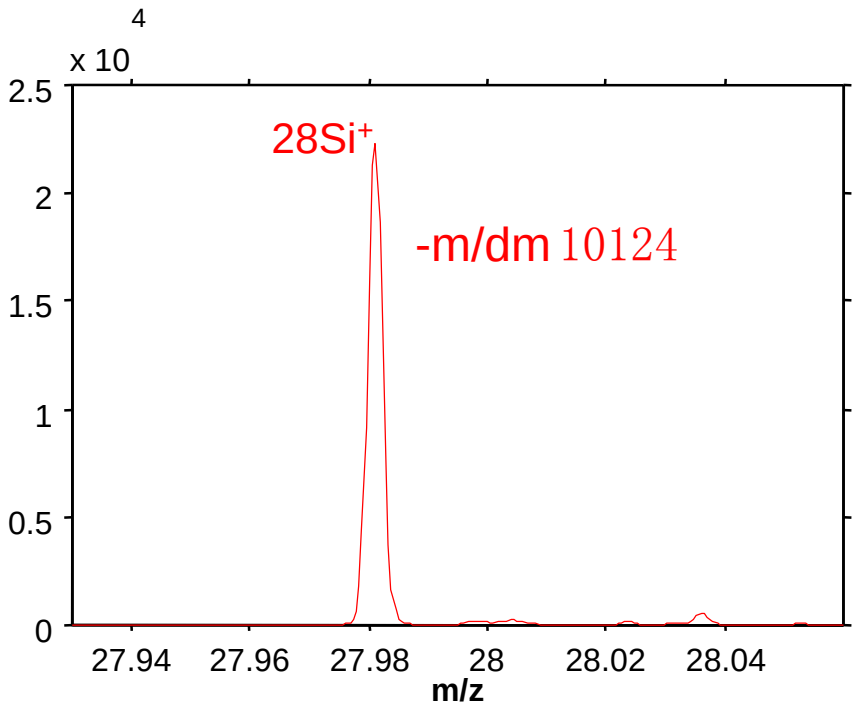
# TOF-SIMS 的分析器

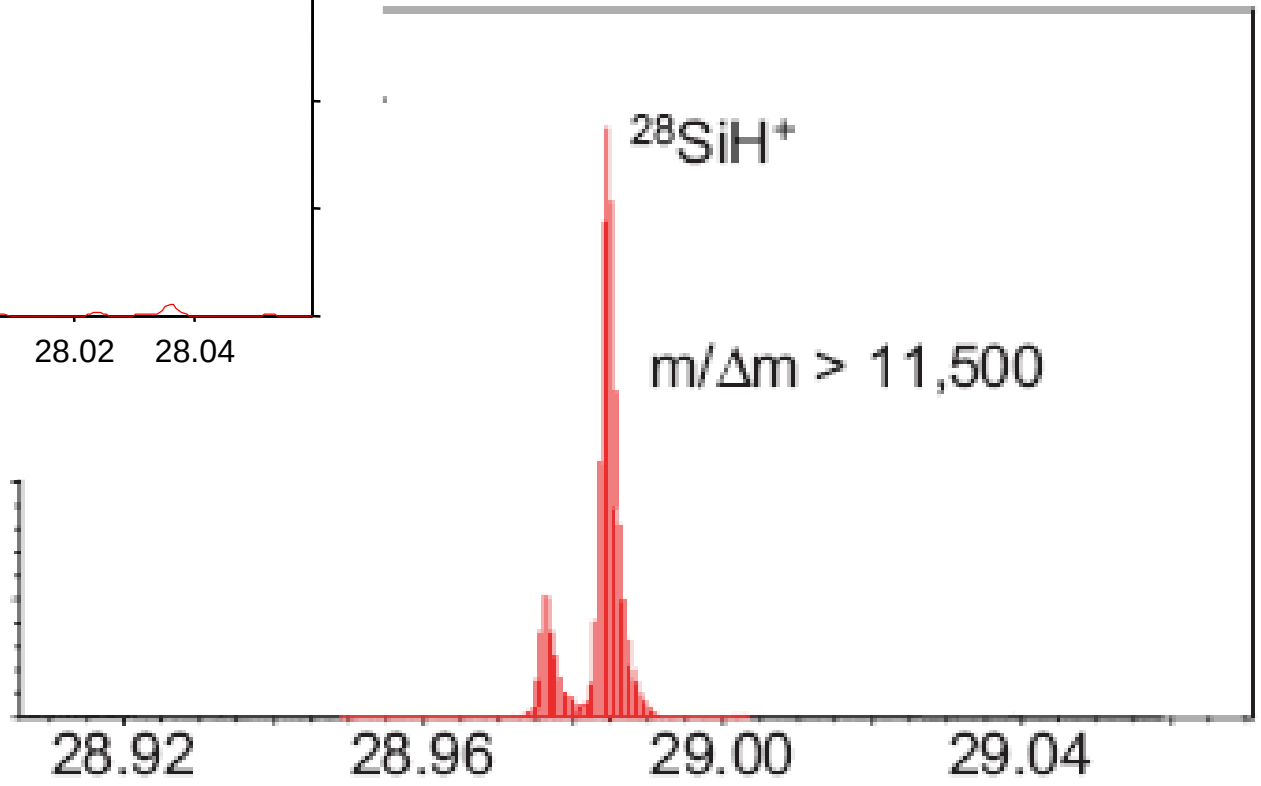


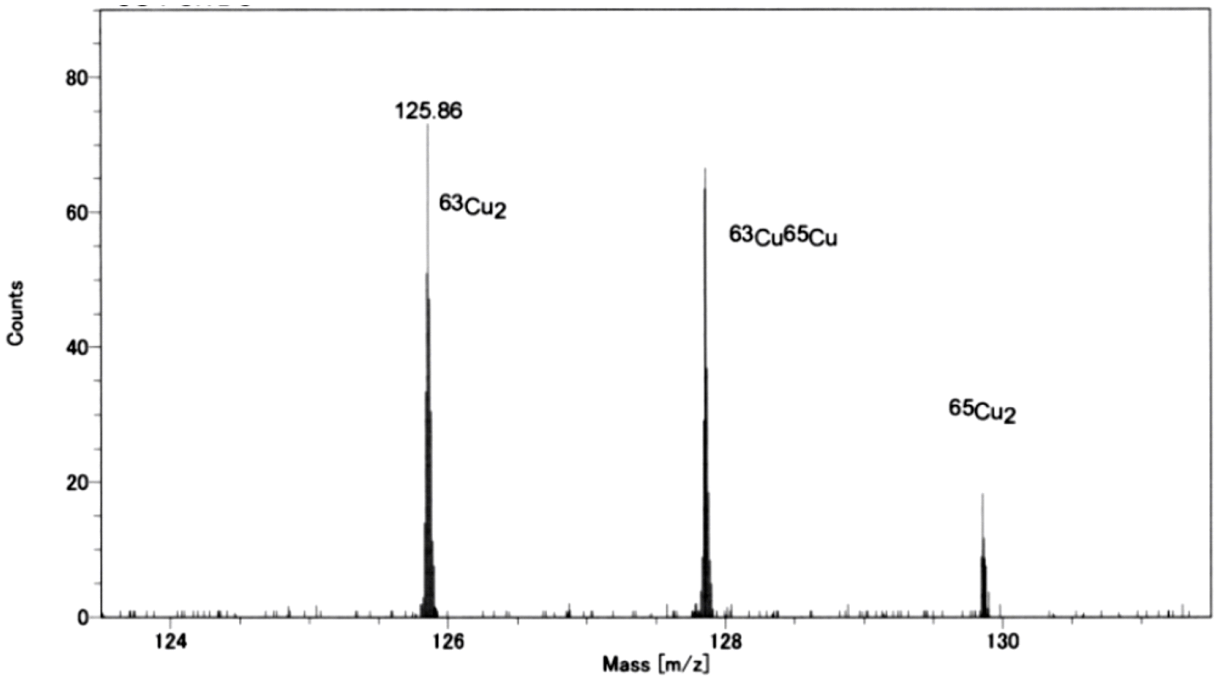
# 质量分辨率超过10000

$^{1}H^{+}$: 1.007825
$^{28}Si^{+}$: 27.976928
$^{29}Si^{+}$: 28.976496
$^{28}Si^{1}H^{+}$: 28.984753



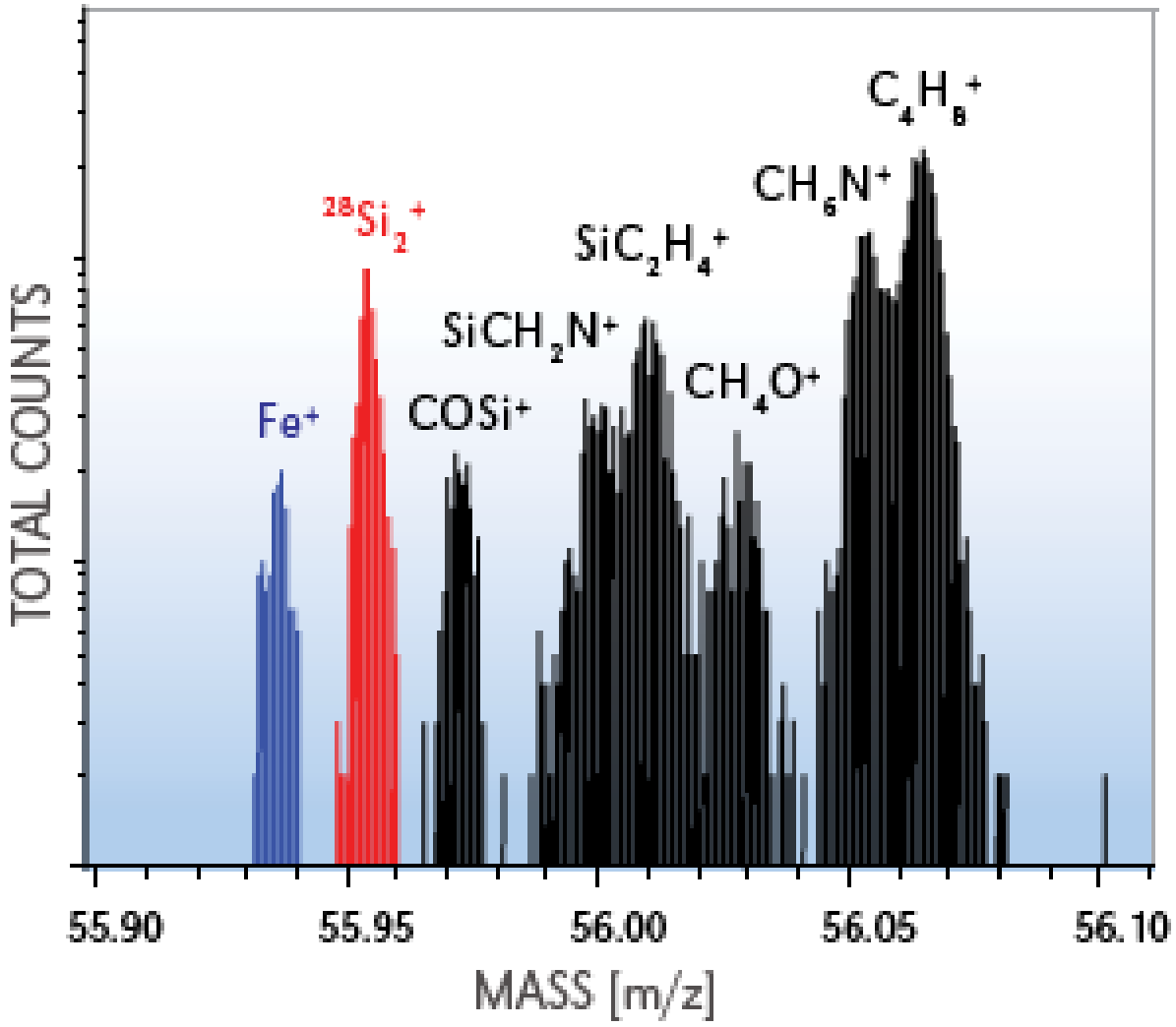
# 質量精度　δM＜1～0.1mamu



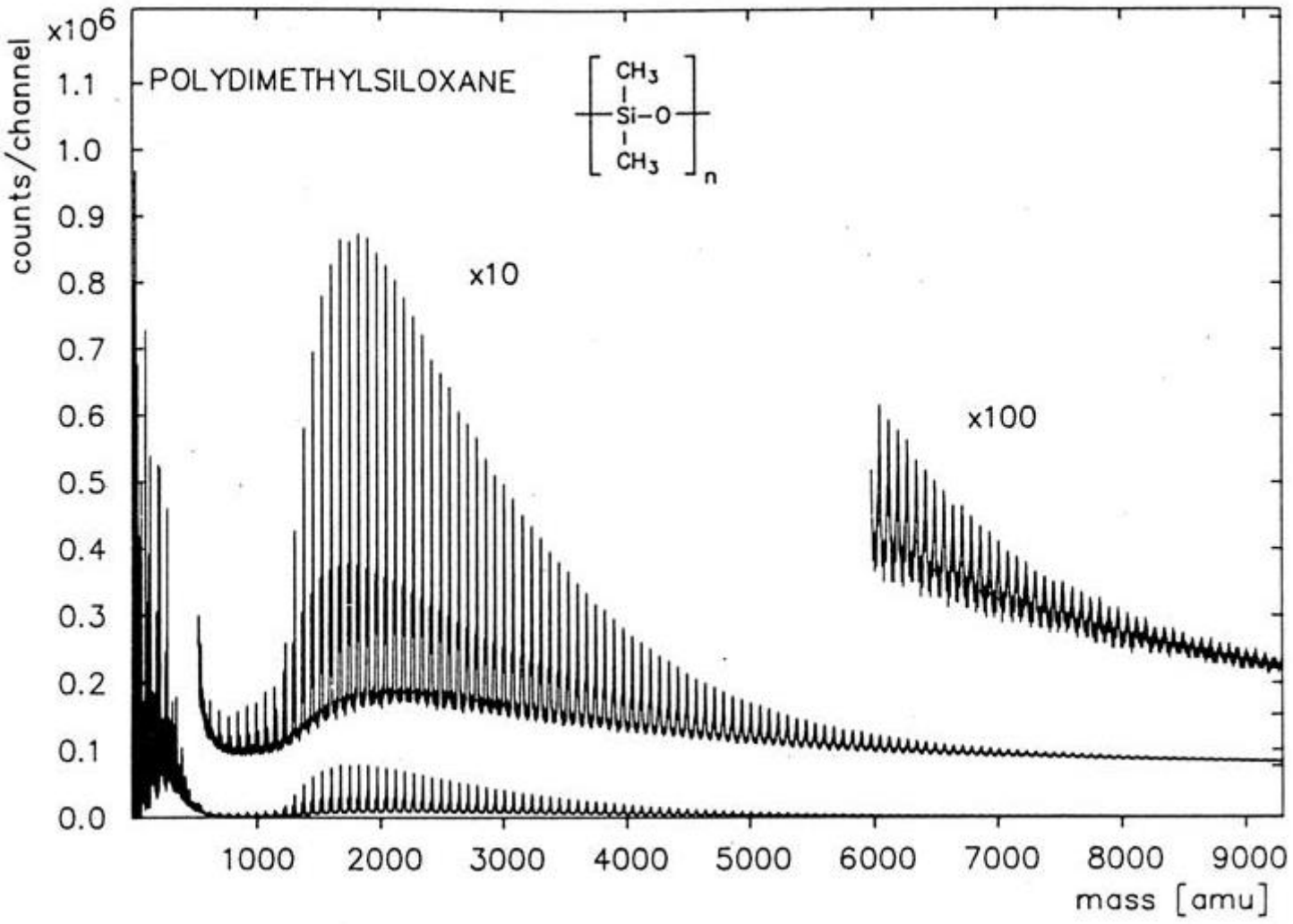

$M_2$(M: 金属) 一番強いフラグメントイオンピーク質量数の実験値と計算値

| Metal | Fragment Ion of $M_2$ | Mass Number(amu) | | Deviation(mamu) |
|---|---|---|---|---|
| | | From Experiment | From Calculation | |
| Al | $^{27}Al_2$ | 53.9612 | 53.9631 | -1.9000 |
| Cu | $^{63}Cu_2$ | 125.8586 | 125.8592 | -0.6000 |
| Si | $^{28}Si_2$ | 55.9535 | 55.9539 | -0.4000 |
| Mo | $^{94}Mo^{98}Mo$ | 191.8120 | 191.8105 | -1.5000 |
| | $^{95}Mo^{97}Mo$ | | 191.8118 | -0.2000 |
| | $^{92}Mo^{100}Mo$ | | 191.8143 | 2.3000 |

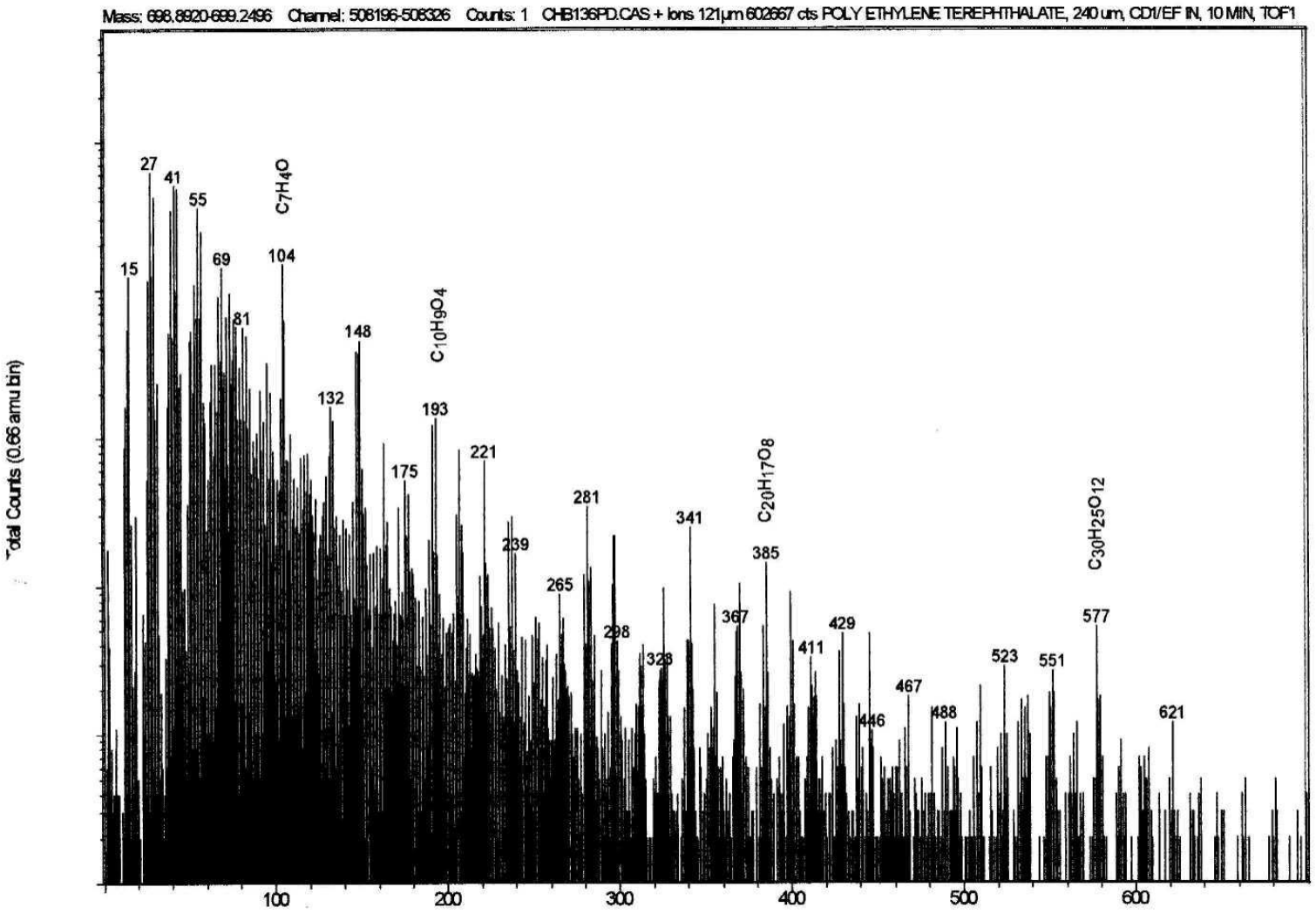
# Si样品上56amu附近的TOF-SIMS质谱



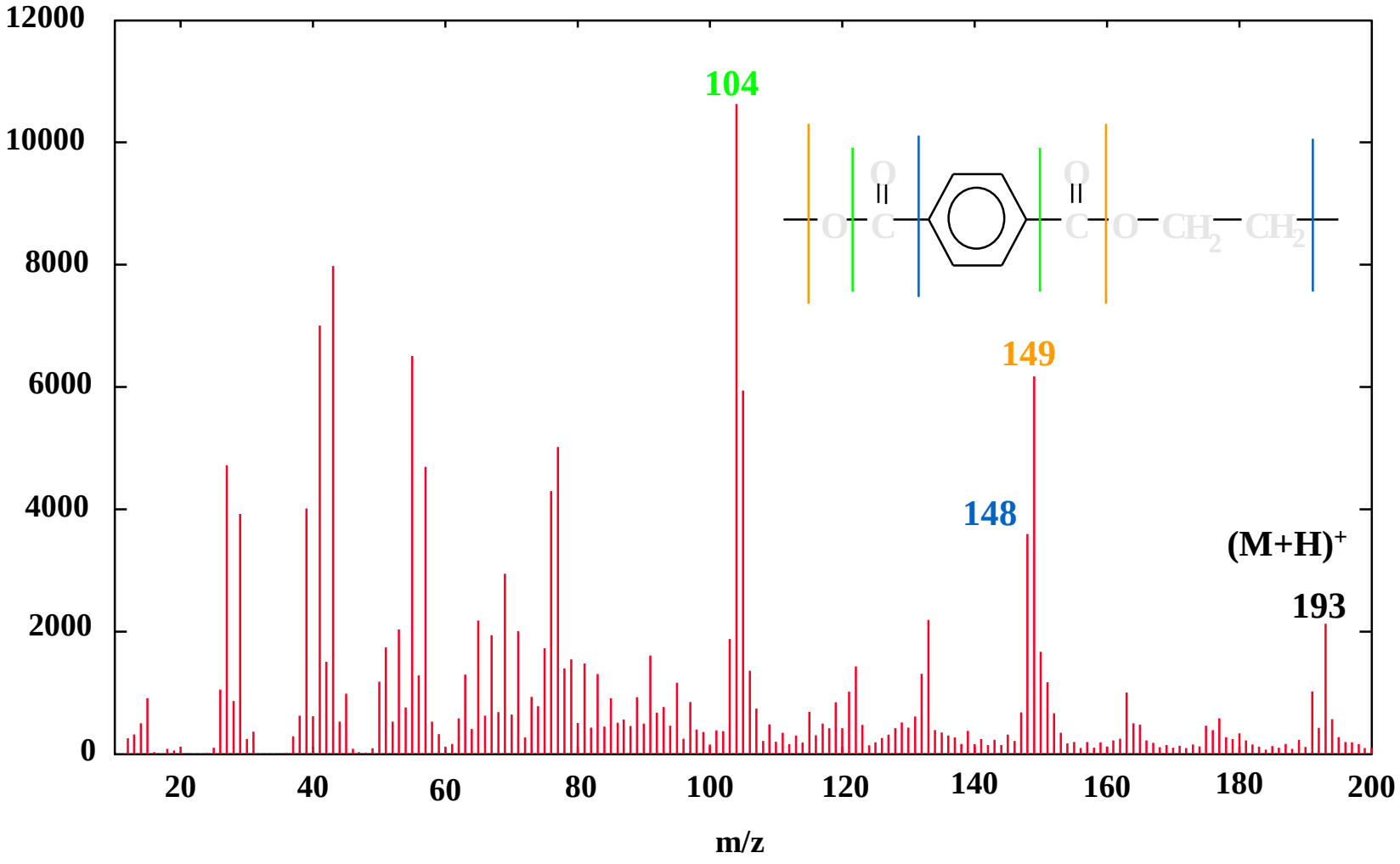
# 質量范围可达上万amu，理论上没有上限(∞)



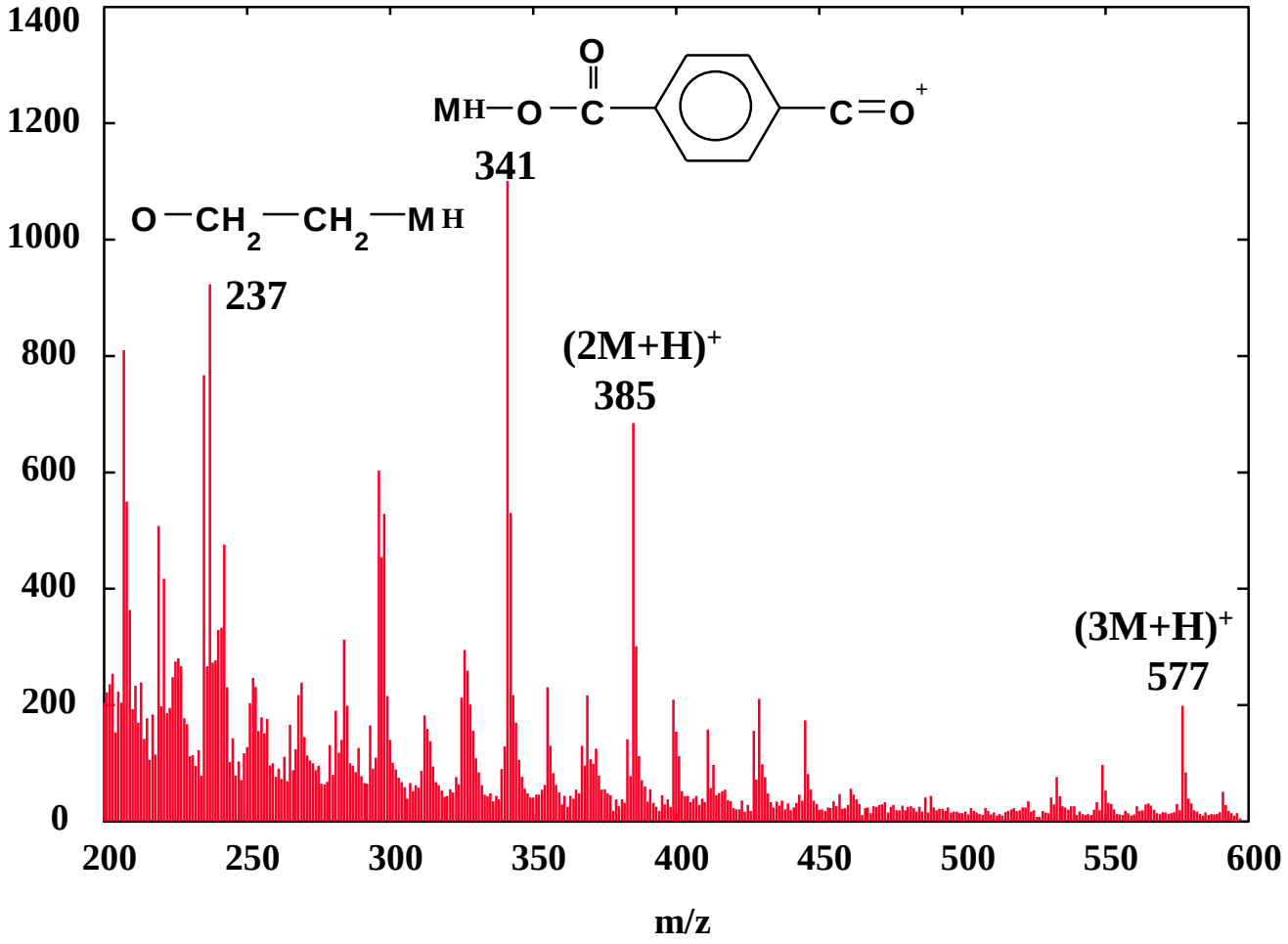
# 涤纶树脂(PET)样品的TOF-SIMS质谱



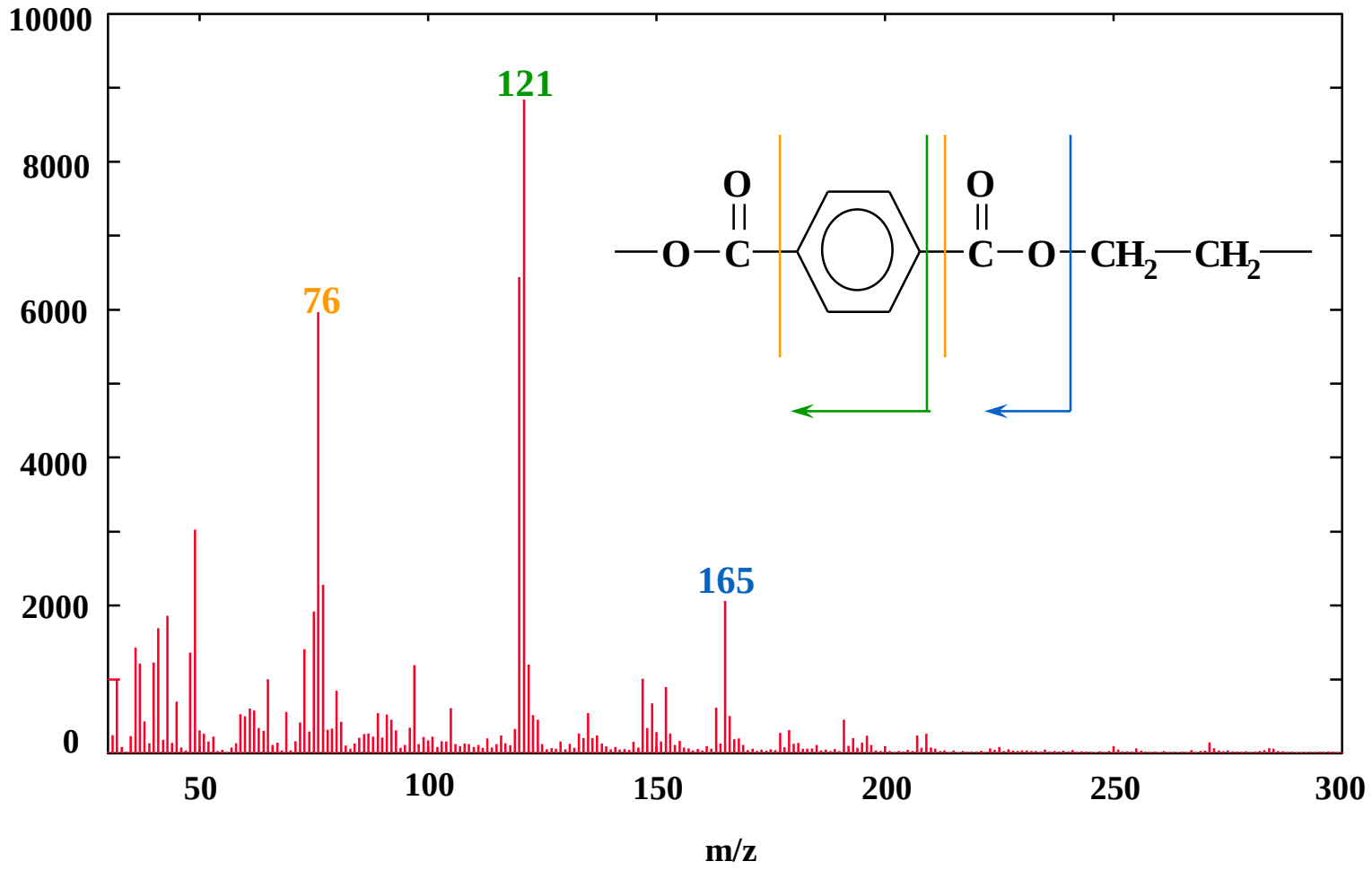
# PET TOF-SIMS 质谱图(+Ion)



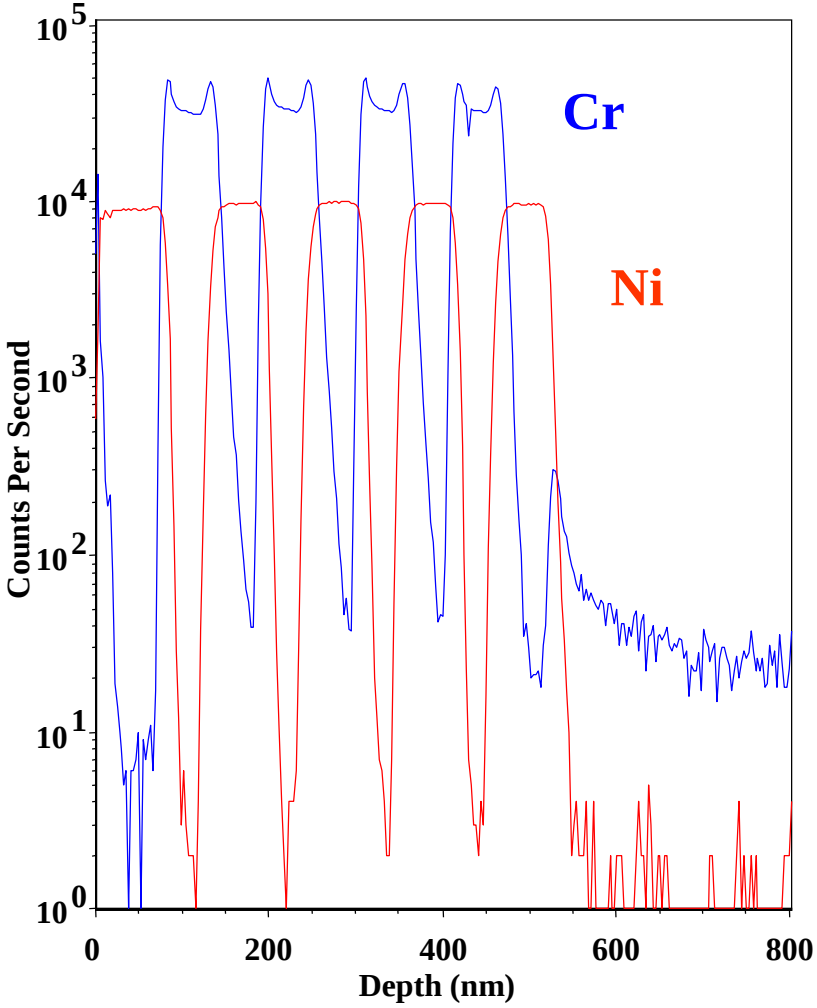
# PET TOF-SIMS 质谱图(+Ion)

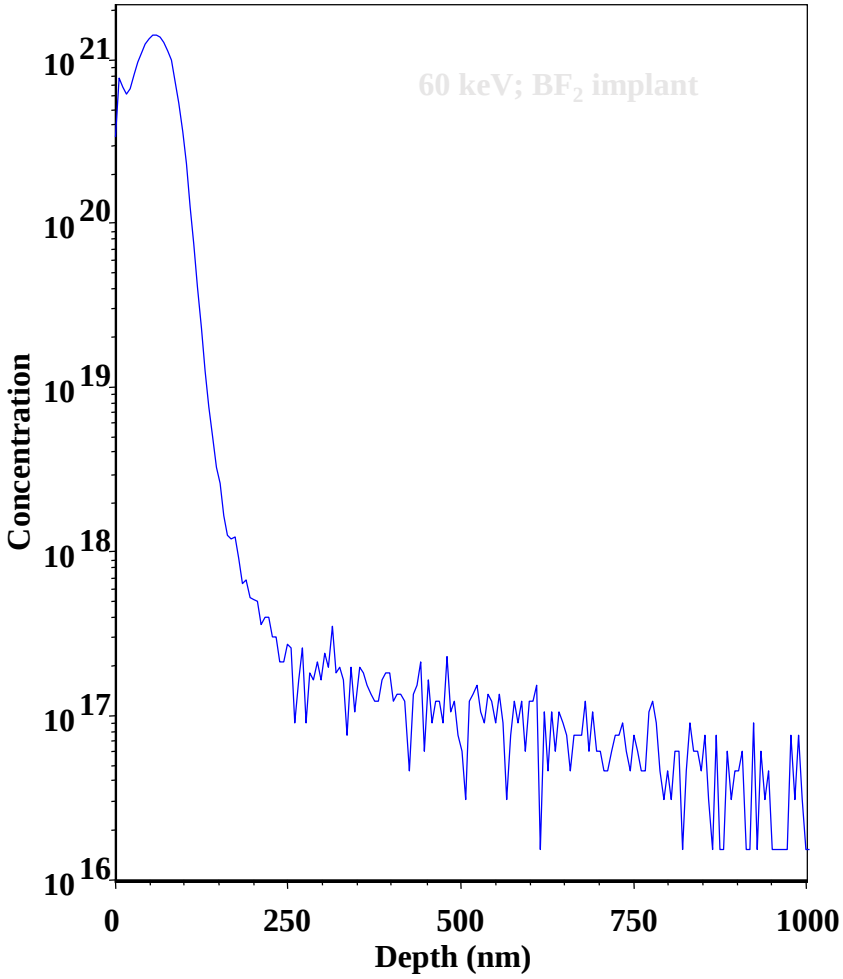


# TOF-SIMS 质谱图(-Ion)



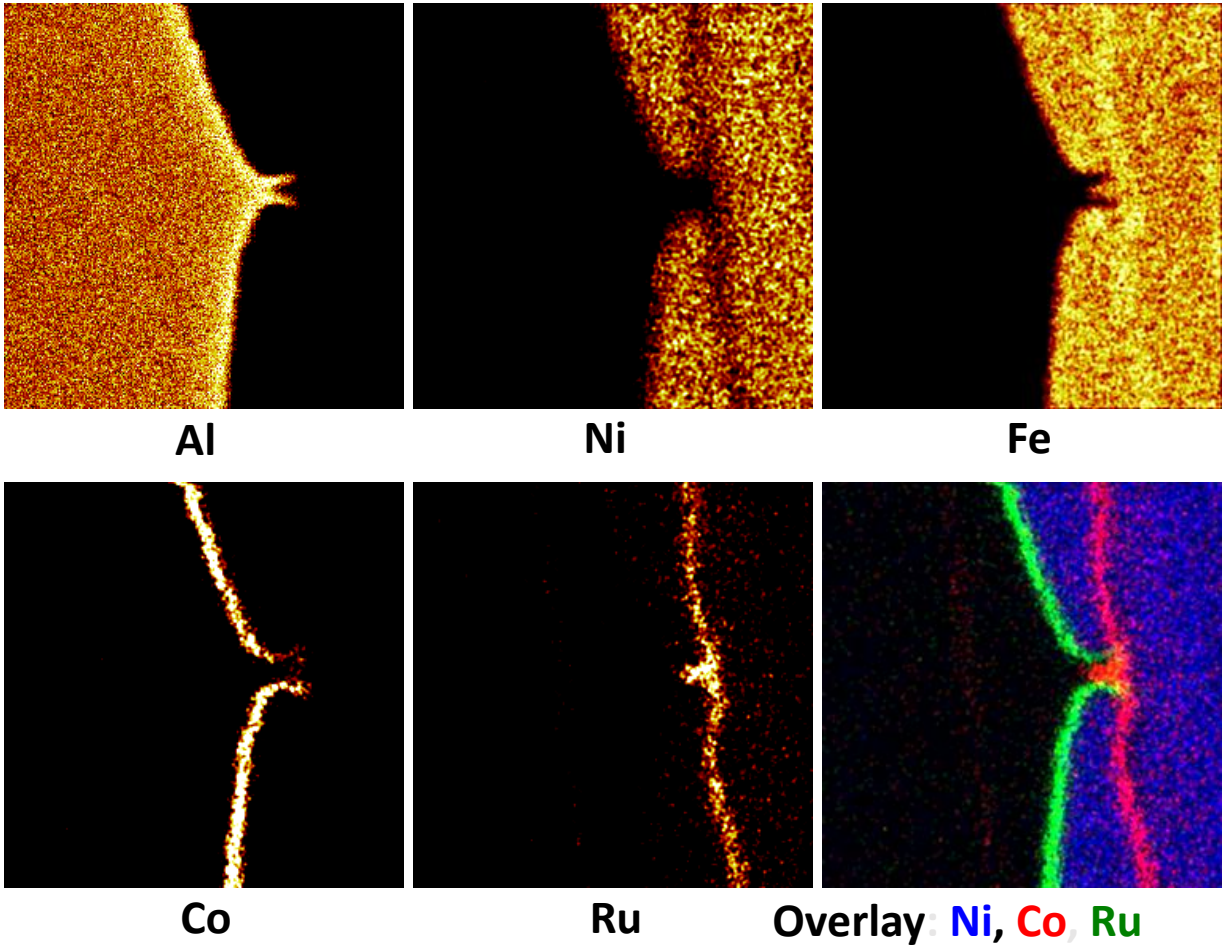
# TOF-SIMS 的深度剖面分析



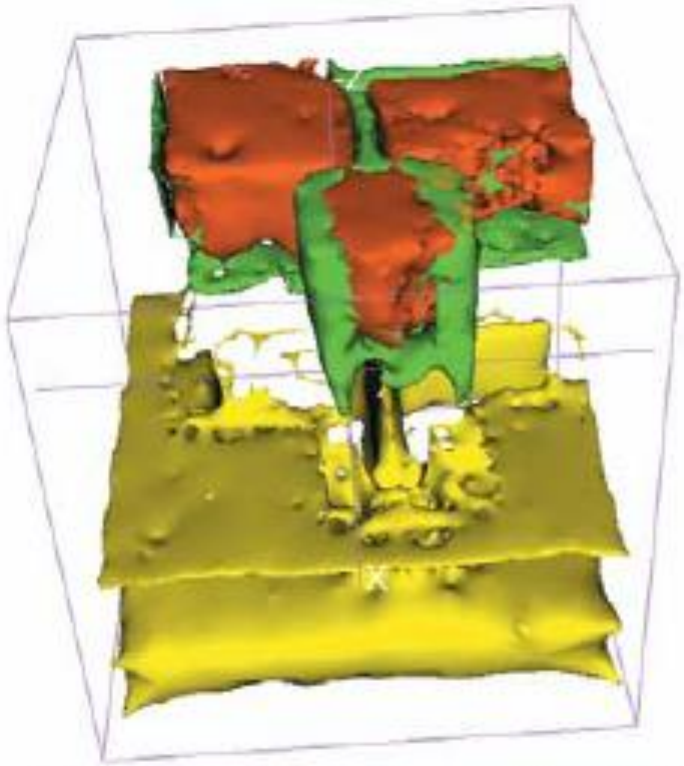



# TOF-SIMS的成像分析

## Hard Disk Read / Write Head Structure



# TOF-SIMS的三维成像分析



Al = Red, AlO = Green, Si = Yellow

**3D Imaging**

# TOF-SIMS在各个领域的应用

a）无机

b）有机

c）电子

c）生物

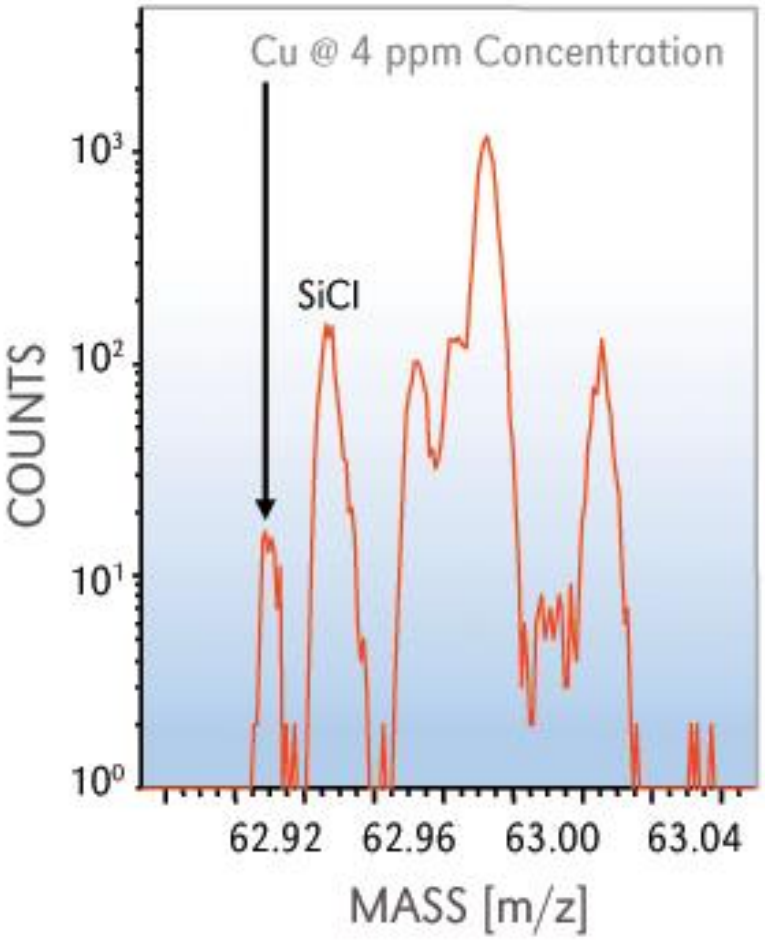
e）医学

f）自组装膜

• • • • • •

# TOF-SIMS 应用实例

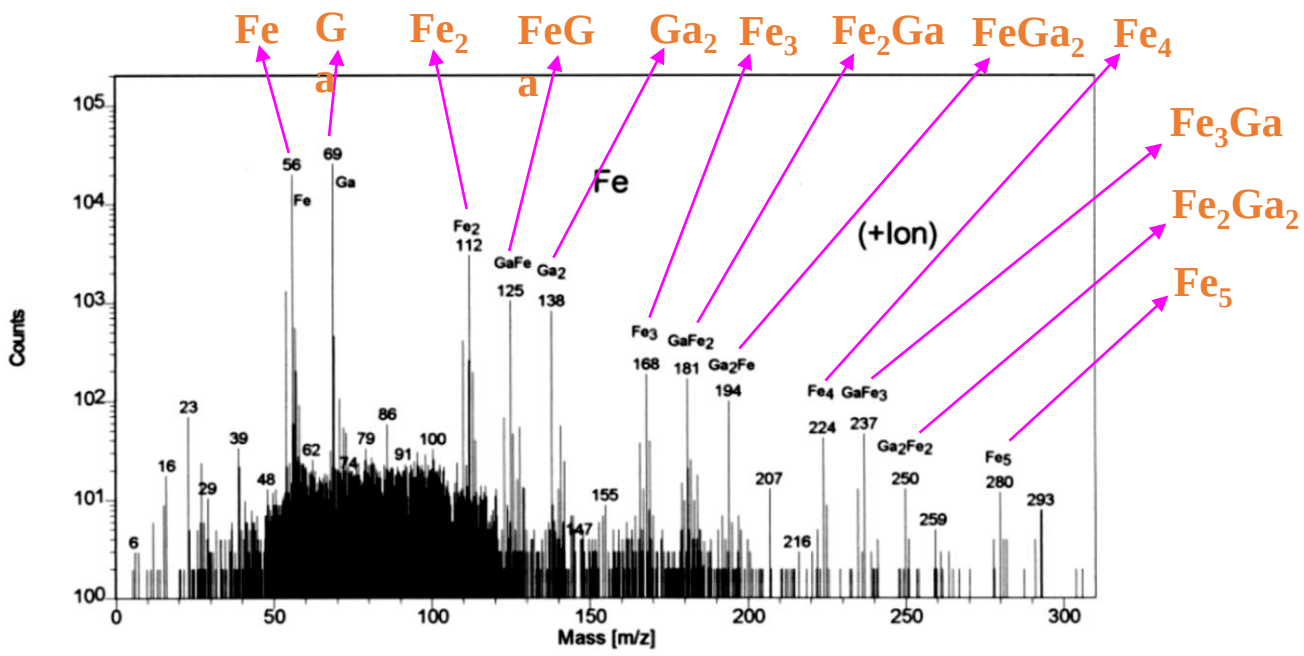
# 4ppm Cu的TOF-SIMS信号



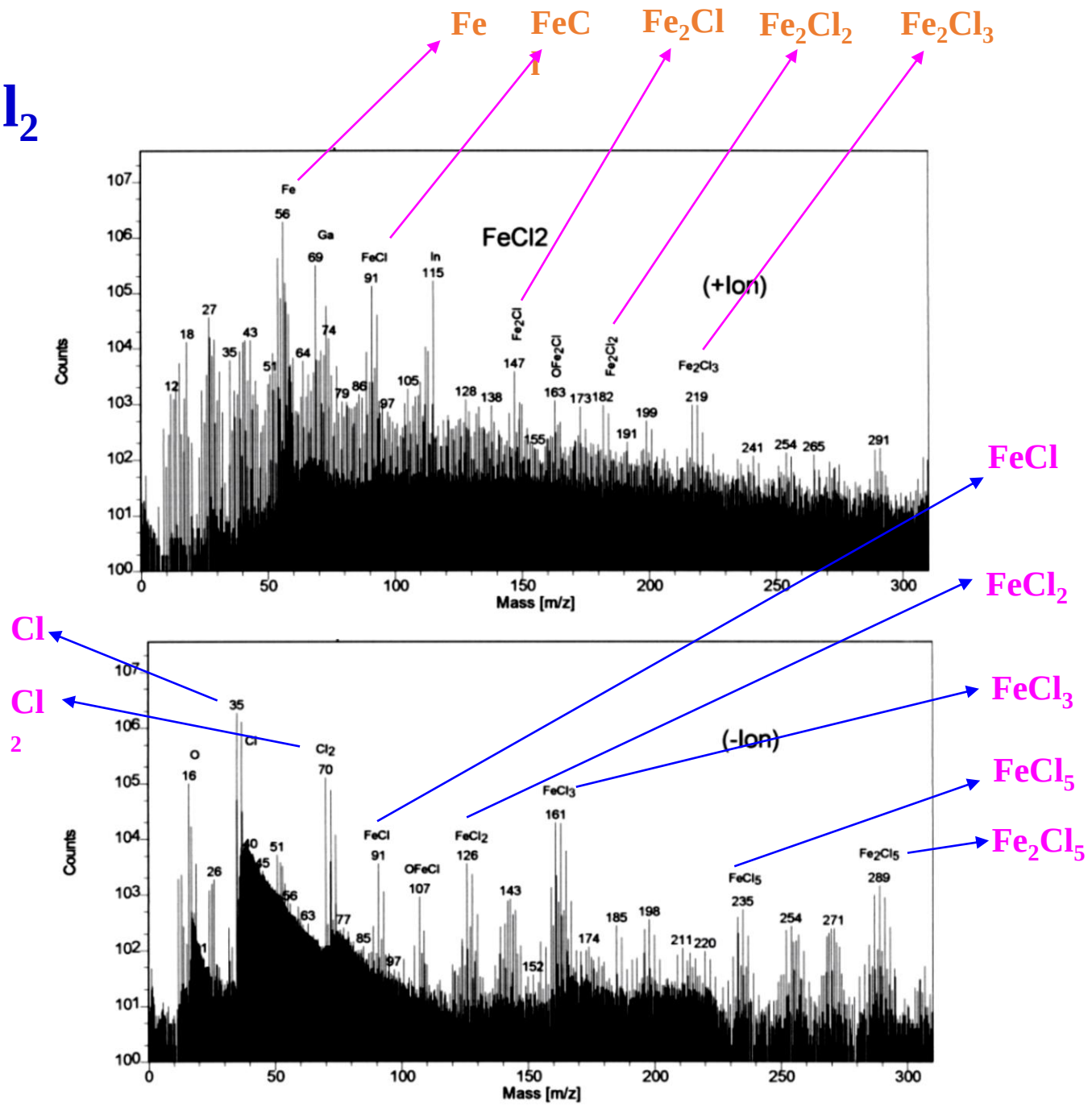
# 金属及其化合物的鉴别

# TOF-SIMS 的无机物的鉴别：Fe、$FeCl_2$

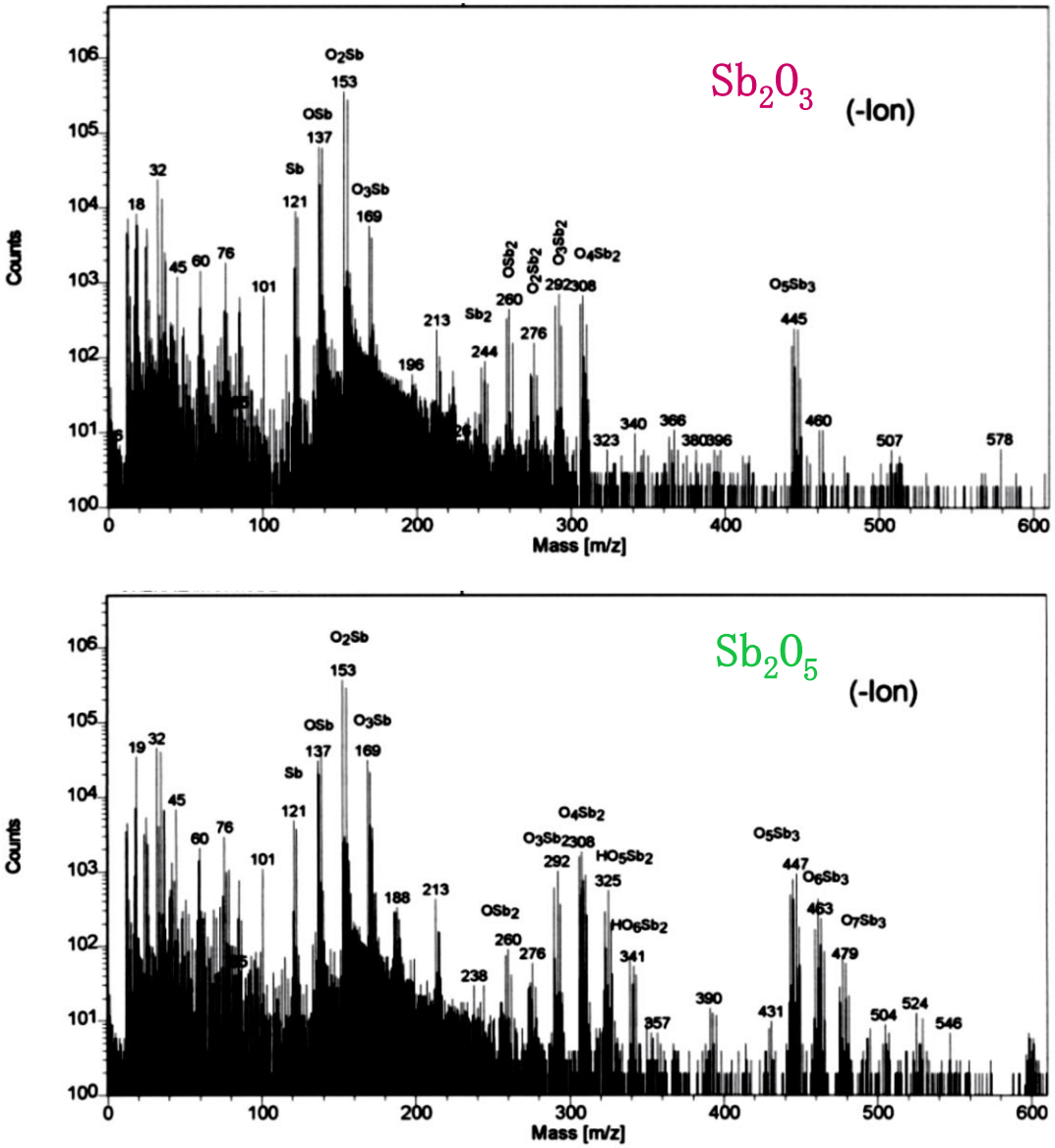
Fe



# $FeCl_2$

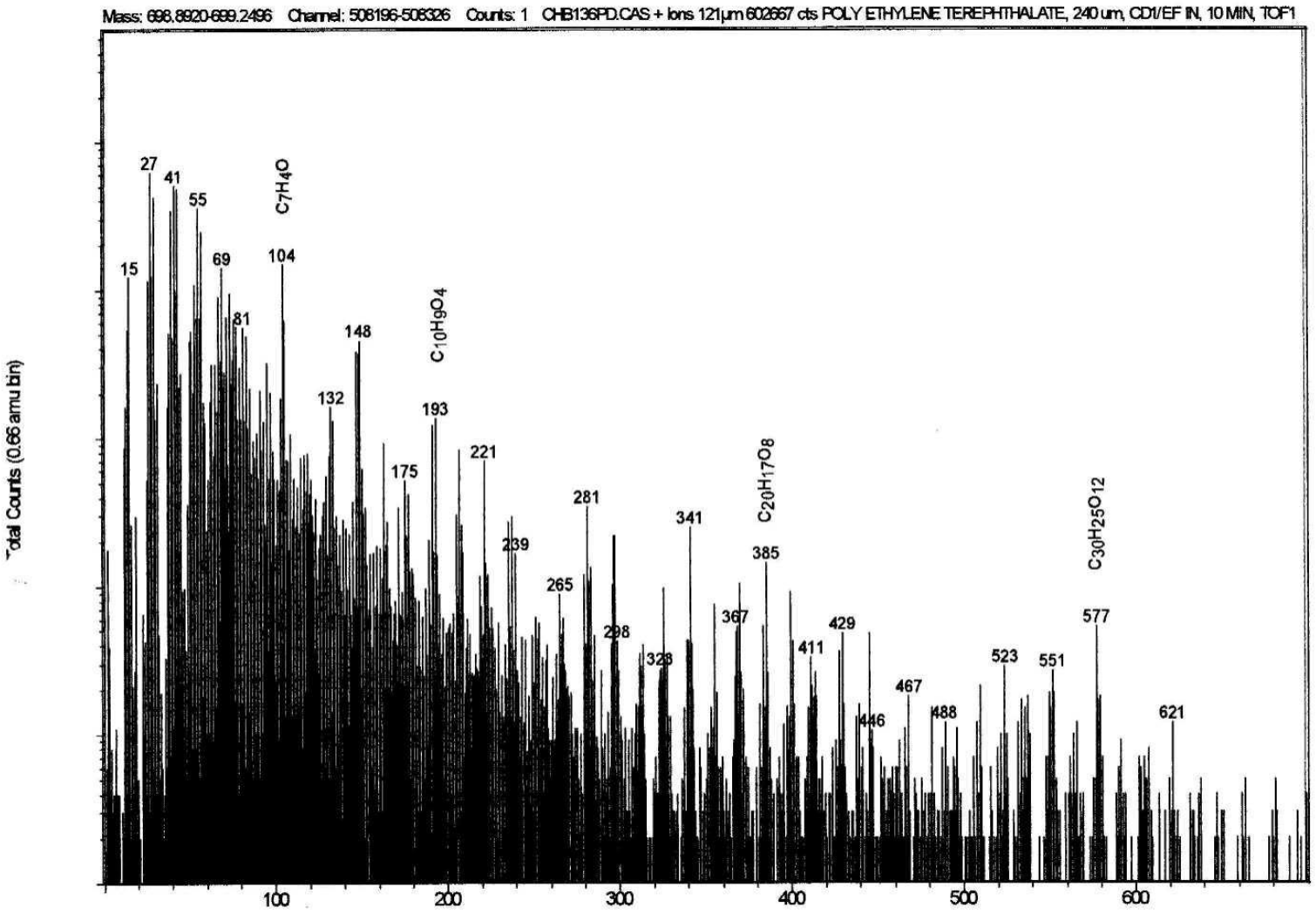

Fe
FeC
Fe2Cl
Fe2Cl2
Fe2Cl3
FeCl2
(+Ion)
Counts
Mass [m/z]
FeCl
FeCl2
Cl
Cl2
FeCl3
(-Ion)
FeCl5
Fe2Cl5

# TOF-SIMS 无机物的鉴别：$Sb_2O_3$、$Sb_2O_5$

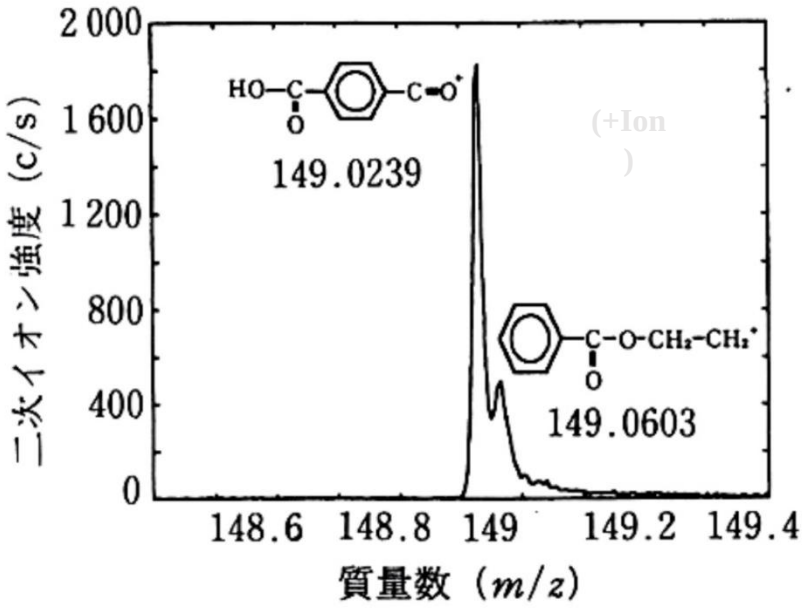


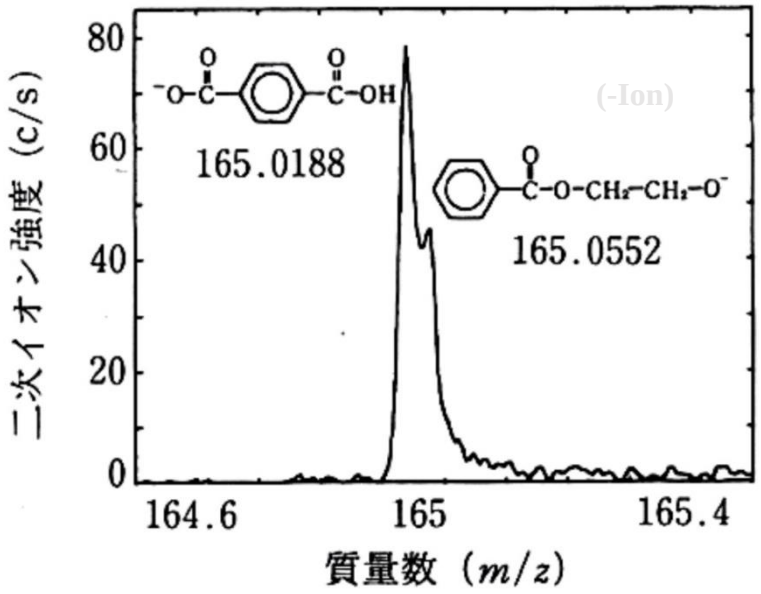
# PET样品有机结构信息的获取

# 涤纶树脂(PET)样品的TOF-SIMS质谱



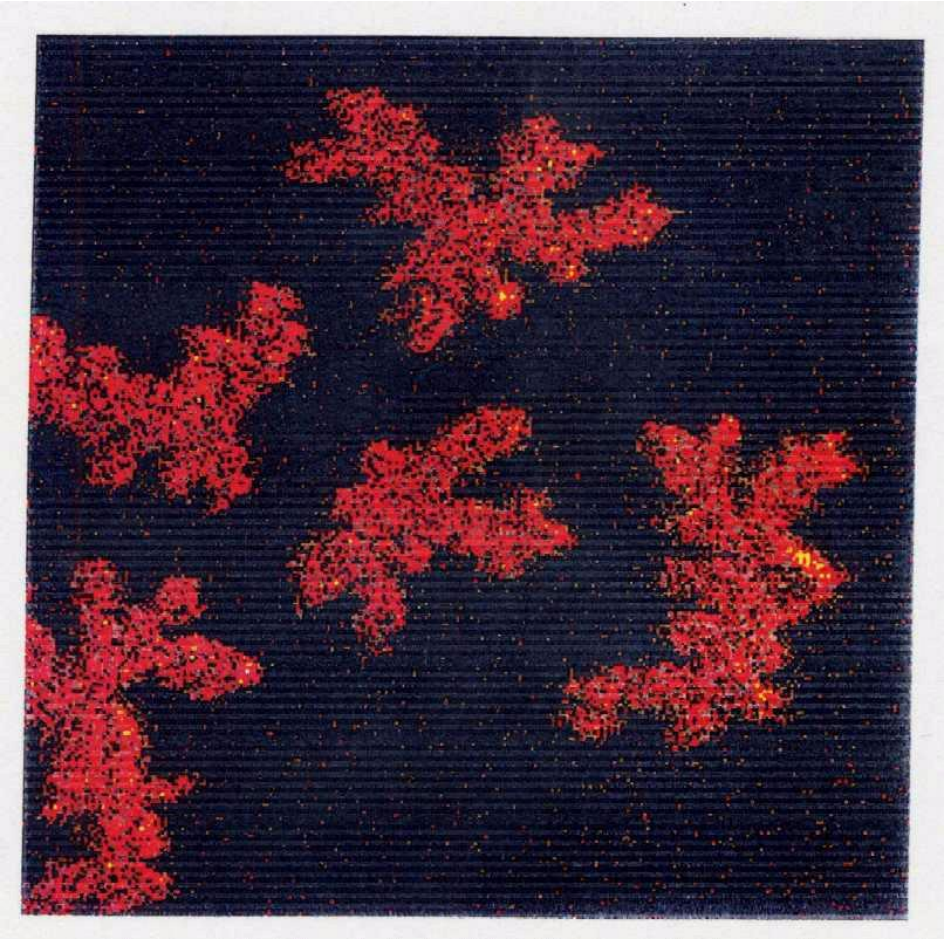
# PET 样品149amu(+Ion)、165amu(-Ion) TOF-SIMS 携带的化学信息



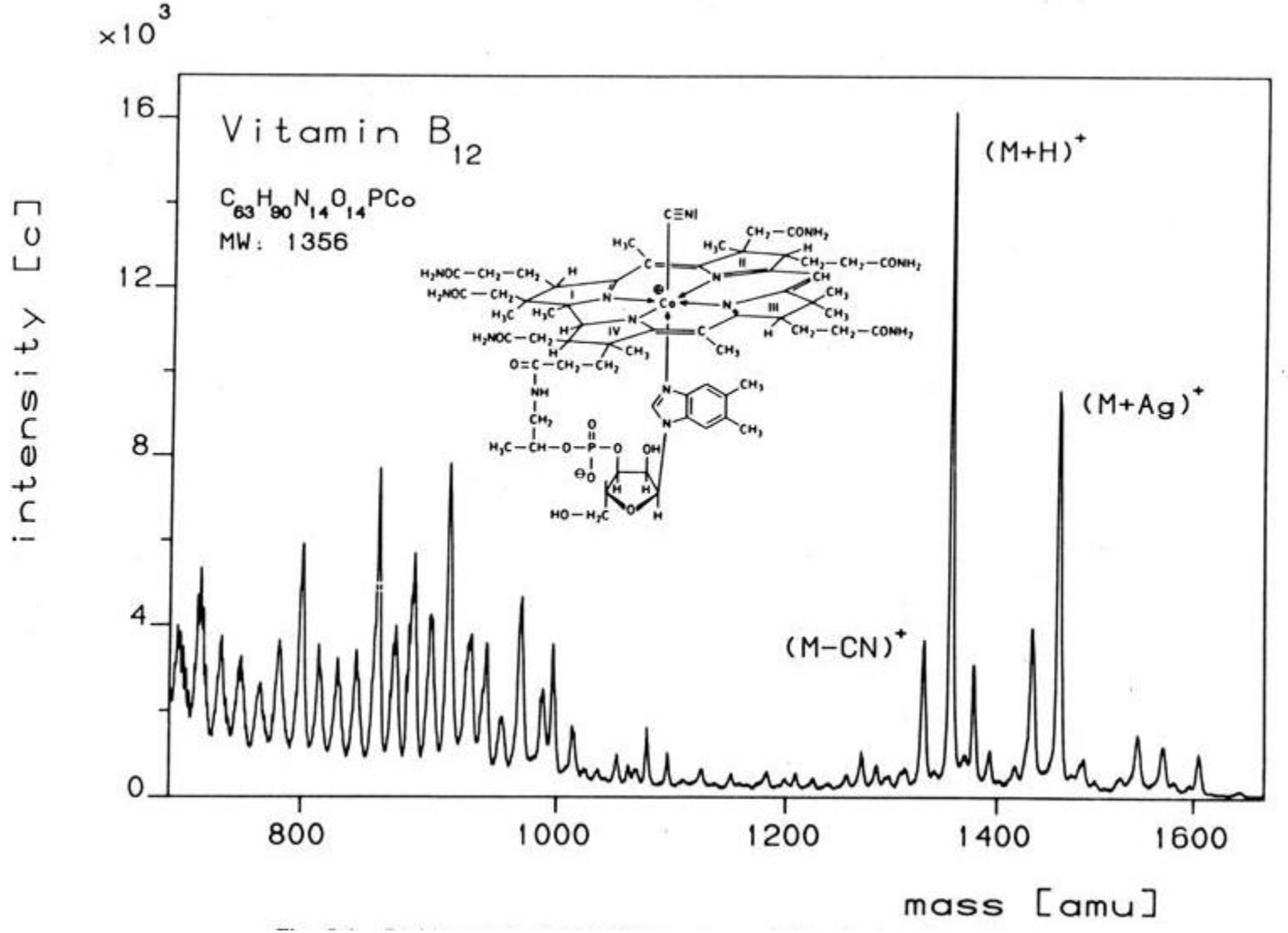




ポリエチレンテレフタレート(PET)で測定された高質量分解能スペクトル

# TOF-SIMS的成像分析：退火涤纶树脂(PET)样品的“鱼眼”(3倍体结晶)



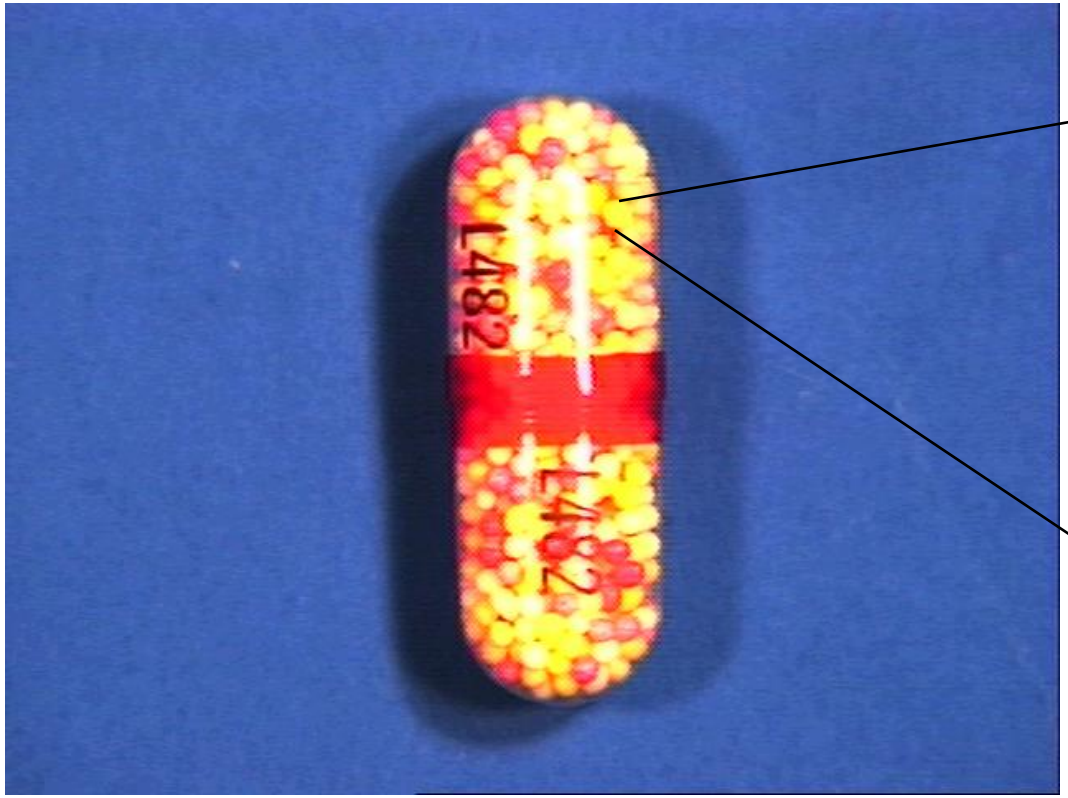
# TOF-SIMS Spectrum of Vitamin B12

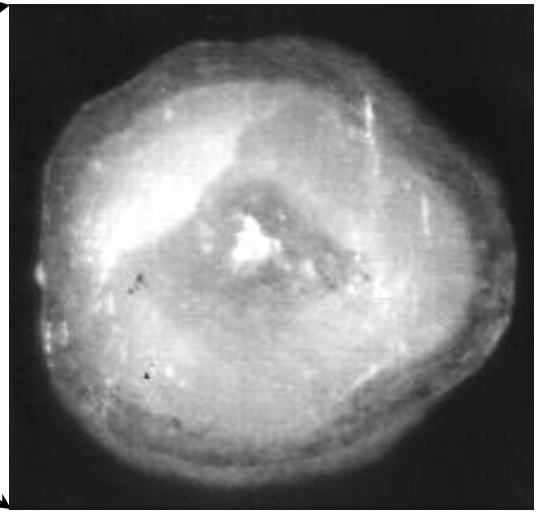


# 胶囊成像的应用

- Patent Protection（专利保护）
- Competitor Analysis（竞争对手分析）
- Layer Thicknesses and Continuity（药层厚度及连贯性）
- Drug Distribution（药物分布）

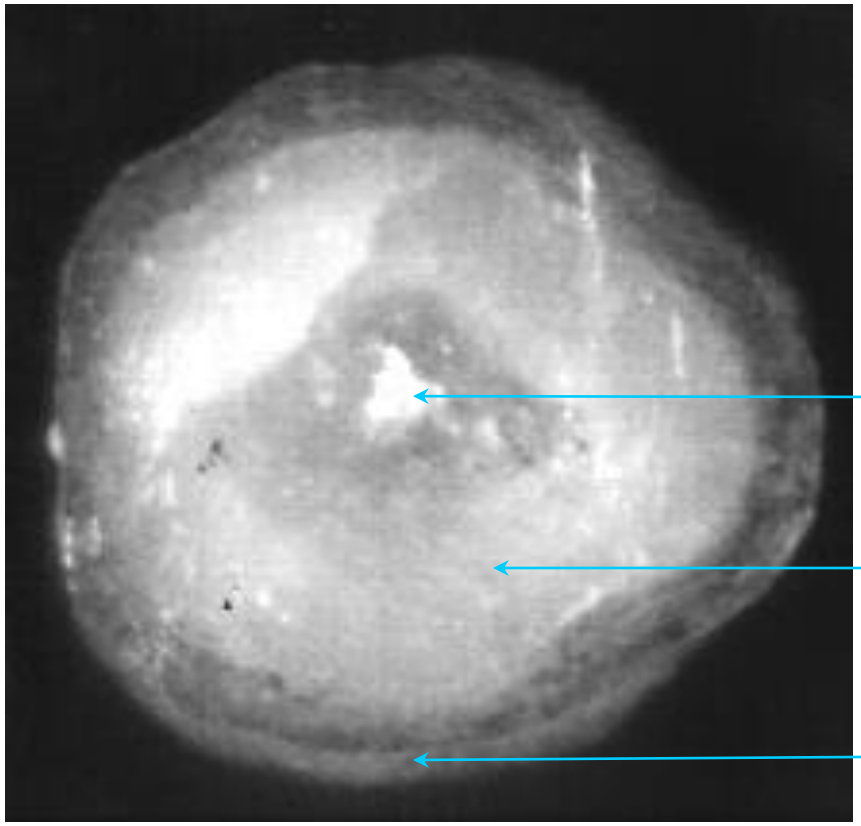
# 胶囊的光学像



**Optical Micrograph of a single pellet cross-section**

# 胶囊截面的光学像



$SiO_2$ Core

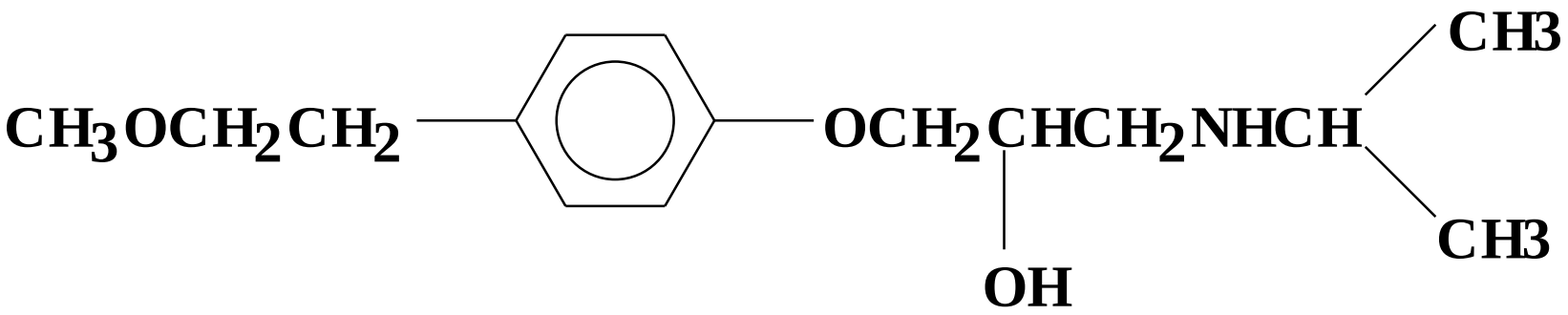
Metoprolol
Drug Layer

Ethylcellulose
Film Coating

# 化学结构

## Metoprolol(美托洛尔)

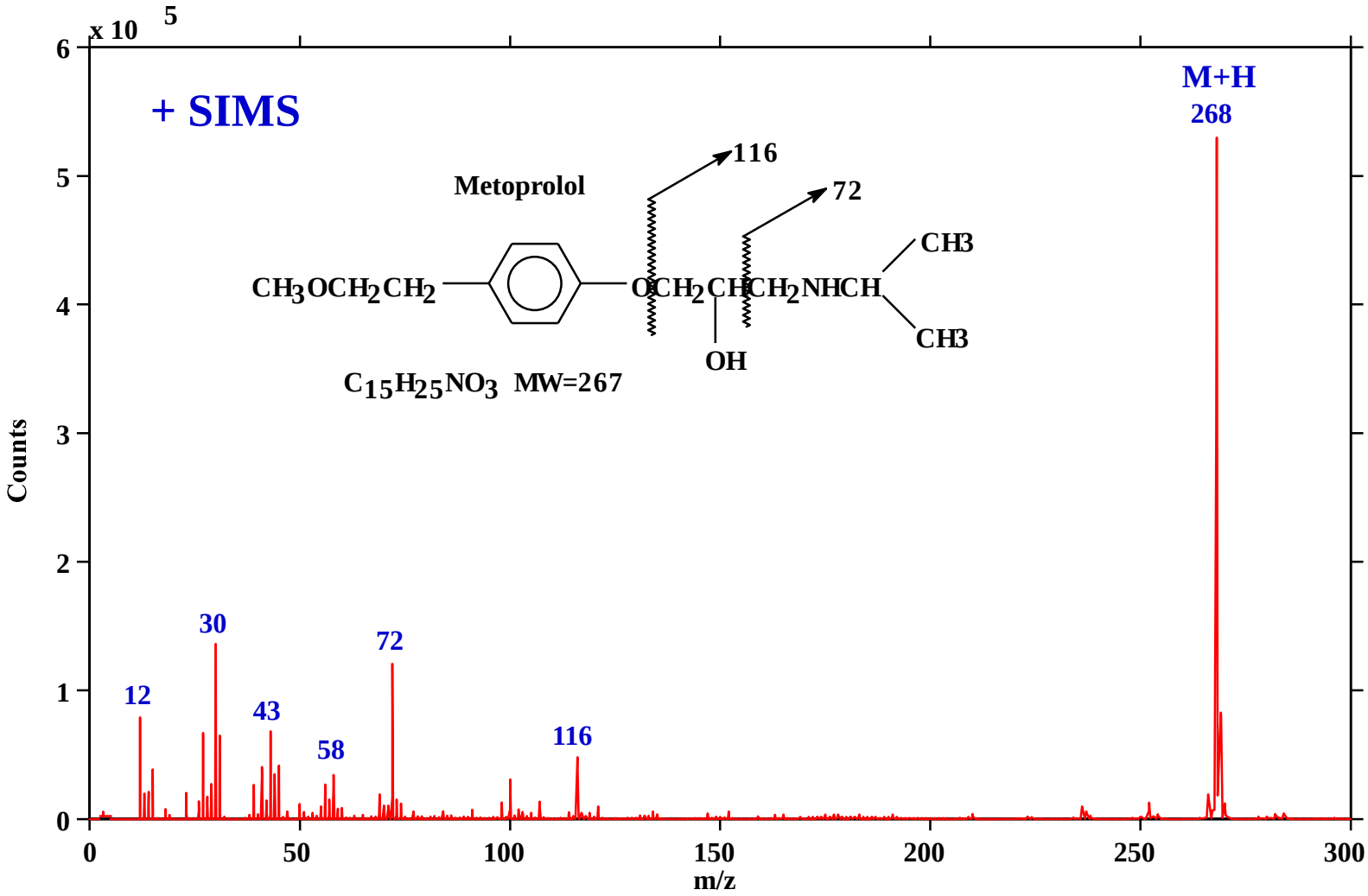
**(Heart Medication)**




$C_{15}H_{25}NO_3$

**M.W. = 267 Da**

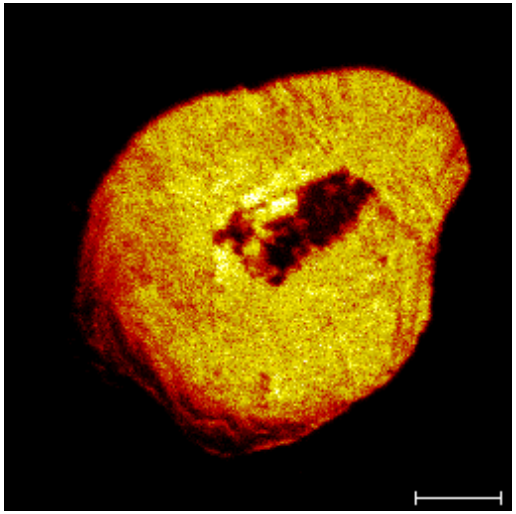
# 胶囊截面的TOF-SIMS谱图

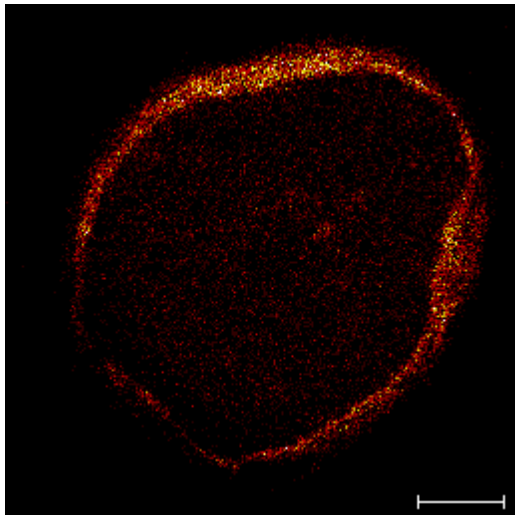


# 胶囊截面的TOF-SIMS像

Drug Molecular Ion $[M+H]^+$

m/z 268



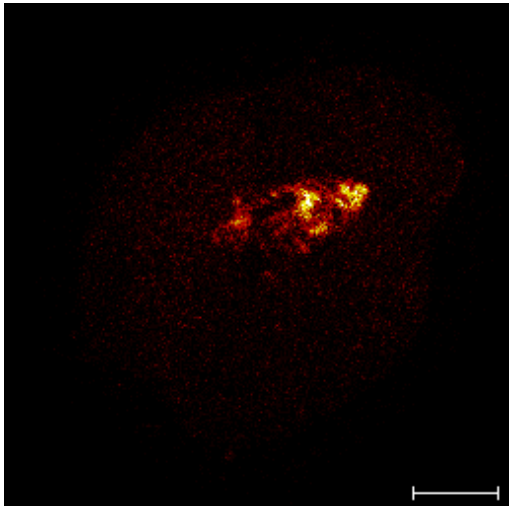


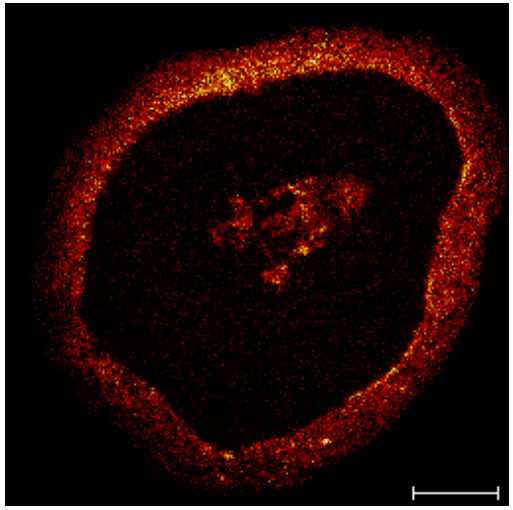
Ethylcellulose Film Coating

$C_3H_7O^+$

Silica Core

$Si^+$





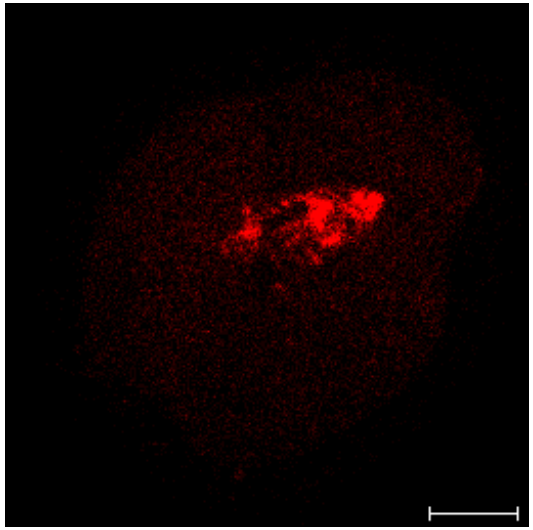
$Na^+$

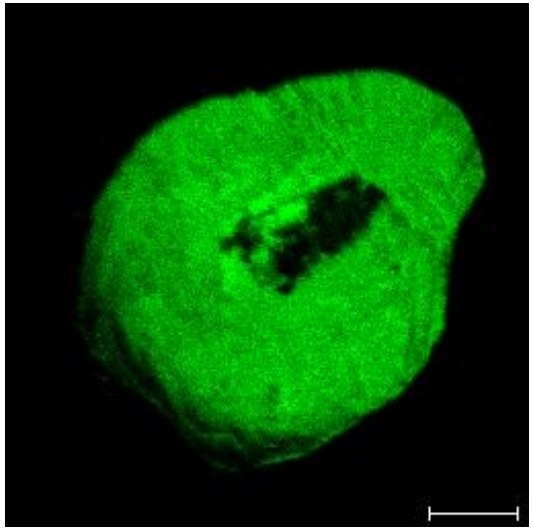
100 μm

# RGB 重叠的TOF-SIMS图像

Silica Core

$Si^+$



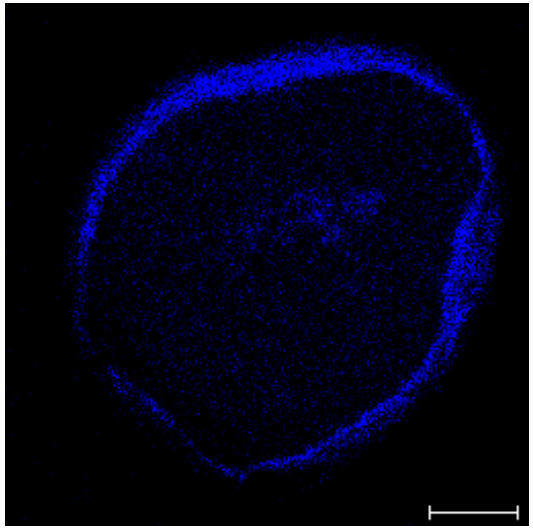


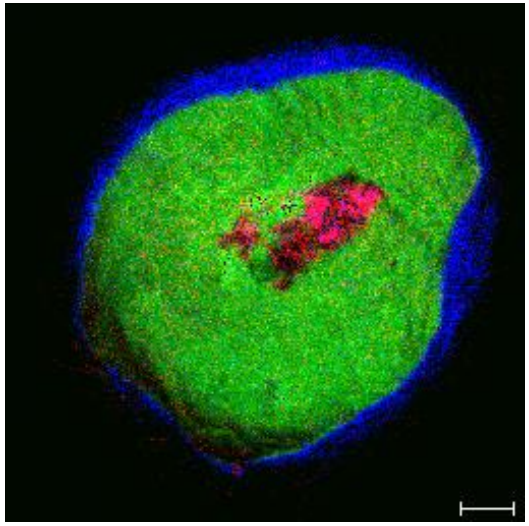
Drug
Molecular Ion

$[M+H]^+$
m/z 268

Ethylcellulose
Film Coating

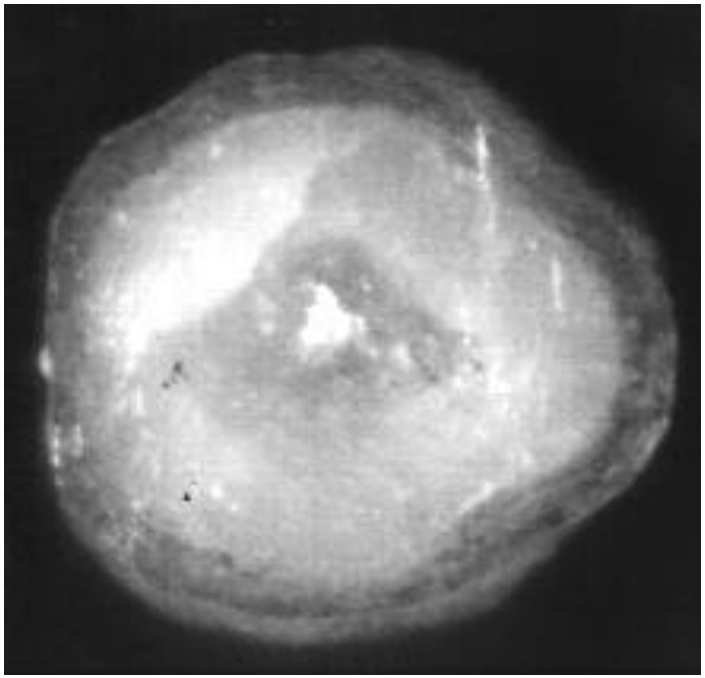
$C_3H_7O^+$



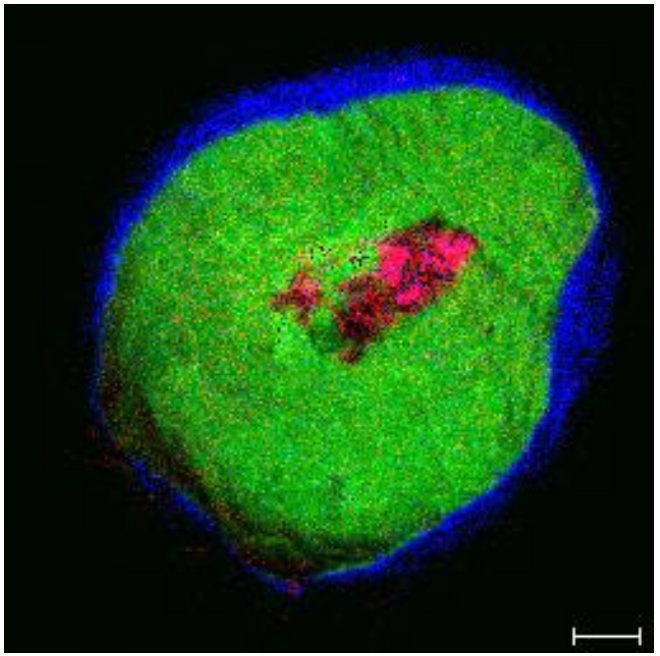


Overlay

# 光学像与TOF-SIMS重叠像比较



Optical Micrograph



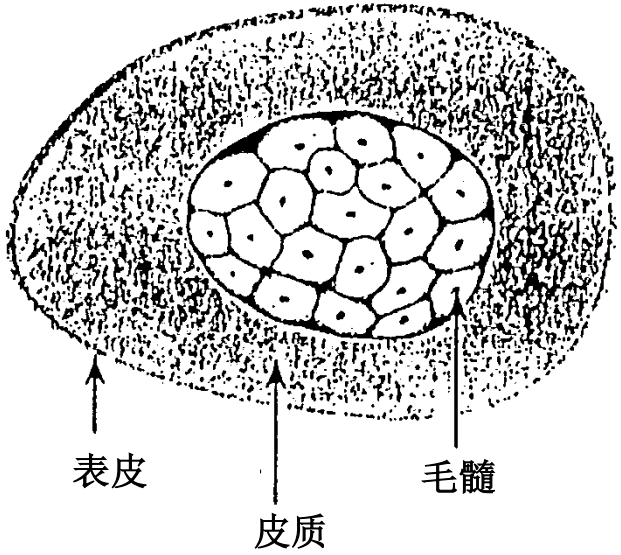
Blue = $C_3H_7O^+$

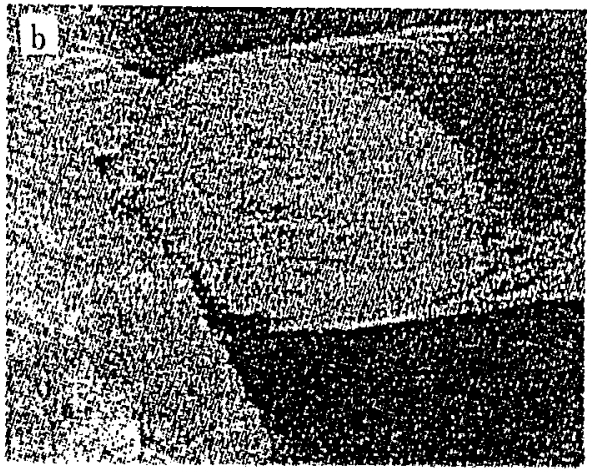
Green = Drug $[M+H]^+$

Red = $Si^+$

100 μm

# 洗发液、护发液在头发中的渗透



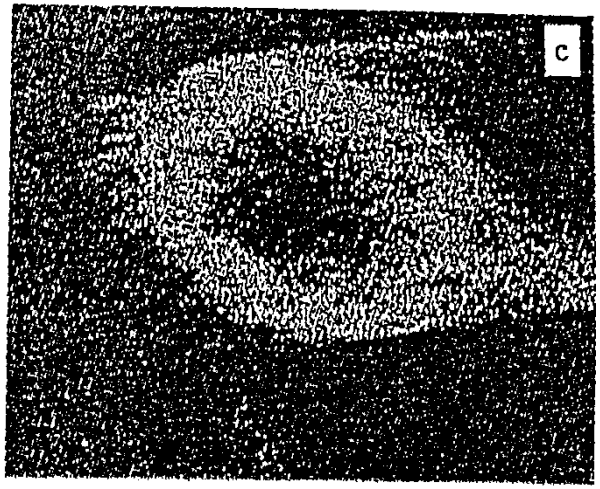




b. 全イオン像




洗发剂

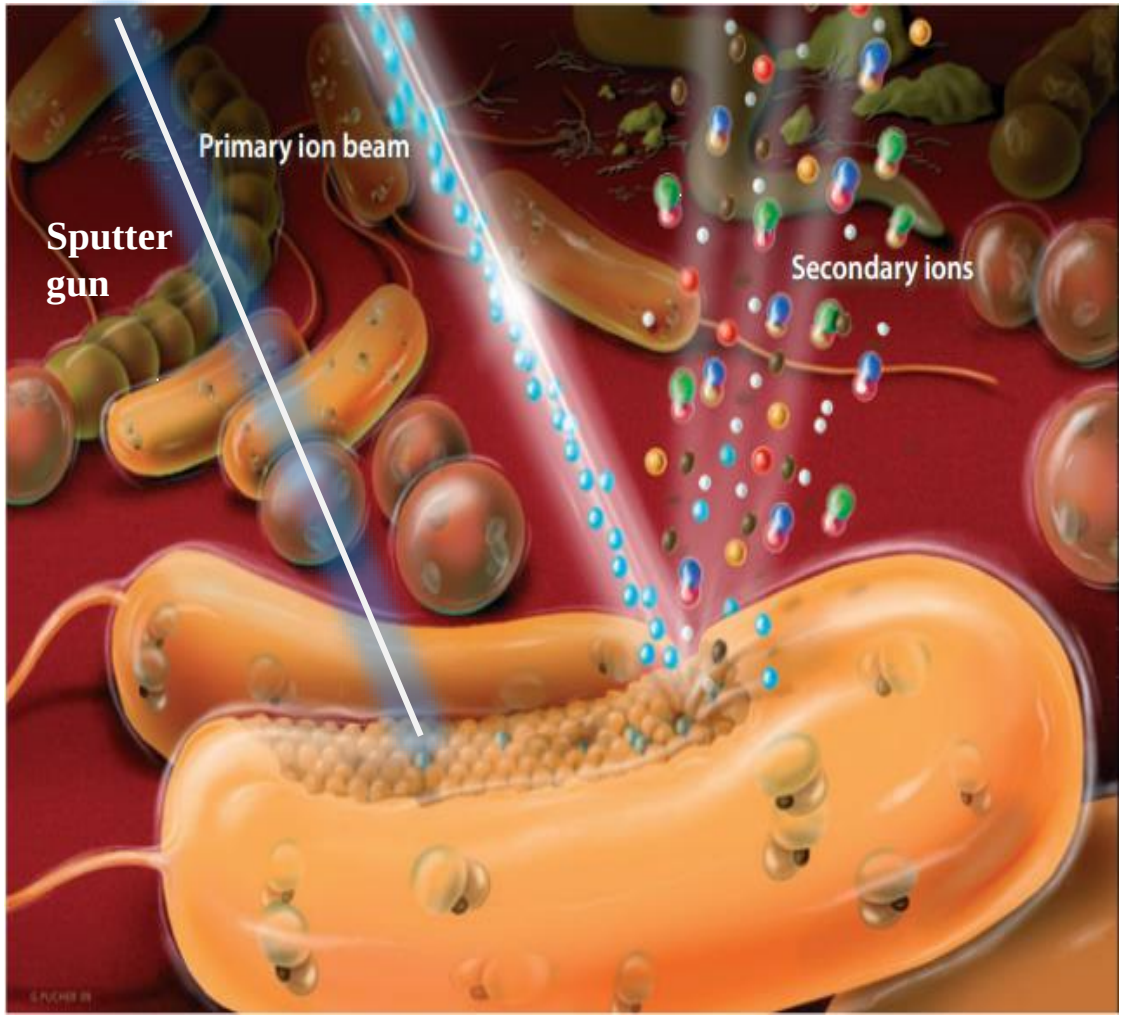
d. シリコーンの分布像



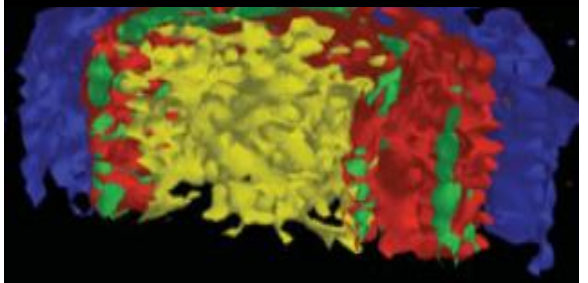

护发剂

c. リンス剤の分布像

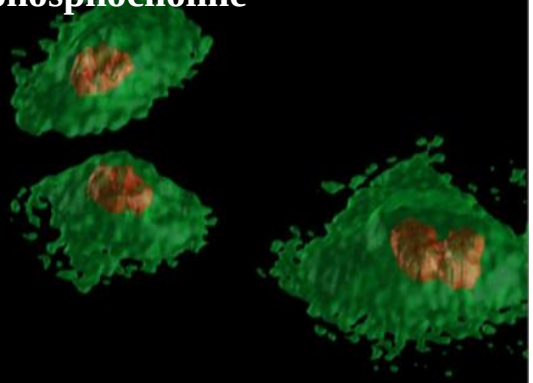
# TOF-SIMS在生物领域的应用



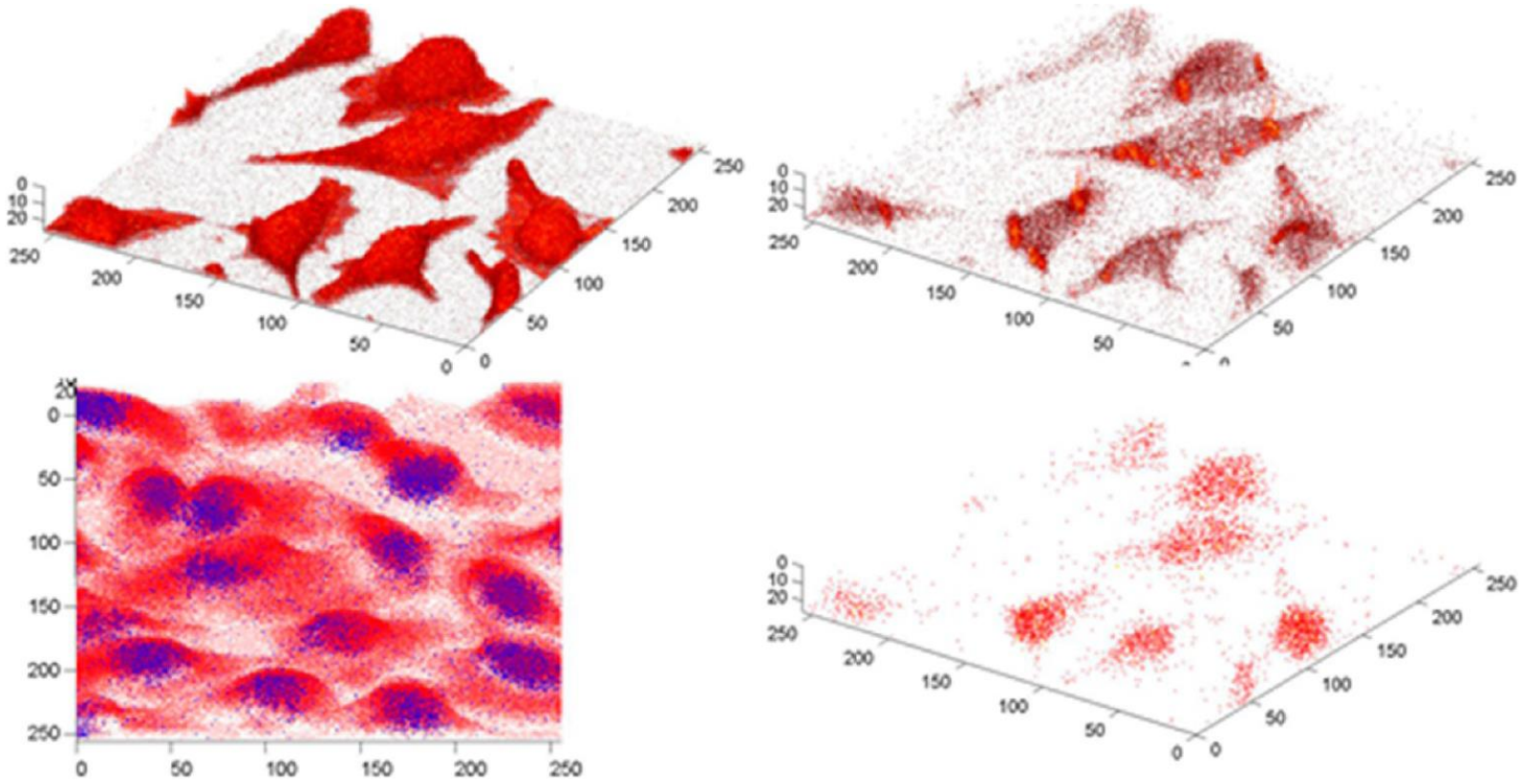

**Blue Na, m/z 23 Green K, m/z 39**
**Yellow lipid fragment, m/z 86**
**Red lipid head group, m/z 184.1,**



**red, m/z 136.1, adenine**
**green, m/z 184.1, phosphocholine**



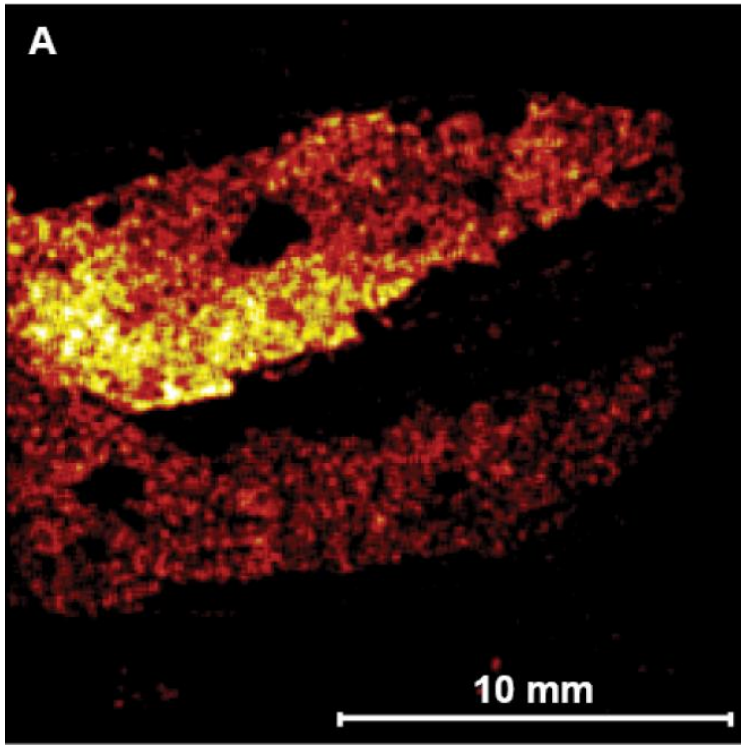
# 海拉(HeLa)细胞内在物质和外来物质的TOF-SIMS 3D 成像

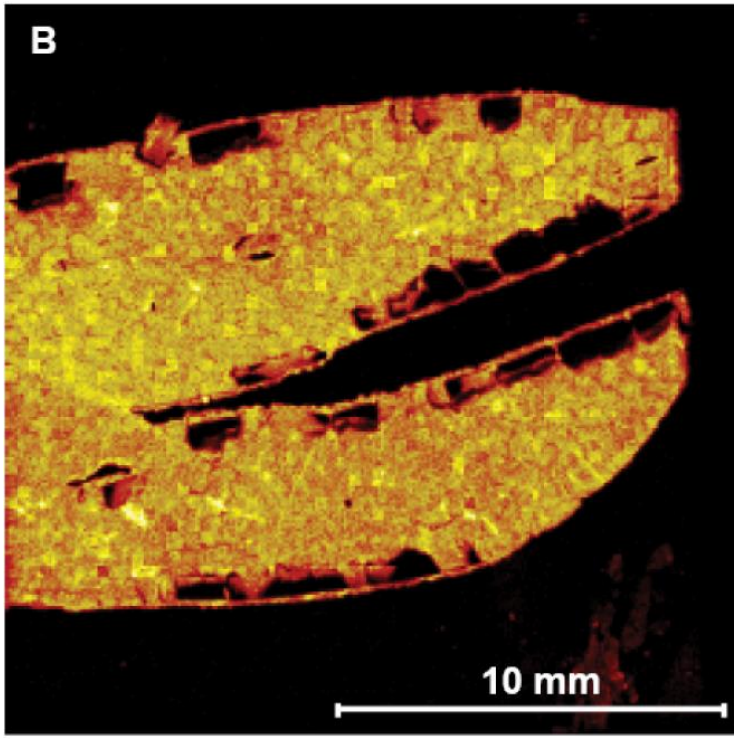


左下: **$BrdU^-$ (Bromodeoxyuridine 溴脱氧尿苷)(blue)和$\Sigma CxHyOz^-$ 信号(red)重叠图;**
**左上: $CN^- + CNO^-$; 右上: $\Sigma CxHyOz^-$; 右下: $\Sigma BrdU^-$**

# TOF-SIMS mosaic images (19 mm x 19 mm) of a liver cross-section of rat

## 肝脏截面的TOF-SIMS图像分析的拼接图(19mm×19mm)




**Lateral distributions of drug**

**Lateral distributions of phospholipids(磷脂)**

# TOF-SIMS在刑侦、考古领域的应用

# TOF-SIMS分析指纹及其外在污染物

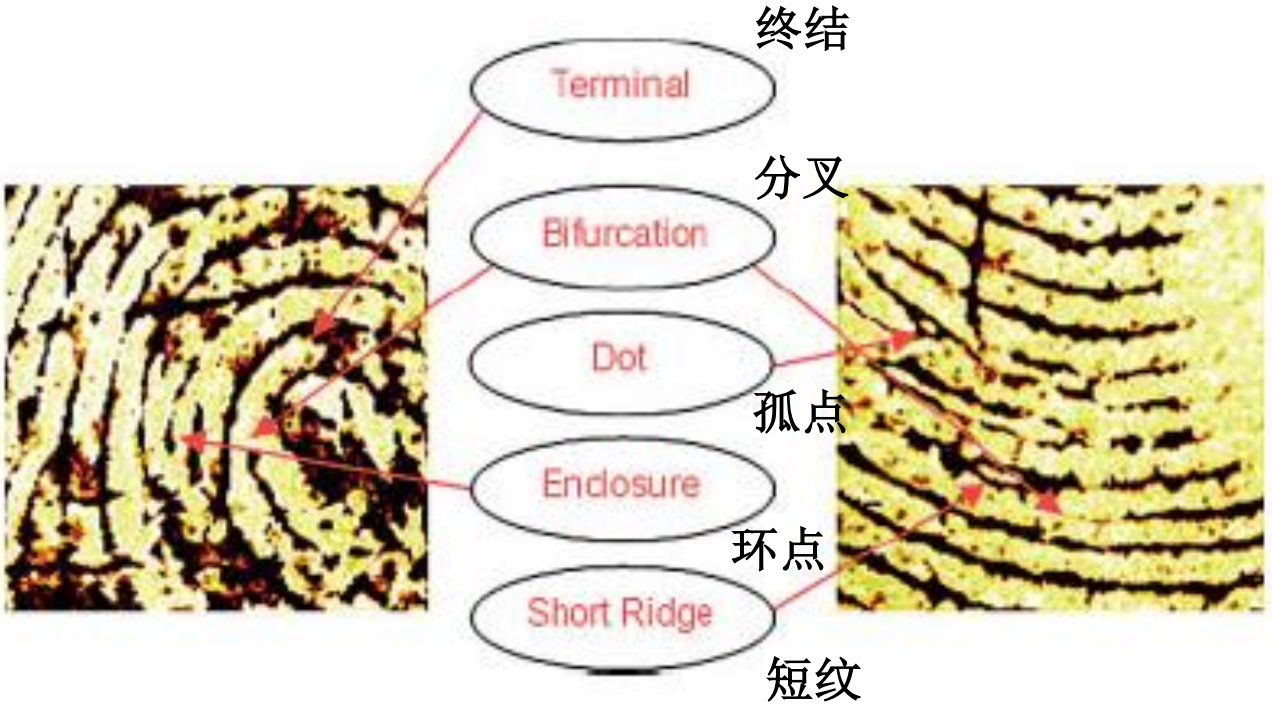
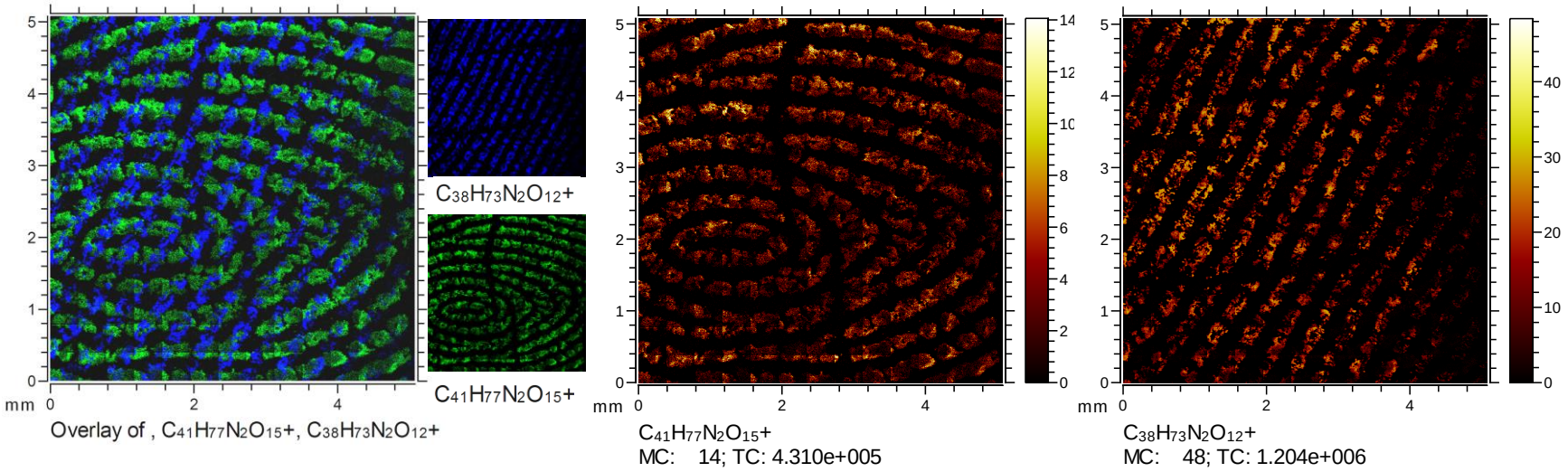
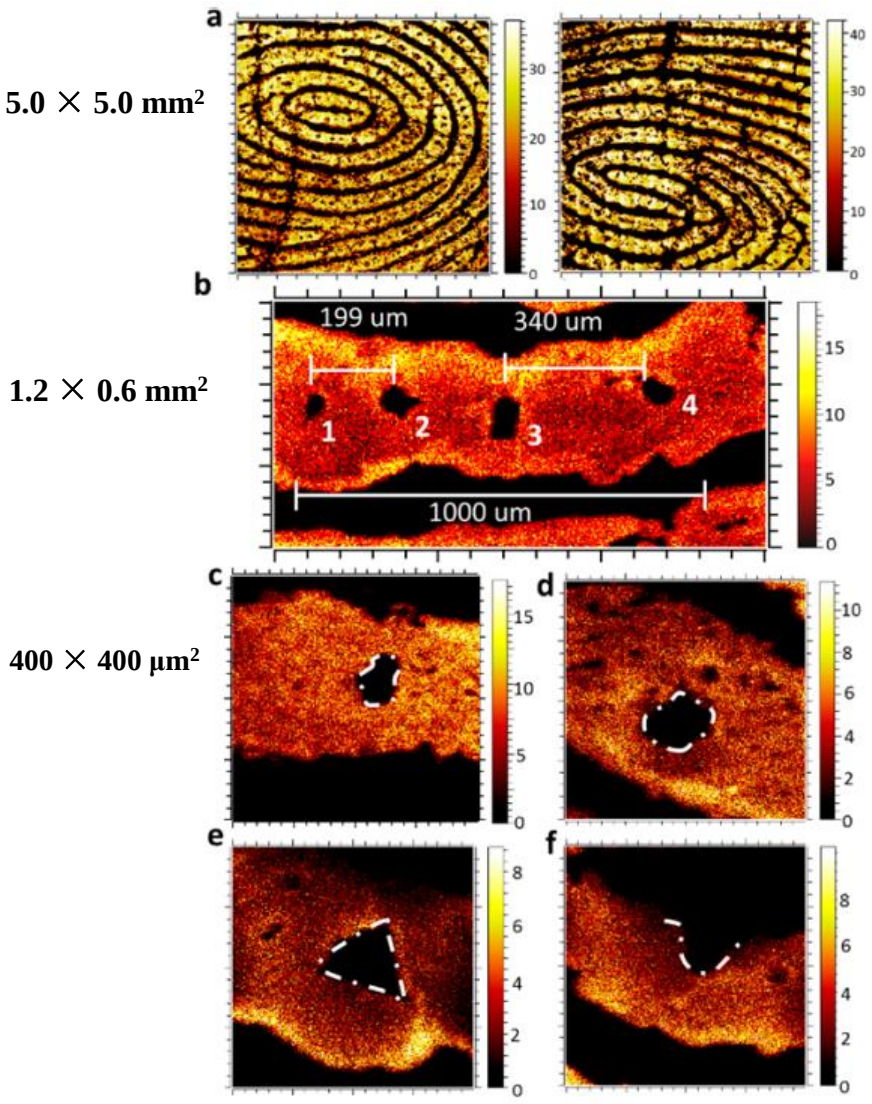



**Figure 1.**
显示出用于法庭指纹分析所必需细节的 $5\times5mm^2$ 指纹摩擦背脊(fingerprint friction ridges ) SIMS 像很引人注目。
Courtesy(允许) of Dr S. J. Hinder and Prof. J. F. Watts, University of Surrey, UK.

# Detection of overlaying fingerprints contact with different Antibiotics



# Images of molecular ions of roxithromycin at m/z 837.5 in fingerprints contaminated by roxithromycin solution




a) showed pattern features of two fingerprints, revealing two pattern types, loop (left) and double loop (right).

b) the number and position of pores could be easily observed in a ridge and the precise distance between two pores could also be obtained.

c–e) shows three closed pores.

c),d) the pores are nearly rounds with different sizes.

e) The pores is nearly a triangle.

f) shows an open pore, which is located at the edge of a ridge.

# 用TOF-SIMS调查出自韩国的天然染料和古代纺织品

Table I　从植物中提取染料的细节以及从它们的TOF-SIMS质谱中得到的特定的离子

**Table 1**
Details of the dyes extracted from the plants and specific ions obtained from their TOF-SIMS spectra

| Name of plant botanical (English) | Part employed for dyeing | Major dye component | MW | Form and solubility[a] | Peak mass for positive ion ($m/z$) | |
|---|---|---|---|---|---|---|
| | | | | | Molecular ion, $M^+$ | Fragment ion |
| *Curcuma longa* L. (turmeric) | Rhizome | Curcumin | $C_{21}H_{20}O_6$ 368.38 | Orange-yellow crystal pow. w i, eth i, al s, aa s | $(M+H)^+$, 369 | $C_{20}H_{19}O_5$ (339), $C_{19}H_{17}O_4$ (309), $C_{10}H_9O_3$ (177) |
| *Gardenia jasmoides Ellis* (gardenia) | Ripe fruits | Crocin | $C_{44}H_{64}O_{24}$ 976.96 | Brownish red needles hot w s, al s, eth s | – | $C_{12}H_{22}O_{10}$ (326), $C_{12}H_{22}O_9$ (310), $C_{11}H_{20}O_9$ (296) |
| *Carthamus tinctorius* L. (safflower) | Flowers | Safflor yellow carthamin | $C_{43}H_{42}O_{22}$ 910.79 | Dark red powder, dil alkali solution s, eth s | – | $C_{37}H_{37}O_{21}$ (817), $C_{31}H_{26}O_{16}$ (654), $C_{22}H_{21}O_{11}$ (461), $C_{21}H_{22}O_{11}$ (450), $C_{12}H_{13}O_8$ (285), $C_{12}H_{13}O_7$ (269) |
| *Rubia cordifolia* L. *Rubia akane nakai* (madder) | Root | Purpurin Alizarin | $C_{14}H_8O_5$ 256.21<br>$C_{14}H_8O_4$ 240.21 | Long orange needles hot w s, al s(red),eth s(int. yellow) | $(M+H)^+$, 257<br>$(M+H)^+$, 241 | $C_{14}H_8O_4$ (240)<br>$C_{13}H_9O_4$ (229) |
| *Caesalpinia sappon* L. (sapponwood) | Wood | Brazilin | $C_{16}H_{14}O_5$ 286.27 | Yellow crystals, turn orange in air and light, w s, al s, eth s | $(M+H)^+$, 287 | $C_{16}H_{13}O_4$ (269) |
| *Lithospermum erythror-hizon Shieb. et zucc.* (gromwell) | Root | Shikonin | $C_{16}H_{16}O_5$ 288.29 | Dark purple powder, w s, organic sol s | $(M+H)^+$, 289 | $C_{16}H_{15}O_4$ (271), $C_{10}H_5O_4$ (189) |
| *Indigofera tinctoria* L. (Indigo) | Leaves or whole plant | Indigotin or indigo | $C_{16}H_{10}N_2O_2$ 262.26 | Dark blue powder, w, al, eth, dil acid i | $(M+H)^+$, 263 | $C_{16}H_{11}N_2O$ (247), $C_{15}H_{11}N_2O$ (235) |

[a] w, water; eth, ether; al, alcohol; aa, acetic acid; s, soluble; I, insoluble; int, intense.

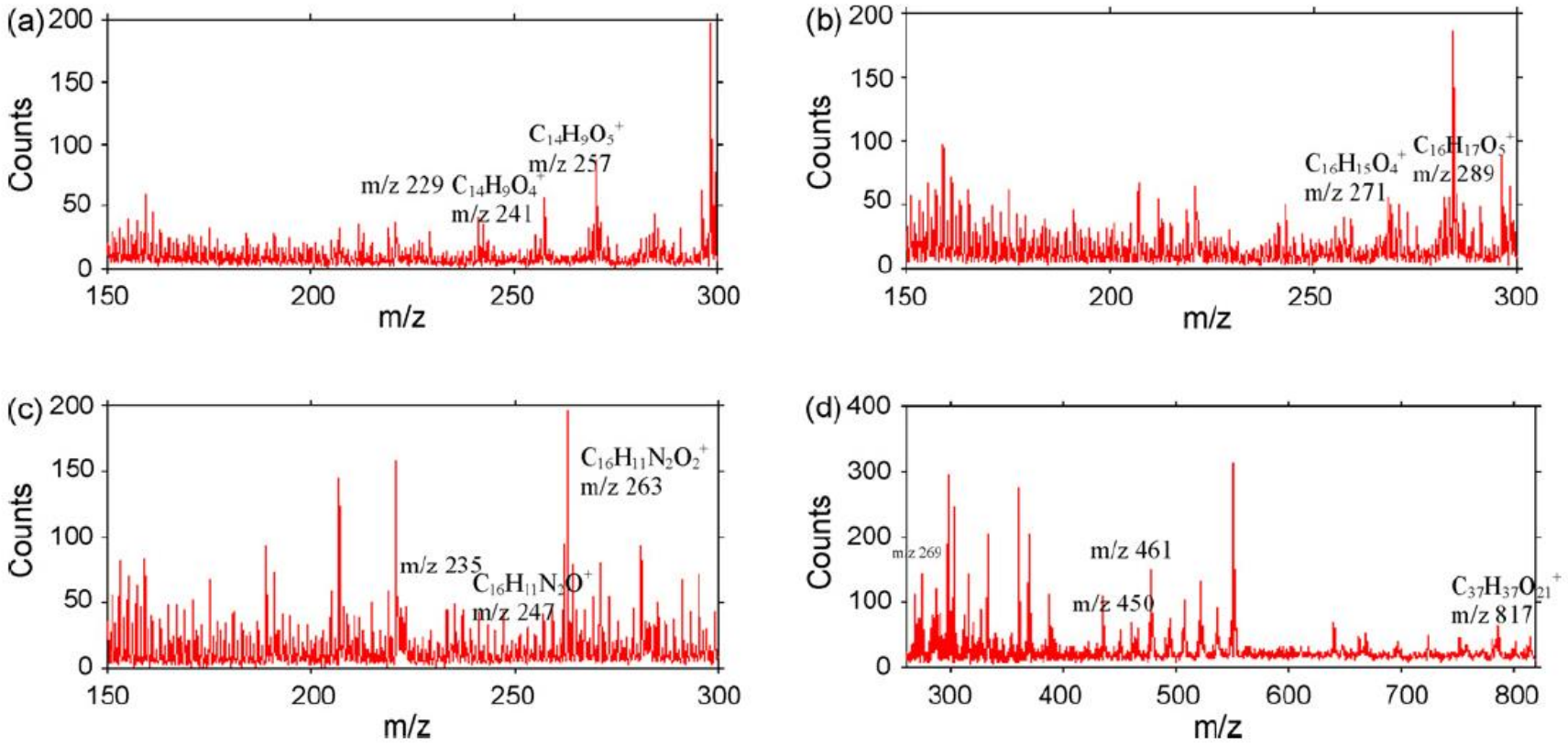



**Fig. 1**

**分别用(a)洋西草(madder)、(b)紫草(gromwell),、(c)靛青(indigo) 、(d)红花 (safflower)染料染的纺织品的正离子TOF-SIMS质谱。**

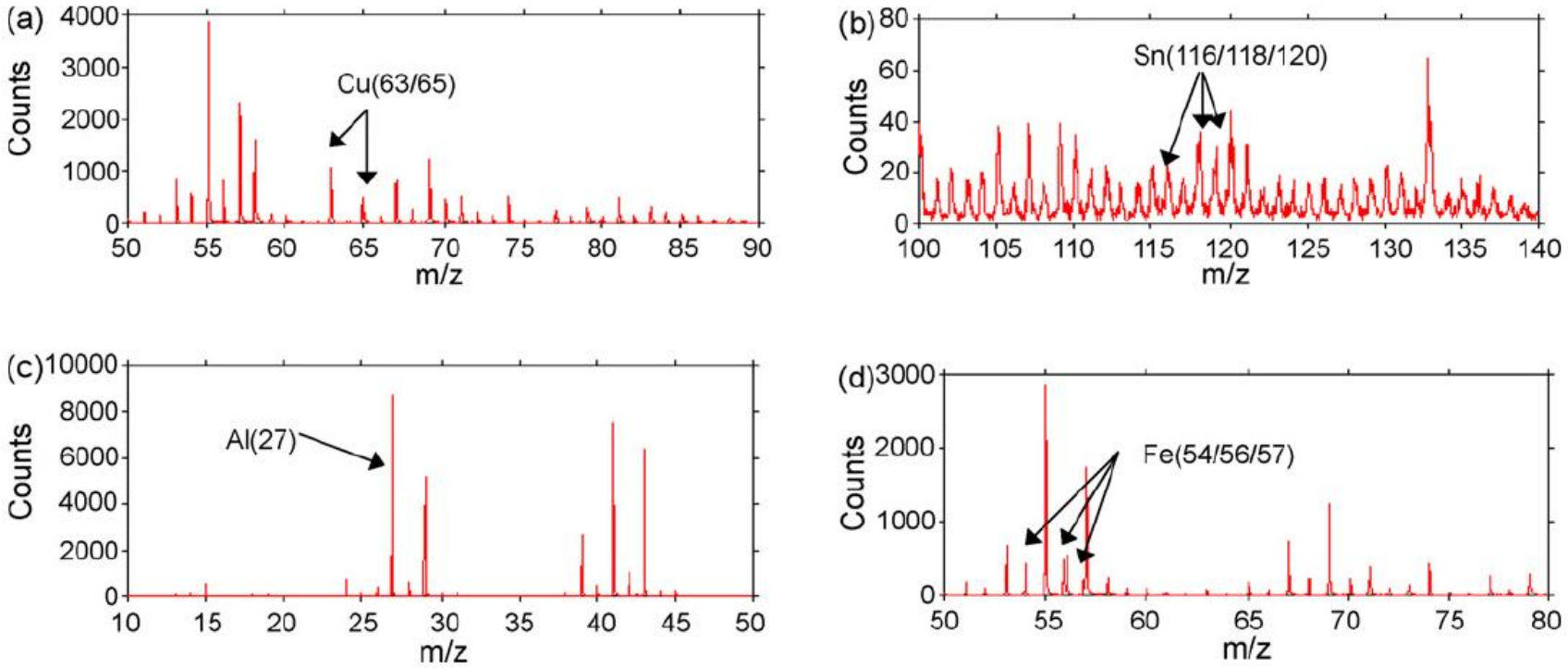



**Fig. 2.**

**分别用(a) sapponwood 和 $CuSO_4$、 (b) 姜黄(turmeric) 和 $SnCl_4$、(c) (turmeric)和 $KAl(SO_4)_2$、 (d) 红花(safflower) 和$FeSO_4$染料作为媒染剂( mordant)染的纺织品的正离子TOF-SIMS质谱。**

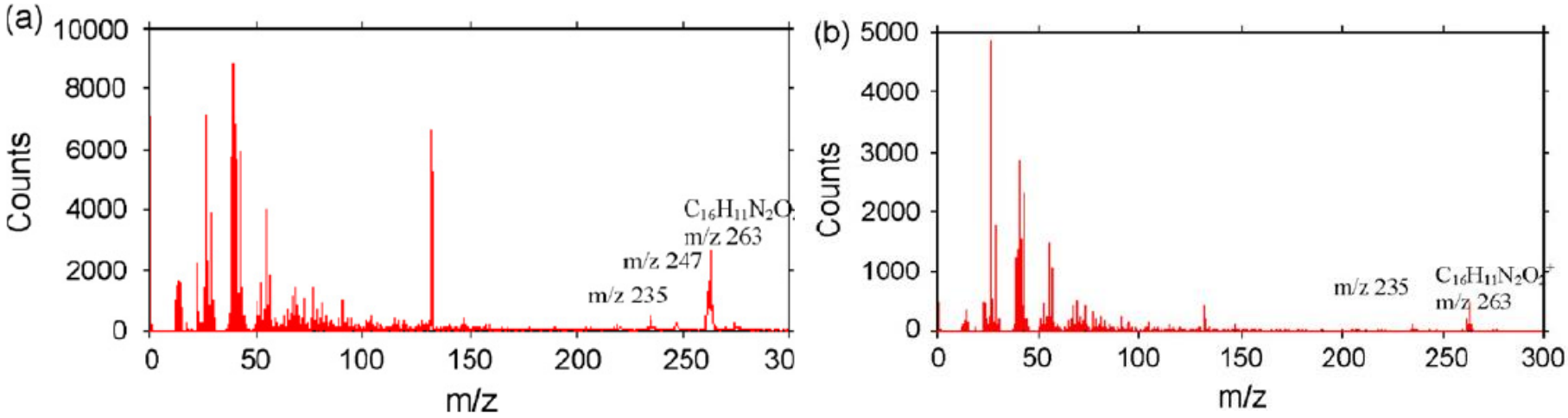


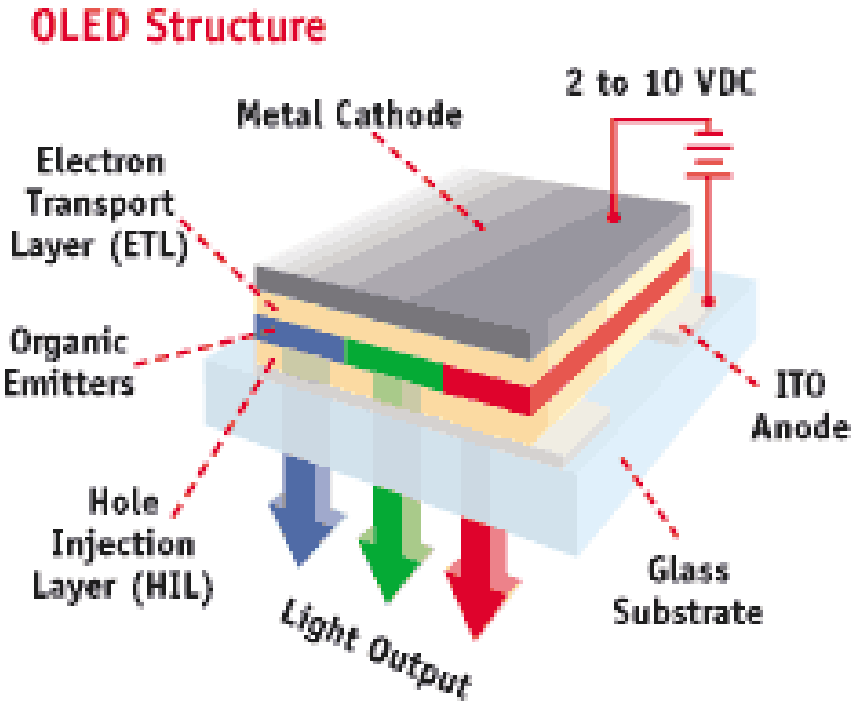

**Fig. 3. Positive ion TOF-SIMS spectra of antique textiles known as (a) Kim-Wak and (b) Gurye Sonssi.** 著名**Kim-Wak (a)**和**Gurye Sonssi (b)**古代纺织品的正离子**TOF-SIMS**质谱。

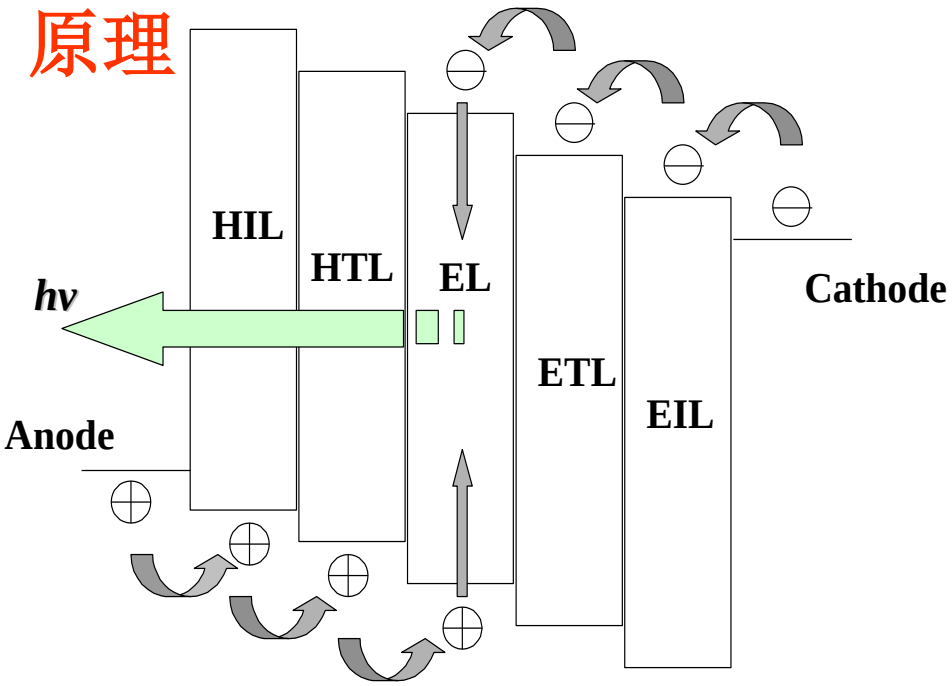
一个考古样品是取自**17**世纪的官服**(Kim-Wak)**

另一个考古样品是取自**16**世纪男人高丽服装的一小片**(Gurye Sonssi)**

# 用TOF-SIMS研究在OLED器件中多层有机层的结构

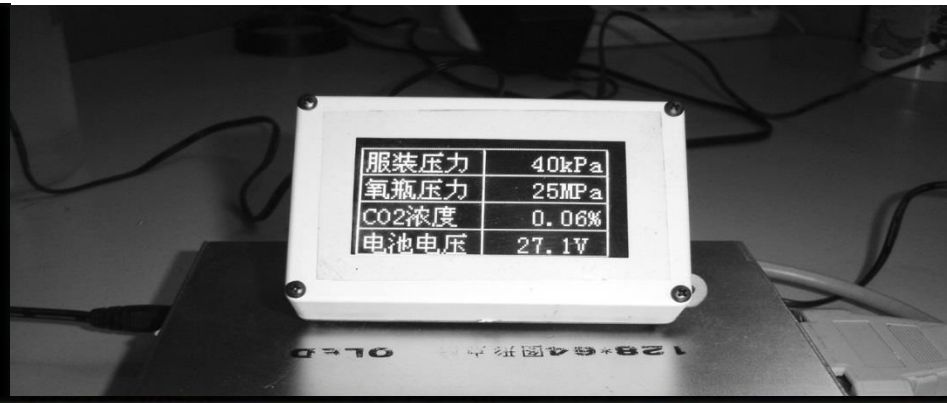
## OLED简单结构和原理



# “神七”航天员胸前发绿光的显示屏

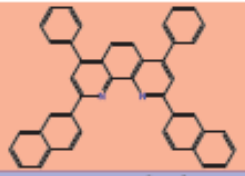
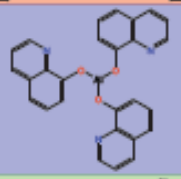
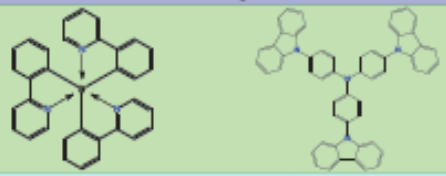
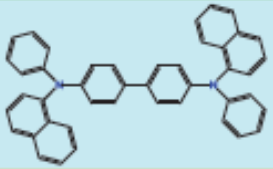
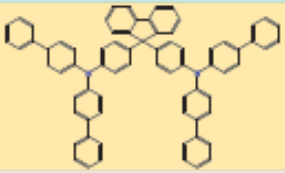




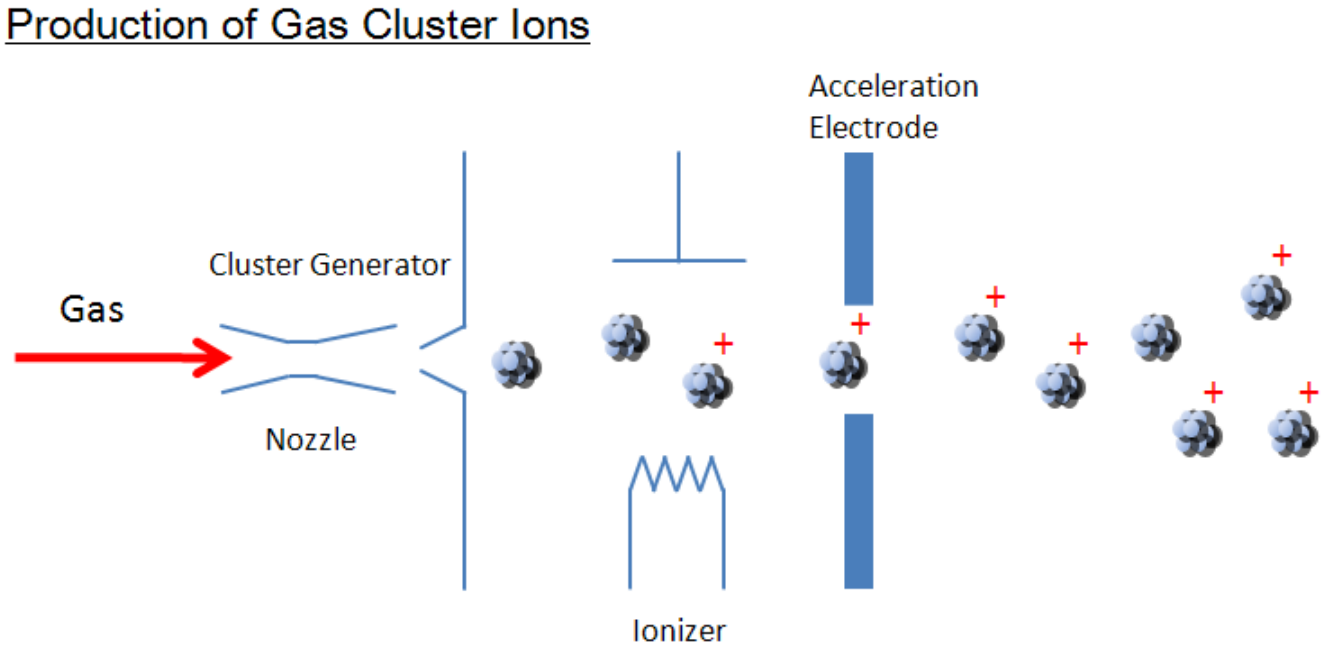


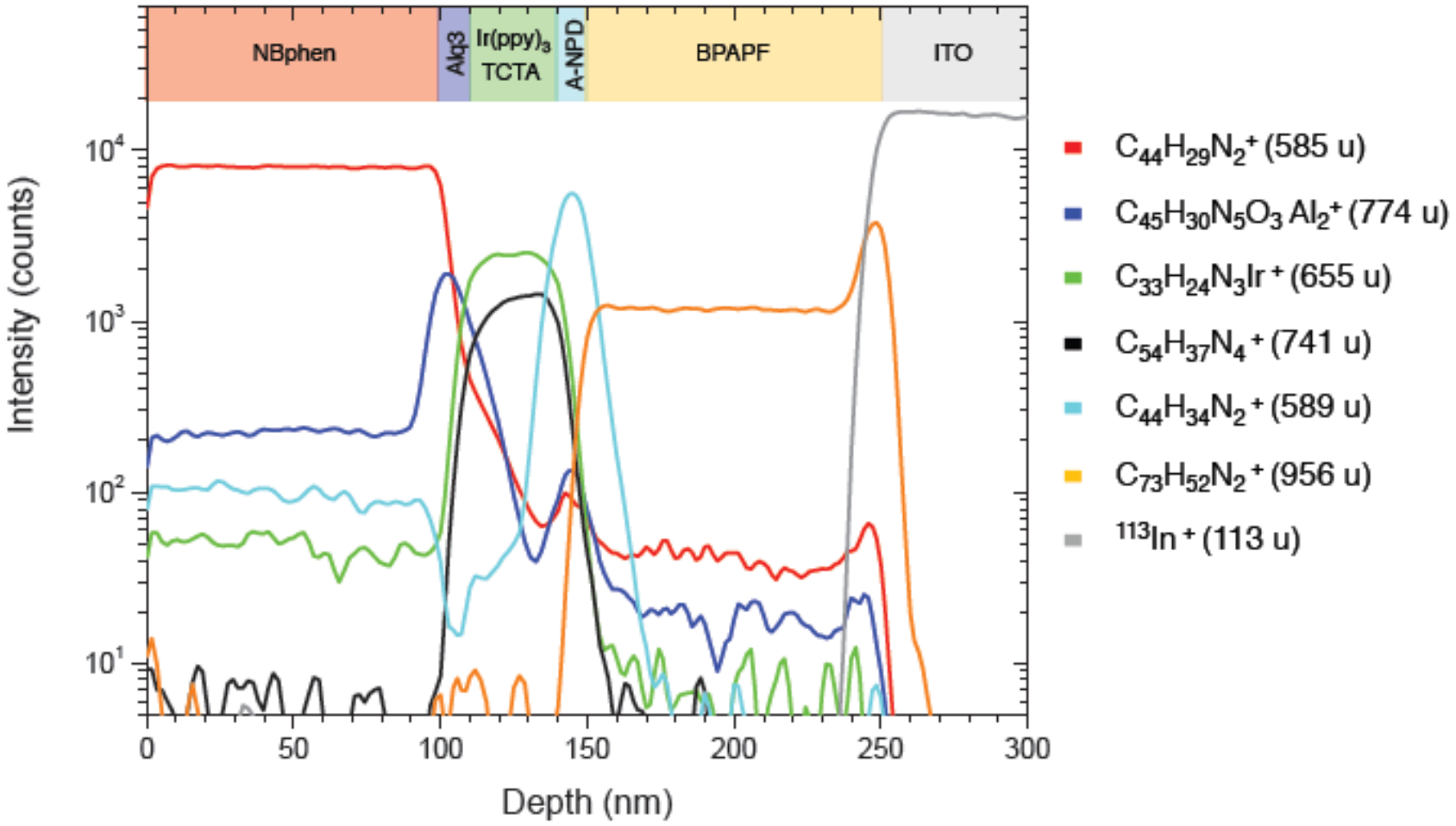


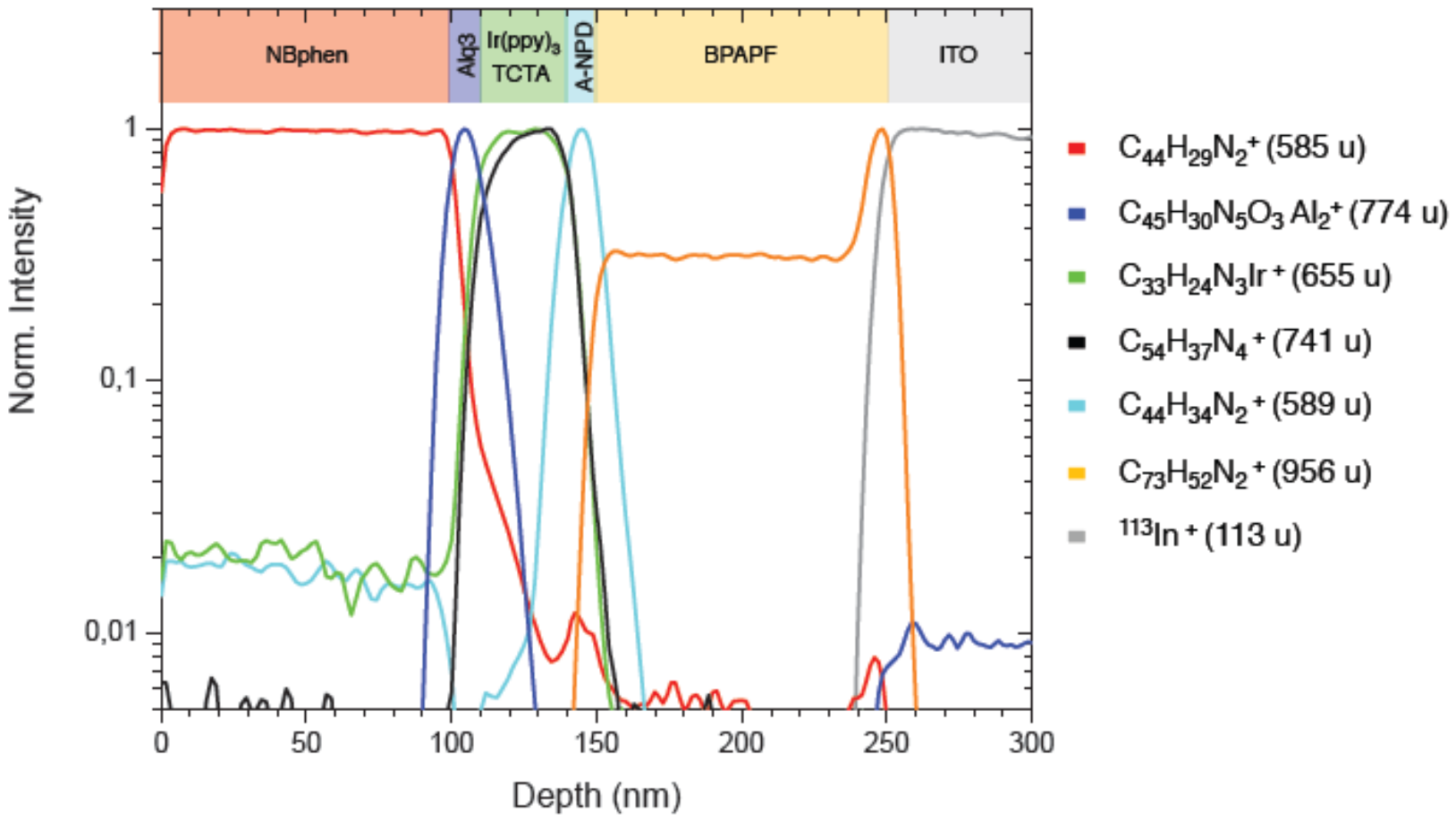
开国际先例，把OLED运用到“神七”航天服上：2008年9月27日，当“神七”航天员翟志刚打开载人飞船轨道舱舱门，迈出中国人在太空的第一步时，可看到翟志刚胸前的显示屏发出的绿光。

| layer construction | | layer composition | characteristic SI signal |
|---|---|---|---|
| 100 nm NBphen | ETL / HTL (electron transport- / hole blocking layer |  | $(M+H)^+$ $C_{44}H_{29}N_2^+$ 585 u |
| | |  | $(2M-R)^+$ $C_{45}H_{30}Al_2N_5O_5^+$ 774 u |
| 10 nm Alq3 | Flourescent Host | | |
| 30 nm $Ir(ppy)_3$ / TCTA | Green Dopant / Phosphorescent Host |  | $Ir(ppy)_3$: $(M)^+$: $C_{33}H_{24}IrN_3^+$: 655 u TCTA: $(M+H)^+$: $C_{54}H_{37}N_4^+$: 741 u |
| 10 nm a-NPD | HTL | | |
| 100 nm BPAPF | HTL (hole transport layer) |  | $(M+H)^+$ $C_{44}H_{33}N_2^+$ 589 u |
| | |  | $(M)^+$ $C_{73}H_{52}N_2^+$ 956 u |
| ITO | Substrate | $In_2O_3$ / $SnO_2$ | $^{113}In^+$ 113 u |

# Ar团簇离子枪（GCIB）

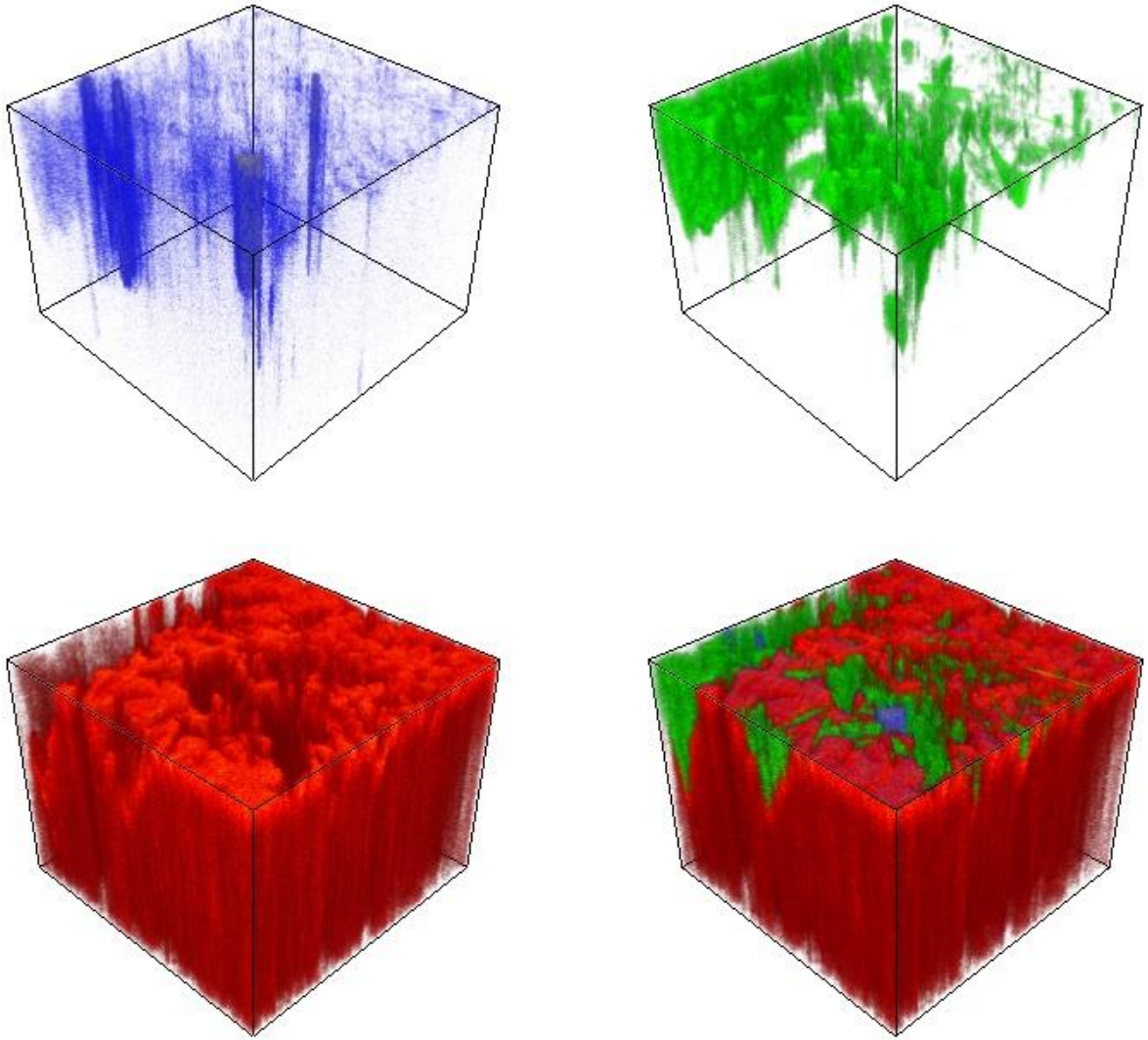



NBphen
Alq3
Ir(ppy)3
TCTA
A-NPD
BPAPF
ITO
Intensity (counts)
10^4
10^3
10^2
10^1
0
50
100
150
200
250
300
Depth (nm)
C44H29N2+ (585 u)
C45H30N5O3Al2+ (774 u)
C33H24N3Ir+ (655 u)
C54H37N4+ (741 u)
C44H34N2+ (589 u)
C73H52N2+ (956 u)
113In+ (113 u)


NBphen
Alq3
Ir(ppy)3
TCTA
A-NPD
BPAPF
ITO
Norm. Intensity
1
0,1
0,01
0
50
100
150
200
250
300
Depth (nm)
C44H29N2+ (585 u)
C45H30N5O3 Al2+ (774 u)
C33H24N3Ir+ (655 u)
C54H37N4+ (741 u)
C44H34N2+ (589 u)
C73H52N2+ (956 u)
113In+ (113 u)

# Polymer Additive in Polypropylene

## 3D Analysis of an Antioxidant in PP





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# 国内外二次离子质谱学术活动

# 国际二次离子质谱学会议

- 国际二次离子质谱学会议由A. Benninghoven教授于1977年创立，在德国Muenster大学召开了第一届国际二次离子质谱学会议(SIMS- I)，从此开始，两年一届的国际SIMS系列会议从来没有中断过，至今已经开过二十届，历届国际二次离子质谱学会议概况如下表所示。
- 最近一届会议(SIMS- XX)于2017年9月10日到9月15日在波兰的克拉科夫(Krakow, Poland)召开。



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| | | Summary of international SIMS conference | | | |
|---|---|---|---|---|---|
| Conference | Year | Country | City | Publication house | China |
| SIMS-I | 1977 | Germany | Munster | | |
| SIMS-II | 1979 | USA | Stanford | Springer-Verlag | |
| SIMS-III | 1981 | Hungary | Budapest | Springer-Verlag | |
| SIMS-IV | 1983 | Japan | Osaka | Springer-Verlag | 3(2) |
| SIMS-V | 1985 | USA | Washington | Springer-Verlag | 1(1) |
| SIMS-VI | 1987 | France | Paris | JohnWiley & Sons | 1(1) |
| SIMS-VII | 1989 | USA | Califarnia | JohnWiley & Sons | 1(1) |
| SIMS-VIII | 1991 | Netherlands | Amsterdam | JohnWiley & Sons | 2(2) |
| SIMS-IX | 1993 | Japan | Yokohama | JohnWiley & Sons | 4(3) |
| SIMS-X | 1995 | Germany | Munster | JohnWiley & Sons | 2(1) |
| SIMS-XI | 1997 | USA | Orlando | JohnWiley & Sons | 3(2) |
| SIMS-XII | 1999 | Belgium | Brussels | ELSEVIER | 2(2) |
| SIMS-XIII | 2001 | Japan | Nara | Applied Surface Science | 1(1) |
| SIMS-XIV | 2003 | USA | San Diego | Applied Surface Science | |
| SIMS-XV | 2005 | UK | Manchester | Applied Surface Science | |
| SIMS-XVI | 2007 | Japan | Kanazawa | Applied Surface Science | 5(5) |
| SIMS-XVII | 2009 | Canada | Toronto | Applied Surface Science | 3(3) |
| SIMS-XVIII | 2011 | Italy | Riva del Garda | Surface and Interface Analysis | |
| SIMS-XIX | 2013 | Korea | Jeju | Surface and Interface Analysis | |
| SIMS-XX | 2015 | USA | Seattle | Vacuum Science & Technology B | |
| SIMS-XXI | 2017 | Poland | Krakow | Vacuum Science & Technology B;Biointerphases | |

# 中国二次离子质谱学会议

- 中国二次离子质谱学会议已召开过四次，发展到由中国大陆、香港和台湾联合召开。
- 第一届中国二次离子质谱学会议(SIMS I，China)于1993年5月31- 6月5日在北京清华大学召开。
- 1998国际二次离子质谱学研讨会和第二届中国二次离子质谱学会议(ISSIMS 98/SIMS II, China)于1998年4月6-10日在北京清华大学召开。
- 2005全球华人二次离子质谱学及相关课题会议(WCSIMS)于2005年1月15-10日在新竹清华大学召开。
- 第四届中国二次离子质谱学会议暨2008北京二次离子质谱学国际研讨会(SIMS IV/ISSIMS 08 BEIJING)于2008年10月26-29日在北京矿业大学召开。

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# 第五届中国二次离子质谱学会议

## The 5th Chinese National Conference on Secondary Ion Mass Spectrometry

## SIMS China V

会议时间：2014 年10 月18–21 日

会议地点：北京，中国科学院地质与地球物理研究所

以后两年一届，在偶数年份召开

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# 第七届中国二次离子质谱学会议

## The 7th Chinese National Conference on Secondary Ion Mass Spectrometry

## SIMS China VI

会议时间：2018 年10月8–12 日

会议地点：苏州，中国科学院苏州纳米技术与纳米仿生研究所


西交利物浦国际会议中心欢迎您

TOF-SIMS能以极高的灵敏度(ppm～ppb)包括H在内的所有元素及其化合物信息。它是一种普适的分析技术，适用于有机、无机、生物、医学、电子、地质/考古、环境等各种领域的各种固体材料的分析。

TOF-SIMS具有以下几个特点：

1）能检测出包括氢在内的所有元素及其同位素，
2）对所有元素具有ppm～ppb(因元素而异)的检测灵敏度，
3）能测定从表面到深至数十μm的微量元素及化合物浓度布，
4）能观测各种元素及化合物的二维以及三维的分布，
5）能鉴定有机化合物分子。

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# 谢谢！

